

Shell-and-Tube Heat Exchangers

Technical Ballot Draft

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DRAFT FOR TENTH EDITION,



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Introduction

It is necessary that users of this standard be aware that further or differing requirements can be needed for individual applications. This standard is not intended to inhibit a vendor from offering, or the purchaser from accepting, alternative equipment or engineering solutions for the individual application. This can be particularly applicable where there is innovative or developing technology. Where an alternative is offered, the vendor should identify any variations from this standard and provide details.

This standard requires the purchaser to specify certain details and features.

A bullet (●) at the beginning of a section indicates a requirement for the purchaser to make a decision or provide information (for information, a checklist is provided in Annex B).

In this standard, where practical, U.S. Customary (USC) or other units are included in parentheses for information.

Shell-and-Tube Heat Exchangers

1 Scope

This standard specifies requirements and gives recommendations for the mechanical design, material selection, fabrication, inspection, testing, and preparation for shipment of shell-and-tube heat exchangers for the petroleum, petrochemical, and natural gas industries.

This standard is applicable to the following types of shell-and-tube heat exchangers: heaters, condensers, coolers, and reboilers.

This standard is not applicable to vacuum-operated steam surface condensers, waste heat boilers and feedwater heaters. For supplemental informative guidance on waste heat boilers and design features (including firetubes, watertubes, ferrules, channel side refractory lining, internal bypass systems, etc.) see API Recommended Practice 669 - *Heat Recovery Systems*.

This standard is intended for the supply and fabrication of new shell-and-tube heat exchangers. The applicability of this standard to the modification, alteration, or replacement of existing shell-and-tube heat exchangers or components, including tube bundles is subject to agreement between the purchaser and the vendor, provided that any applicable jurisdictional requirements are met.

2 Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

AMPP SP21534, *Material and Fabrication Requirements for Carbon Steel Pressure Vessels in Petroleum Refining Wet H₂S Environments*

API Technical Report 938-C, *Use of Duplex Stainless Steels in the Oil Refining Industry*

API Standard 521, *Pressure-relieving and Depressuring Systems*

API Recommended Practice 941, *Steels for Hydrogen Service at Elevated Temperatures and Pressures in Petroleum Refineries and Petrochemical Plants*

API Recommended Practice 582, *Welding Guidelines for the Chemical, Oil, and Gas Industries*

ASME B16.5 ¹, *Pipe Flanges and Flanged Fittings: NPS 1/2 through NPS 24 Metric/Inch Standard*

ASME PCC-1, *Pressure Boundary Bolted Flange Joint Assembly*

ASME PCC-2, *Repair of Pressure Equipment and Piping*

EJMA ², *Standards of the Expansion Joint Manufacturers Association*

NACE MR0103 ³, *Materials Resistant to Sulfide Stress Cracking in Corrosive Petroleum Refining Environments*

¹ ASME International, 2 Park Avenue, New York, New York 10016-5990, www.asme.org.

² Expansion Joint Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591, www.ejma.org.

³ NACE International (formerly the National Association of Corrosion Engineers), 1440 South Creek Drive, Houston, Texas 77084-4906, www.nace.org.

NACE MR0175, *Petroleum and natural gas industries—Materials for use in H₂S containing environments in oil and gas production—Parts 1, 2 and 3*

TEMA⁴, *Standards of the Tubular Exchanger Manufacturers Association*, Tenth Edition

3 Terms and Definitions

For the purposes of this document, the following definitions apply.

3.1

annular distributor

An additional chamber incorporated into a shell side nozzle to evenly distribute shell side fluids entering or exiting the tube bundle.

3.2

category A welded joint

Longitudinal welded joint within the main shell, communicating chambers, nozzles, or transitions in diameter; or any welded joint within a sphere or within a formed or flat head; or circumferential welded joint connecting hemispherical heads to main shells, to transitions in diameters, or to communicating chambers.

3.3

category B welded joint

Circumferential welded joint within the main shell, communicating chambers, nozzles, or transitions in diameter including joints between the transitions and a cylinder at either the large or small end; or circumferential welded joint connecting formed heads, other than hemispherical to main shells, to transitions in diameter, to nozzles, or to communicating chambers.

3.4

communicating chamber

Heat exchanger appurtenance that intersects the shell or heads of the heat exchanger and forms an integral part of the pressure-containing envelope.

EXAMPLE Sump, Annular Distributor.

3.5

cyclic service

Services which as part of their intended operation are subjected to the application of fluctuating mechanical and/or thermal loads, which must be considered in the mechanical design.

EXAMPLE: Exchangers in batch operating service, exchangers in intermittent service, exchangers used as part of reactor catalyst regeneration systems, exchangers with different modes of operations, etc.

3.6

effective heat transfer area

Outside surface area of the tubes that contributes to heat transfer including finned surface (if any).

3.7

flat bar U-bend support

A type of vibration support consisting of a system of flat bars used in a U-tube bundle with a square tube layout. Flat bar support strips extend through the bend field and are either bolted together or welded to a peripheral flat bar outer ring.

NOTE 1 The system only provides out-of-plane support to the U-bends but allows for thermal expansion of all tube rows.

NOTE 2 See Figure 1 for details.

⁴ Tubular Exchanger Manufacturers Association, 25 North Broadway, Tarrytown, New York 10591, www.tema.org.

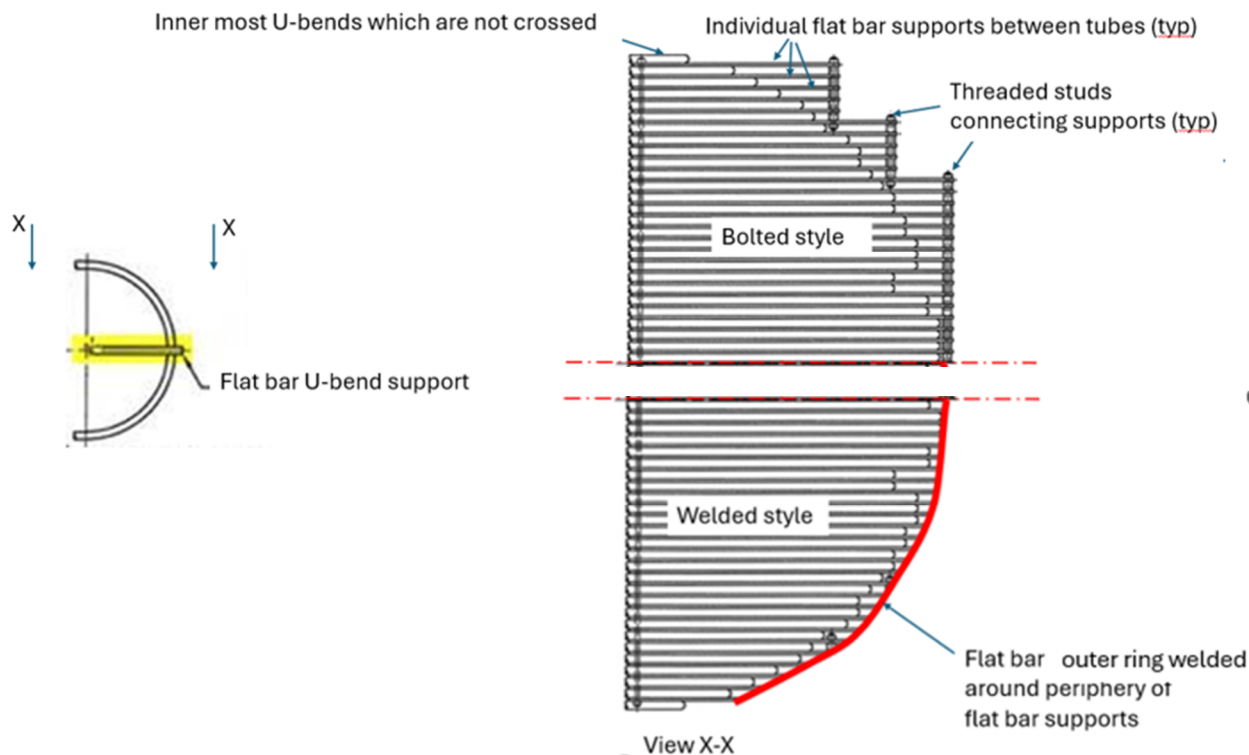


Figure 1 – Flat bar U-bend Support Examples

3.8

full-penetration weld

Welded joint which results in weld metal through the entire thickness of the components being joined.

3.9

heat exchanger unit

One or more heat exchangers arranged in series or parallel for a specified service that operate together to perform the intended duty.

3.10

high temperature hydrogen attack

HTHA

Damage mechanism affecting carbon and low-alloy steels due to the exposure to hydrogen at elevated temperatures and pressures, resulting in the loss of carbides and strength of the materials and cracking.

3.11

high temperature hydrogen service

HTHS

Services with operating hydrogen partial pressure above 350 kPa (50 psi) absolute and operating temperatures above 204 °C (400 °F).

3.12

hydrogen charging service

Service in which the diffusion of atomic hydrogen can occur in the steel

NOTE Hydrogen charging services include wet hydrogen sulphide, sour service, hydrofluoric acid service or hydrogen service where the operating temperature is greater than 204 °C (400 °F).

3.13

hydrogen service

Process streams containing hydrogen at a partial pressure greater than 350 kPa (50 psi) absolute.

3.14**item number**

Purchaser's identification number for a heat exchanger unit.

3.15**minimum design metal temperature****MDMT**

Lowest metal temperature at which pressure-containing components can be subjected to design pressure.

EXAMPLE Ambient temperature or process fluid temperature.

3.16**nubbin**

Projection on the flange gasket surface, positioned at the center of the gasket, used to concentrate the bolt load on the gasket.

3.17**pressure design code**

Recognized pressure vessel design code or standard as specified or agreed by the purchaser.

EXAMPLE ASME BPVC, Section VIII, Division 1, EN 13445 (all parts).

3.18**seal-welded**

Tube-to-tubesheet joint weld of unspecified strength applied between the tubes and tubesheets for the sole purpose of reducing the potential for leakage.

3.19**strength-welded**

Tube-to-tubesheet joint welded so that the design strength is equal to, or greater than, the axial tube strength specified by the pressure design code.

3.20**tie plate U-bend support**

A type of vibration support consisting of plate drilled with the same pattern as the tube layout, cut into strips along the tube row centerlines and reconstructed sequentially with each row of U-tubes as they are stabbed into the baffle cage.

NOTE 1 Strips are welded to a peripheral support bar.

NOTE 2 See Figure 2 for details.

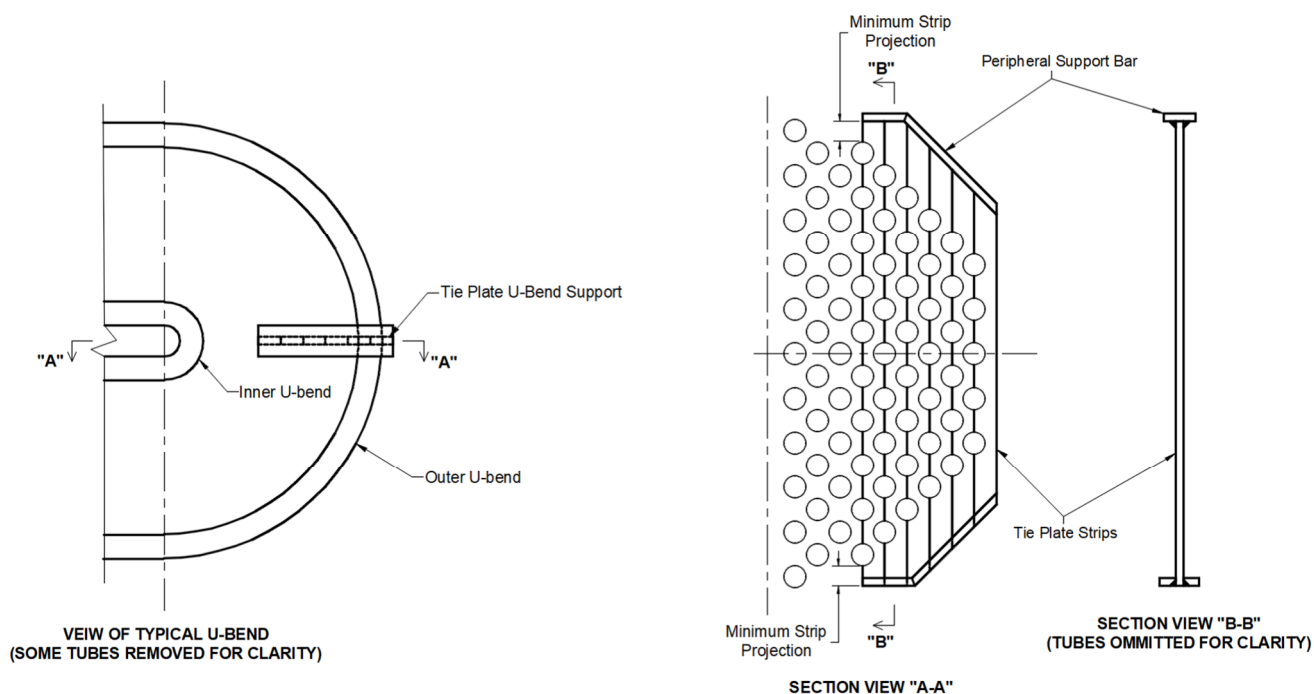


Figure 2 –Tie Plate U-bend Support Example

3.21

vibration support

Full or partial support designed to mitigate vibration issues within a tube bundle. They can be located at any point within a tube bundle, e.g. inlet or U-bend.

4 General

- **4.1** The pressure design code shall be specified or agreed by the purchaser. Pressure components shall comply with the pressure design code and the supplemental requirements given in this standard.
- 4.2** Heat-exchanger construction shall conform to TEMA, Class R, unless another TEMA class is specified.
- **4.3** The vendor shall comply with the applicable local regulations specified by the purchaser.
- 4.4** Annex A includes some recommended mechanical and design details for information.
- 4.5** Annex B provides a checklist that can be used by the purchaser to ensure that bulleted items in this standard are addressed.
- 4.6** Annex C provides examples of datasheets.
- **4.7** The purchaser shall specify if either stream has fluid characteristics requiring special considerations e.g. slurry, entrained particulates, or other certain types of fouling mechanisms.
- **4.8** The purchaser shall specify if cyclic service design is required. See Annex A.2.1 for additional guidance.
- 4.9** If cyclic service is specified, the purchaser shall specify the type and magnitude of variation in pressure, temperature and flow rate, the time for the variation (hours, weeks, months, etc.) and the number of cycles or frequency for this variation expected during the life of the equipment. The extent and acceptance criteria of any required analysis shall be subject to the agreement of the purchaser. A.2.1 for guidance on cyclic service.

- **4.10** The purchaser shall specify if the service is designated as sour or wet hydrogen sulfide service. Identification of the complete set of materials, qualification, fabrication, and testing specifications to prevent in-service environmental cracking is the responsibility of the purchaser. See A.2.2 for guidance on sour or wet hydrogen sulfide service.

Note: General definitions for sour or wet H₂S service can be found in NACE MR0103, AMPP SP21534 and MR0103.

- **4.11** The purchaser shall specify if the shell and/or tube side is in hydrogen service.
- **4.12** For single tube pass floating-head and all fixed tubesheet heat exchangers, the purchaser shall specify the data required to determine the need for an expansion joint. This shall include all intended operating conditions defined on the expansion joint datasheet shown in Annex C. See A.2.3 for additional information.
- 4.13** For multiple tube pass heat exchangers, tube mean metal temperatures for each individual pass shall be specified on the datasheet whenever the tube side operating temperature range exceeds 110 °C (200 °F) per shell.
- 4.14** For shell side crossflow (TEMA X-shells) the exchanger configuration should ensure distribution of the fluid across the entire length of the tube bundle, except for cases with boiling or condensing of single (pure) component fluids. See A.2.4 for guidance.
- **4.15** The purchaser shall specify design loads and load combinations to be used for the analysis of heat exchanger supports.
- 4.16** For flow induced and acoustic vibration considerations, see A.2.5.
- **4.17** The purchaser shall consider if a tube rupture scenario is to be taken into consideration in the selection of the design pressure of the low-pressure side of the heat exchanger, which would otherwise require a provision for pressure relief in accordance with API Standard 521. See A.2.6 for guidance.
- **4.18** The purchaser shall specify if the service is in HTHS and the material selection temperatures for each side.

5 Proposal Information Required

5.1 For each heat exchanger unit, the vendor's proposal shall include completed datasheets, similar to those given in Annex C, or if a datasheet is included in the inquiry, a statement indicating complete compliance with that datasheet.

5.2 The vendor's proposal shall include sketches that are sufficient to describe the construction for the following, if provided:

- a) design features that are not fully defined by the nomenclature in TEMA, Section 1;
- b) preliminary dimensions for the annular distributor;
- c) floating-head designs which include a tail-pipe and expansion joint;
- d) expansion joints for fixed tubesheet exchangers. Including the location, material and type;
- e) high pressure closure including channel to tubesheet attachment, shell to tubesheet attachment, and welded diaphragm or welded lip-seal construction.

5.3 The proposal shall include a detailed description of all exceptions to the requirements of the purchaser's inquiry, including a setting plan when provided.

5.4 For stacked heat exchangers, the vendor shall supply the following components unless otherwise specified by the purchaser:

- a) bolts, nuts, and gaskets for interconnecting nozzles;

- b) shims or spacers and bolting for interconnecting supports.

5.5 The vendor shall provide a separate quotation for the following items, unless otherwise specified by the purchaser.

- a) A test component consisting of a test ring and gland, in accordance with TEMA, Figure E-4.13-2 or equivalent, for each heat exchanger or group of similar heat exchangers with floating heads.
- b) One spare gasket for each external girth flange, floating-head flange (if applicable), and internal pass-partition flange (if applicable) per heat exchanger.

5.6 When the pressure design code does not include rules necessary for the design of all components the method of analysis to be applied shall be agreed with the purchaser. This includes, for example, thin flexible tubesheets or large untubed areas of a tubesheet.

5.7 When thermal and hydraulic design is in the scope of supply of the vendor, the proposal shall include calculations to support the design, including flow induced and acoustic vibration analysis.

6 Drawings and Other Required Data

6.1 Outline Drawings and Other Supporting Data

6.1.1 The vendor shall submit, for review by the purchaser, outline drawings for each heat exchanger unit. The drawings shall include at least the following information:

- a) service, item number, project name and location, purchaser's order number, vendor's shop order number, and other special identification numbers;
- b) design pressure, test pressure, maximum design temperature, minimum design metal temperature (MDMT), and any restriction on testing or operation of the heat exchanger;
- c) maximum allowable working pressure (MAWP) in the corroded condition and at the design temperature for the shell side and tube side;
- d) connection sizes, location, orientation, projection, direction of flow and if flanged, the rating and facing, or if welded to the connecting piping, the weld bevel preparation;
- e) coupling sizes, rating, and orientation;
- f) dimensions, orientation and location of supports, including bolt holes and slots, and the stacking arrangement;
- g) overall dimensions of the heat exchanger;
- h) tube bundle removal and component dismantling clearance;
- i) mass of the heat exchanger, empty and full of water, and of removable components with a mass greater than 25 kg (60 lb), e.g. removable tube bundle, channel, channel cover, and shell cover. When the operating fluid density is greater than water, the density of the process fluid should be used when calculating the full of fluid mass;
- j) specified corrosion allowance for each side of the heat exchanger (see A.3.1);
- k) references to the applicable code and the purchaser's specification;
- l) requirements for postweld heat treatment;
- m) requirements for non-destructive examination (NDE);
- n) requirements for material impact testing;

- o) requirements for surface preparation and painting;
 - p) gasket materials;
 - q) insulation thickness;
 - r) location of expansion joints, annular distributors, and any other special components or closures;
 - s) location and orientation of nameplates, lifting lugs, grounding clips or other attachments;
 - t) location of the center of gravity of the heat exchanger (empty and full of water);
 - u) forces and moments on connections as specified by the purchaser (see 7.6.10);
 - v) material specifications and grades for all components.
 - w) tabulation of required bolt loadings (bolt torque with assumed nut factors or bolt tension for bolts which are tensioned) along with any requirements for washers for girth flanges, channel covers, and floating-head flanges.
- **6.1.2** When thermal and hydraulic design is in the scope of supply of the vendor, or when a check-rating of the exchanger designs provided by the purchaser is required, the vendor shall submit thermal and hydraulic design calculations including flow induced and acoustic vibration analysis. See A.3.2.

6.1.3 The vendor shall recommend the tools required for the assembly and maintenance of the heat exchanger. If torquing or tensioning of bolts is required, the vendor shall provide applicable procedures.

6.1.4 Reaction loads at base of supports for specified load cases.

6.2 Information Required After Outline Drawings Are Reviewed

6.2.1 The vendor shall submit gasket details, including type and material, on a separate drawing. This drawing shall not be marked with any restrictions for use.

6.2.2 The vendor shall submit copies of applicable welding procedure specifications and welding procedure qualifications for review or record, unless otherwise agreed.

6.2.3 Upon receipt of the purchaser's review comments on the outline drawings, the vendor shall submit copies of all detailed drawings and documentation for the purchaser's review. These shall fully describe the heat exchanger and shall include at least the following information:

- a) full views and cross-sectional views with all dimensions and materials sufficient for stress calculations for each part;
- b) tube bundle details, including the following:
 - tube layout;
 - tube description and number in each pass;
 - number of baffles, type and description (for segmental baffles include cross-baffle cut, layout, and orientation in a view that shows the cuts);
 - details and locations of all sealing and sliding strips;
 - details and locations of tie-rods and spacers;
 - details and locations of support plates;
 - details of tubesheet and tube holes, including cladding or weld overlay if required;

- details of pass-partition plates;
 - impingement protection device details, if applicable;
 - U-tube bend schedule, if applicable.
- c) details of each pressure-retaining weld, including weld material, weld nominal thickness, weld location, and applicable non-destructive examination method;
 - d) details of each weld and weld nominal thickness for nonpressure attachments welded to pressure parts and for all load bearing attachments;
 - e) complete bills of materials, including the material specification;
 - f) expansion joint details, including bellows manufacturers drawing if applicable;
 - g) details of cladding and weld overlay;
 - h) weld map for each heat exchanger showing the weld joints, including welding procedure number(s);
 - i) details of tube-to-tubesheet joints, including procedures for installation, welding, expansion, inspection, and testing;
 - j) flange-face finish;
 - k) special installation and maintenance instructions including lifting and handling;
 - l) total number of heat exchangers and their arrangement;
 - m) conditions for hydrogen, caustic or wet sour service (if applicable);
 - n) hydrostatic test pressures;
 - o) leak testing requirements;
 - p) lifting lugs;
 - q) test equipment which is shipped with the equipment, for example test covers and backing flanges;
 - r) nameplate and bracket details;
 - s) reference list of sub-vendor drawings such as bellows expansion joints, valves, separation equipment, spray nozzles, etc.;
 - t) U-bend heat treatment procedures (if applicable).

6.2.4 The vendor shall submit for the purchaser's review the following documentation, unless otherwise agreed. The scheduling of the documentation shall be agreed between the purchaser and the vendor:

- a) Mechanical design calculations for all the heat exchanger pressure-retaining components. If calculations are made using computer software, all input and output data shall be detailed so as to facilitate an understanding of the calculation procedures. The formulas in the applicable sections of the pressure design code and TEMA shall be referenced.
- b) Provide required bolt loadings on gasket drawing (bolt torque or tension) along with any requirements for spring washers.
- c) Design calculations based on seismic, wind, transportation, and piping loads, if these loads are provided by the purchaser.

- d) Proposed procedures for assembly of flanged joints.
- e) Design calculations for thermal loads imposed on nozzles of stacked heat exchangers, when the shell mean metal temperature differential between the stacked shells exceeds 110 °C (200 °F) for ferritic steels and 75 °C (135 °F) for non-ferritic steels.
- f) Design calculations for expansion joints, if applicable.
- g) Design calculations for the thickness of pass partition plates and longitudinal baffles, if applicable.
- h) When specified as required, the details of any specialized bundle handling devices.
- i) Inspection, testing and fabrication procedures, as applicable.

6.2.5 The vendor shall submit calculations verifying the design for the supports, and any lifting and pulling devices specified by the purchaser.

6.2.6 The vendor shall submit calculations verifying the design for the diameter of the anchor bolt based on the structure or foundation type, coefficient of friction, anchor bolt material and allowable stresses specified by the purchaser.

6.2.7 After final review the vendor shall revise all the required drawings and welding procedures and submit each with the following text marked on every sheet separately and dated: "CERTIFIED FOR CONSTRUCTION".

6.3 Final Reports and Records

- After the heat exchanger is completed, the vendor shall furnish the purchaser with the following documents in the format and quantities specified by the purchaser:
 - a) "as-built" datasheet;
 - b) all outline and detail drawings, marked "CERTIFIED AS-BUILT";
 - c) certified record of all impact tests performed;
 - d) certified material test reports for all pressure parts, including tubes and load bearing attachments welded to the pressure boundary, e.g. pass partitions, lifting lugs, and saddle wear plates. Each material test report shall be identified by a part number;
 - e) welding consumable certificates;
 - f) complete certified bill of materials suitable for obtaining all replacement parts, including quantity, description, material specification, and identification of each part;
 - g) temperature charts of all postweld heat treatments;
 - h) completed manufacturer's data report in accordance with the pressure design code;
 - i) nameplate rubbing or a facsimile;
 - j) all mechanical design calculations, marked "CERTIFIED AS-BUILT";
 - k) non-destructive examination (NDE) map;
 - l) all associated NDE reports, including radiographic, magnetic-particle, liquid-penetrant, ultrasonic, hardness, impact, positive material identification (PMI), production tests and any other reports as applicable;
 - m) tube-to-tubesheet leak-test results;
 - n) hydrostatic test records in the form of a chart or certification;

- o) tube wall reduction records;
- p) when thermal, hydraulic, and vibration calculations are included in the vendor's scope of supply, the calculations shall be revised based on the as-built configuration;
- q) tabulation of required bolt loadings bolt torque with assumed nut factors or bolt elongations for bolts which are tensioned, along with any requirements for washers for girth flanges, channel covers, and floating-head flanges;
- r) For bundle of TEMA type C or D type head, bundle drawing showing center of gravity with and without channel cover included.

7 Design

7.1 Design Temperature and Design Pressure

- **7.1.1** All heat exchangers shall have at least two design temperatures for each side, a maximum design temperature and a minimum design metal temperature (MDMT), as specified by the purchaser. For additional information see A.4.1.

NOTE: The design temperature chosen may not be the same as the material selection temperature for high temperature hydrogen service.

7.1.2 Metal temperature reductions below the MDMT in cases of low temperature scenarios having coincident conditions of reduced operating stress shall not be taken unless specified or agreed by the purchaser. For additional information, see A.4.1.

Note: These conditions may occur, for example, during auto-refrigeration.

7.1.3 The design temperature of a component (including external bolting) influenced by both the shell side and tube side fluids shall be the more severe of either the shell side or tube side design temperature.

7.1.4 Differential pressure design shall not be used, unless otherwise specified by the purchaser.

7.2 Cladding for Corrosion Allowance

7.2.1 If cladding (including weld overlay) is used, the cladding thickness including weld overlay restoration shall be used only as corrosion allowance and not as the pressure retaining envelope, unless otherwise specified or approved by the purchaser.

7.2.2 Weld overlay thickness (including weld overlay restoration) shall comply with the following (see also 9.3.2):

- a) have sufficient thickness to provide the specified chemical composition to a depth of at least 1.5 mm ($1/16$ in.) from the finished surface, unless otherwise specified by the purchaser;
- b) have a minimum overall thickness of 3 mm ($1/8$ in.);
- c) pass partition grooves of tubesheets and gasket contact surfaces shall also comply with the requirements of 7.2.2 (a) & (b) after final machining;
- d) radius or profiling at nozzle connections shall not reduce any clad thickness below the specified minimum value. See 9.3 for details.

7.2.3 The cladding (including weld overlay) thickness at the tube side face of a tubesheet shall not be less than 10 mm ($3/8$ in.). For additional information see A.4.2.

7.2.4 The cladding (including weld overlay) thickness on the shell side face of a tubesheet shall not be less than 10 mm ($3/8$ in.).

7.2.5 Where either face of a tubesheet is clad (including weld overlay), any tapped holes (for pulling eyes, impingement rods, tie rods, etc.) shall be of the same nominal chemistry as the respective face. Weld overlay shall be deposited in an

over-sized hole. A minimum of 3 mm (1/8 in.) of weld overlay shall remain after drilling and tapping the hole. Alternative configurations may be used if approved by the purchaser.

7.2.6 The minimum cladding or weld overlay thickness at locations where the pass partition plates are welded shall be 5 mm (3/16 in.), whenever the base material requires post-weld heat treatment, unless the welding procedure specification utilized has been demonstrated to not affect the physical properties of the base metal.

7.2.7 Where clad components are used, the additional thickness at the pass partition plate junction may be achieved by weld deposit overlay prior to the heat treatment.

7.2.8 Where seal-welded diaphragm or lip-seal gaskets are used, the weld overlay shall have a minimum thickness of 6 mm (1/4 in.) applied to the gasket seating surface.

7.3 Shell Supports

7.3.1 When the support of a removable-bundle heat exchanger is fixed to the shell, it shall be designed to withstand a longitudinal force equal to 150 % of the bundle mass to account for friction and other loads. This force is to be applied at the heat exchanger bundle centerline. The shear stress for supports shall not exceed 40 % of the yield strength of the material.

7.3.2 Horizontal heat exchangers shall be provided with two or more saddles designed to support the heat exchanger under all specified conditions. Design of the saddles shall be as follows:

- a) saddles shall be attached to saddle bearing-plates;
- b) the bearing surface of the saddles shall be at least one-third of the circumference of the shell;
- c) saddle-bearing plates shall have the same nominal chemical composition as the shell and shall be continuously welded directly to the heat exchanger shells;
- d) saddle-bearing plates shall be provided with vent holes 6 mm (1/4 in.) in diameter, located at the vertical centerline;
- e) saddle-bearing plates shall be at least 6 mm (1/4 in.) thick and shall have all corners rounded to a radius of at least 25 mm (1 in.).

7.3.3 The lower shells of stacked removable-bundle heat exchangers shall be designed to carry the superimposed loads without suffering distortion that could cause binding of the tube bundles.

7.3.4 The vendor's design shall provide for a shim allowance of approximately 6 mm (1/4 in.) between the faces of stacked heat exchanger intermediate supports.

7.3.5 For horizontal heat exchangers, slotted holes shall be provided in the baseplate of all but one of the saddles to allow for longitudinal movement due to thermal expansion or contraction. The width of the slot shall be equal to the anchor bolt diameter plus 8 mm (5/16 in.). The length of the slot shall be equal to the anchor bolt diameter, plus the allowance for longitudinal movement, plus 8 mm (5/16 in.).

7.3.6 Localized stresses caused by concentrated loads on any external structural attachment due to piping reactions, supported equipment, lifting of exchanger, etc., shall be evaluated. Calculations shall be performed in accordance with WRC 297, WRC 537, finite element analysis, or TEMA RGP as applicable. Any other standard or numerical methods may

be used subject to approval by the purchaser. All geometrical limits specified in methods (such as WRC, etc.) used for local load analysis shall be followed. Extrapolation beyond stated limits is not allowed.

7.3.7 All heat exchangers shall be provided with two grounding lugs located on different supports.

7.4 Stationary and Floating Heads

7.4.1 Stiffeners shall not be used for channel cover design as a means of retaining pressure or to prevent deflection.

7.4.2 The pressure differential used to calculate the pass-partition plate thickness in accordance with TEMA, Section 5, Paragraph RCB-9.132 shall be at least 1 1/2 times the allowable pressure drop across each pass-partition plate. The plate thickness shall not be less than two times the tube side corrosion allowance, plus 3 mm (1/8 in.).

7.4.3 A 6 mm (1/4 in.) diameter drain hole shall be provided in all pass partition plates.

7.4.4 Floating-head cover bolting shall comply with TEMA, Section 5, Paragraph RCB-11. Bolt spacing and clearances shall not be less than the minimum recommended by TEMA.

7.4.5 Floating-head cover bolting shall be readily accessible and shall have adequate spanner (wrench) clearance between the floating-head bolts and the shell flange at the cover end when the shell cover is removed.

7.4.6 Packed floating-head tail-pipe and packed floating tubesheet designs (e.g. TEMA types P and W) shall not be used.

7.4.7 Unless otherwise specified by the purchaser, floating-heads shall be designed for design pressure on either side, with atmospheric pressure or, if specified, vacuum on the other side. Examples of acceptable floating-head designs are shown in Figure 3.

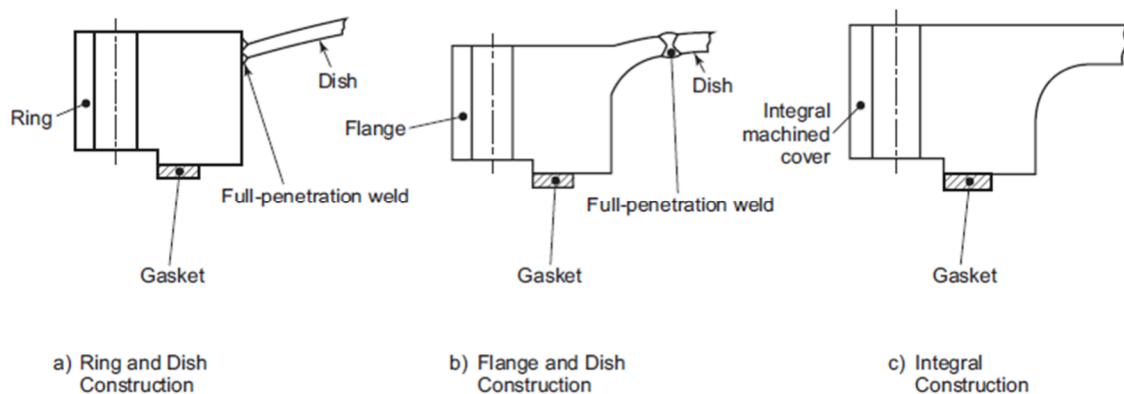


Figure 3 — Typical Designs for Floating-head Covers

7.4.8 Internal floating-head covers shall have the specified corrosion allowance on all wetted surfaces, except gasket seating surfaces. The specified corrosion allowance shall be included on the back side of the floating-head backing device.

7.4.9 Nubbins shall not be used on floating-head flanges.

7.4.10 The shell cover on horizontal S-type floating-head exchangers shall be equipped with a vent and drain.

7.4.11 S-type floating-heads shall utilize style A (dove-tail) split rings per TEMA, Figure RCB-5.141. The dove-tail mating surfaces shall have rounded corners.

7.4.12 For custom-designed girth flanges, channel covers, and floating-head flanges these shall be provided with 3 mm (1/8 in.) future machining allowance on the gasket contact seating surfaces (including pass-partition surfaces). The

additional thickness shall not be used in the calculation of maximum allowable working pressure. This requirement does not apply to clad or weld overlay construction.

7.4.13 When there is a large tube side temperature difference the design should consider the guidance in .

7.5 Tube Bundle

7.5.1 Tubes

7.5.1.1 The minimum outside diameter of the tubes shall be 19.05 mm ($3/4$ in.), unless otherwise specified or approved by the purchaser.

7.5.1.2 The tube wall thickness shall be as listed in Table 1, or thicker if required by the design conditions, including any tube corrosion allowance specified by the purchaser.

Table 1 — Minimum Thickness of Tubes

Dimension in millimeters (inches)

Tube Material	Minimum Thickness ^a
Carbon steel, low-alloy steel (max. 9 % chromium), aluminum, and aluminum alloy	2.11 (0.083)
Copper and copper alloys	1.47 (0.058)
High-alloy [austenitic, ferritic, and austenitic/ferritic (duplex)] steel and other nonferrous materials	1.47 (0.058)
Titanium	1.07 (0.042)
^a Tubes shall be furnished on either a minimum wall basis or an average wall basis, provided the tube thickness is not less than that specified above.	

7.5.1.3 For carbon and low alloy steel low-fin tubing, the wall thickness under the root of the fin shall be in accordance with Table 1 unless otherwise approved by the purchaser. See A.4.3 for guidance.

7.5.1.4 When tubes are both internally and externally enhanced the thickness at all locations shall be agreed by the purchaser.

7.5.1.5 For U-tubes with integral fins, the U-bend plus a minimum of 50 mm (2 in.) of the straight portion from the U-bend tangent line shall not be finned.

7.5.1.6 Unless otherwise specified by the purchaser, integrally finned tubes shall be provided with the same over-fin diameter as the bare tube end.

7.5.1.7 When specified by the purchaser, the vendor may provide bare landings of integrally finned tubes at the baffle contact points.

7.5.1.8 The mean diameter of U-bends shall not be less than 3 times the nominal outside diameter of the tube. For martensitic stainless steels, super ferritic stainless steels (>26 wt % Cr), super austenitic stainless steels (>6 wt % Mo), super duplex stainless steels, titanium, and high nickel alloys (>30 wt % Ni), the mean diameter of U-bends shall not be less than 4.0 times the nominal outside diameter of the tube.

7.5.1.9 Where duplex stainless steel tubes are specified, the mean diameter of U-bends shall not be less than 4.0 times the nominal outside diameter of the tube when heat treatment is performed after bending, and 6.6 times the nominal outside diameter of the tube when no heat treatment is applied.

7.5.1.10 For U-tubes, design calculations shall be based on the reduction in wall thickness associated with bending. The thickness of the tubes need not be increased to meet the requirements in Table 1 provided the wall thickness in the U-bends meet the minimum requirements of the pressure design code and any specified tube corrosion allowance.

7.5.1.11 For U-tube and floating-head type exchangers, the minimum clearance between any part of the U-bend or floating-head cover, and the shell rear head, shall be 38 mm (1½ in.), to accommodate thermal expansion of the tube bundle. Both the crown and knuckle of the head shall be considered.

7.5.1.12 The number of tubes in any pass shall not be greater than 10 % above or below the average number of tubes per pass, unless otherwise approved by the purchaser.

7.5.2 Tubesheets

7.5.2.1 For a vertical heat exchanger where the stationary tubesheet is at the bottom, a suitable means of holding the tube bundle in place shall be provided for when the bonnet or channel is removed. If collar bolts are used, at least four integral (one piece) bolts shall be provided with their location identified on the drawings and by stamped markings on the outside diameter of the tubesheet. The use of drilled and tapped holes is prohibited.

7.5.2.2 The distance between the edge of the tube holes and the edge of all gasket grooves (including pass-partition grooves) shall not be less than 1.5 mm (1/16 in.) for tubesheets with expanded tube-to-tubesheet joints and not less than 3 mm (1/8 in.) for tubesheets with welded tube-to-tubesheet joints.

7.5.2.3 Unless otherwise specified or approved by the purchaser, tubesheets shall be designed for design pressure on either side, with atmospheric pressure or, if specified, vacuum on the other side.

7.5.2.4 A full diameter stationary tubesheet shall be provided for removable tube bundle exchangers with bonnets (see TEMA, Figure N-1.2, Type B stationary head). The tubesheet shall be provided with integral (one piece) collar bolts for a minimum of 25 % of the bolts (four minimum). The tubesheet design shall allow for hydrostatic testing of the shell side with the bonnet removed and all bolting installed. The use of drilled and tapped holes is prohibited.

7.5.2.5 Tubesheets shall be provided with 3 mm (1/8 in.) future machining allowance on the tube side gasket contact seating surface (including pass-partition surfaces). The additional thickness shall not be used in the calculation of maximum allowable working pressure.

NOTE: This requirement does not apply to clad or weld overlay construction.

7.5.2.6 For vertical fixed tubesheet exchangers, a means shall be provided to completely vent and drain the shell.

- **7.5.2.7** The purchaser shall specify when integrally-hubbed tubesheets are required. See A.4.4 for guidance.

7.5.2.8 Differential thermal expansion shall be considered in the selection of integral tubesheet-to-cylinder joints when the differential thermal expansion between the integral tubesheet and attached cylinder exceeds a ratio of 0.001.

7.5.2.9 The use of gasketed tubesheets with a high temperature difference (>110 °C [>200 °F]) applications shall be limited to a maximum diametrical flange differential thermal expansion between the tubesheet and either flange of 2.5 mm (0.1 in.) unless otherwise approved by the purchaser. For flange and bolted joints with a high temperature differential, other designs should be considered for higher reliability, see A.4.6 for additional information.

7.5.2.10 The tube and tube-to-tubesheet joint stresses shall be analysed for multi-pass fixed tubesheet or floating head heat exchangers whenever the tube mean metal temperature difference between adjacent passes exceeds 60 °C (108 °F).

7.5.3 Baffles and Support Plates

7.5.3.1 The thickness of transverse baffles and support plates, including detuning baffles, shall be the larger of twice the specified corrosion allowance or as specified by TEMA.

7.5.3.2 To facilitate drainage of the shell, transverse baffles and support plates shall have notches that are 6 mm (1/4 in.) in height for shell diameters up to and including 406 mm (16 in.), and 10 mm (3/8 in.) in height for larger shell diameters.

7.5.3.3 When required at the inlet or outlet baffle spacing, partial tube support plates shall:

- a) extend a minimum of 3 full tube rows;

- b) support impingement rods (if included in the design).

Note: See A.4.5.1 for additional guidance.

7.5.3.4 Full Tube Support Baffles

7.5.3.4.1 A full support baffle shall be provided at the nozzle centerline for the following:

- a) inlet and outlet nozzles for TEMA type G and H shells;
- b) at the central nozzle for TEMA type J shells.

Note: Full support baffles at these locations provide the maximum support to tubes at the nozzle locations to minimize tube vibration.

7.5.3.4.2 Unless otherwise specified by the purchaser, a full support baffle shall be provided approximately 50 mm (2 in.) away from the U-bend tangent for the following configurations:

- a) When the shell inlet or outlet nozzle is located before the U-bend.
- b) TEMA type G, H, X and J shells, when U-tubes are utilized. The full support baffle at the U-bend shall not be trimmed for these shell types.

Note: Trimming of full support baffles of TEMA G, H, X, and J shell types may cause a hydraulic imbalance of the shell side fluid.

7.5.3.4.3 Full support baffles shall be trimmed approximately 3 mm (1/8 in) above the top tube rows and 3 mm (1/8 in) below the bottom tube rows, so as to fully support these tubes or as needed to support any impingement rods in all cases except at the U-bends of TEMA type G, H, X, and J shells (see 7.5.3.4.2).

7.5.3.4.4 Floating heads with removable shell covers shall be provided with full support baffle with center cut-out to prevent stagnation between the tubesheet and floating head support baffle. The details of the floating head support baffle shall be agreed upon between the purchaser and supplier.

7.5.3.5 If shell side longitudinal baffles are used, the minimum clearance (ligament) between the transverse baffle edge and the tube holes in the baffles shall be 3 mm (1/8 in.) as shown in Figure 4.

7.5.3.6 Tube to baffle-hole diametric clearance shall not exceed 0.4 mm (1/64 in.) in the following cases:

- a) Kettle-type reboilers and steam generators.
- b) For pulsating services, e.g. where shell side flow is driven by a reciprocating compressor.
- c) Where the thermal design / vibration analysis indicates that the amplitude of vibration, at any point in the tube bundle, exceeds 10 % of the gap between tubes-
- d) When required by TEMA.

Note: See A.2.5.3.1 for more information.

7.5.3.7 The bundle (including baffles and support plates) shall be supported by tie-rods, skid bars, or bundle handling devices, to provide protection from damage during handling.

7.5.3.8 Welded longitudinal baffle thickness shall be calculated as per TEMA R-4.2.2.2 using 1 1/2 times the shell side allowable pressure drop or the TEMA minimum thickness plus a single shell side corrosion allowance; whichever is greater.

7.5.3.9 Tie-rods and spacers shall not obstruct the cleaning lane between tube rows for bundles with square or rotated square tube layouts.

7.5.3.10 U-bend supports shall be provided where they are required to mitigate damaging flow-induced vibration or to meet the 80 % of unsupported tube span limit per TEMA.

7.5.3.11 Flat bar U-bend supports shall be as follows:

- a) be provided if required per 7.5.3.10 for bundles that include a full support and 90° square tube layout;

Note: flat bar U-bend supports may be used where flow exists over the U-bends, but flat bar U-bend supports only restrict in-plane movement of the tube bends by friction, which may not be sufficient to dampen any damaging flow induced vibration, if present.

- b) only extend partially into the bend field if the inner tube rows are crossed because the crossed rows cannot be supported;
- c) be designed such that flat bar support strips that extend through the bend field have a thickness that is sufficient to fill the gap between adjacent tubes, and have a minimum width of 25 mm (1 in.);
- d) be designed such that the flat bar outer ring, if applicable, has a thickness no less than 6 mm (0.25 in.) unless otherwise agreed by the purchaser;
- e) be designed such that the threaded stud, if applicable for the purpose of bolting together the support strips is no less than 10 mm (0.375 in.) nominal diameter and shall be secured at each end by a hex nut tack welded to the stud;
- f) be rigidly attached to the full support plate or last baffle to prevent the system of support strips falling under its own weight unless orientation of the U-bend plane is parallel to grade such that it is impossible for the flat bars to free fall through the bend field. Flat bar U-bend supports meeting this exception may float in the U-bends or be attached to the bundle skeleton by slotted plates or similar means to accommodate thermal expansion of the tubes;
- g) be designed to accommodate no less than the greater of 50 mm (2 in.) or 1.2 times the calculated thermal expansion of the tubes considering worst case conditions when flat bar U-bend support is rigidly attached to full support plate.

7.5.3.12. Tie plate U-bend supports shall be as follows:

- a) be designed such that each strip projects at least 16 mm (0.625 in) beyond the last tube in its row to accommodate welding (see Figure 2), unless otherwise agreed by the purchaser;
- b) include 6 mm (0.25 in.) thick by 25 mm (1 in.) wide peripheral support bars continuously welded on both sides of the strips unless other dimensions are agreed between the purchaser and manufacturer (see Figure 2);
- c) not be rigidly attached to the last baffle or full support. Tie plate U-bend supports may float in the U-bends or be attached to the bundle skeleton by slotted plates or a similar means to accommodate thermal expansion of the tubes;
- d) lock together no less than 5 tube rows;
- e) only support those U-bends required to be supported by 7.5.3.10 except where needed to comply with 7.5.3.12 (d);
- f) not lock together tube rows where the ratio of largest locked bend diameter to smallest locked bend diameter would exceed 3, unless otherwise agreed by purchaser. Tie plate U-bend supports may be provided in non-overlapping sections, positioned slightly out of plane to each other, to avoid over-restraining differential thermal expansion between tube rows of substantially different length. If a conflict exists between 7.5.3.12 (d) and this requirement, 7.5.3.12 (d) shall take precedence.

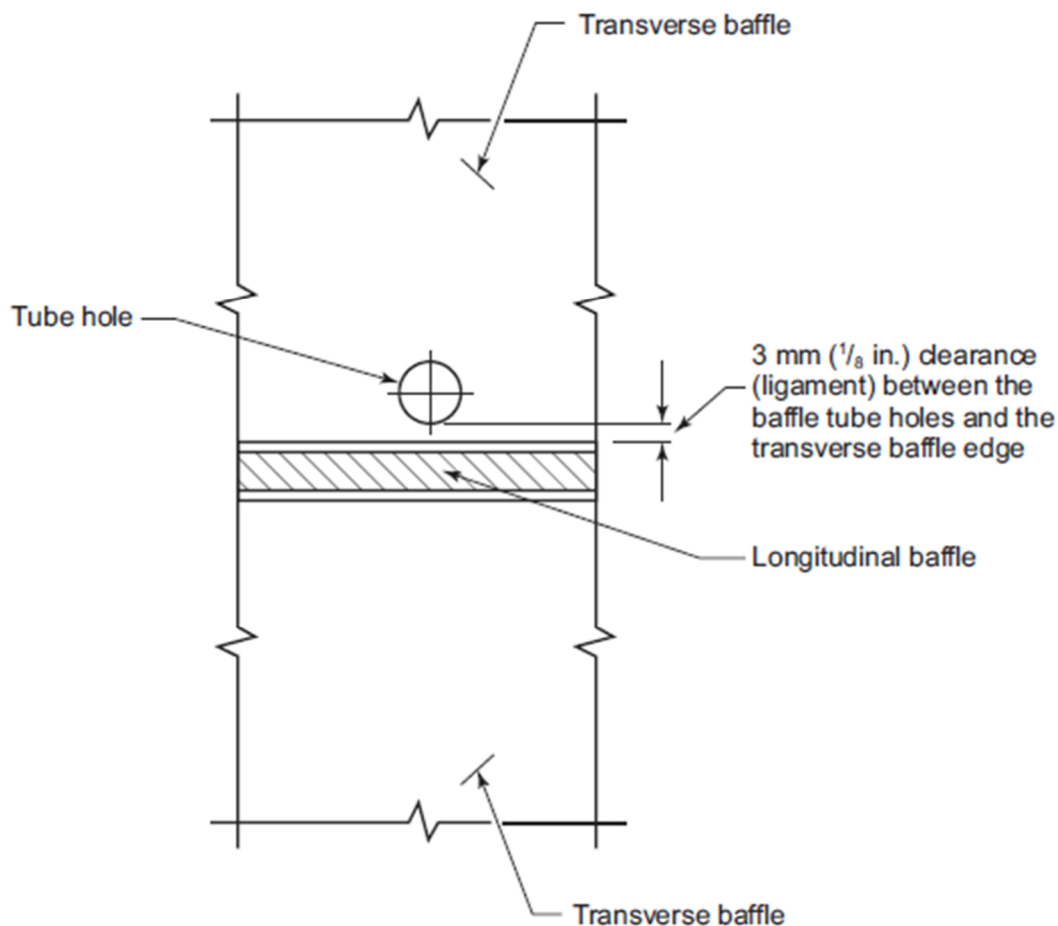


Figure 4 —Clearance Between Tube Holes and Transverse Baffle Edge with Longitudinal Baffle

7.5.4 Impingement Protection

7.5.4.1 If required by TEMA, Section 5, Paragraph RCB-4.61, impingement protection shall be provided by a plate baffle or rods on the tube bundle, an annular distributor, or another means agreed by the purchaser and vendor.

7.5.4.2 If an impingement plate baffle is used, it shall extend at least 25 mm (1 in.) beyond the projection of the nozzle bore (e.g. for a round impingement plate, the plate diameter shall be at least 50 mm (2 in.) larger than the nozzle bore) or 20 % greater than the inlet nozzle inside diameter, whichever is larger.

7.5.4.3 If impingement protection is used, the shell entrance and tube bundle entrance areas (as defined by TEMA, Section 5, Paragraph RCB 4.62) shall not be less than the flow area of the inlet nozzle.

7.5.4.4 The nominal thickness of impingement plate baffles shall not be less than 6 mm (1/4 in.).

7.5.4.5 The impingement plate shall be supported (e.g. by welding to at least two spacers) with a minimum clearance of 3 mm (1/8 in.) to the nearest row of tubes.

7.5.4.6 Perforated impingement plates shall not be used.

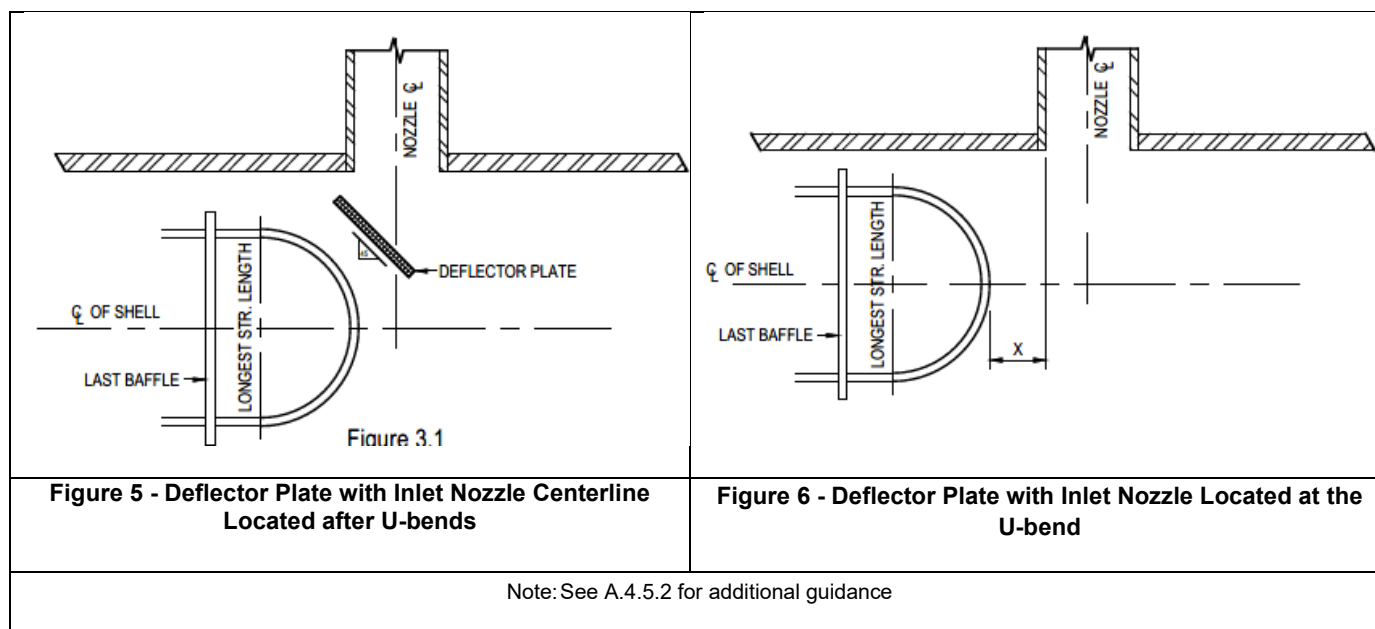
7.5.4.7 Regardless of the requirements of TEMA, the shell side inlet nozzle located at the U-bends, will require a deflector plate or other specialized impingement protection, and shall conform to the following:

- a) A deflector plate as shown in Figure 5 may be used.
- b) The plate shall be securely welded to the shell inside diameter or to support features of the bundle, depending on space constraints.

- c) The size, shape and location of the deflector plate shall be determined by the end user and fabricator.

NOTE See A.4.5.2 for additional guidance.

7.5.4.8 If the shell side inlet nozzle location (dimension "x" in Figure 6) is at least the greater of 50mm (2 in.) or 10% of the inlet nozzle bore beyond the U-bends, impingement protection is not required unless required by TEMA. See A.4.5.2 for additional guidance.



7.5.4.9 If impingement rods are utilized, the following shall be applied:

- a) For triangular or rotated square pitch a minimum of two rows of staggered rods shall be used.
- a) For square pitch one row of rods is acceptable if they are of the same diameter and pitch as the tubes, unless otherwise specified by the purchaser.
- b) The rods shall not have a diameter less than 16 mm ($5/8$ in.).
- c) Rod center-to-center spacing shall be 1.25 to 1.33 times the rod diameter.
- d) The length and width of the rod array shall extend at least 50 mm (2 in.) beyond the projection of the nozzle bore or 20 % greater than the inlet nozzle inside diameter, whichever is larger.
- e) For carbon, low alloy, and austenitic stainless steel materials the rods shall be supplied as solid bars; for other materials tubing can be used if approved by the purchaser.
- f) Impingement rods shall be fixed at both ends. When attached to the tubesheet, impingement rods shall not be attached by welding alone.
- g) An impingement rod's unsupported span shall not exceed the unsupported span of the tubes adjacent to them.
- h) Vibration supports placed near the entrance and exit flow regions shall also support the impingement rods if present.

7.5.4.10 Bundles located in TEMA type k shells, that have an inlet nozzle located normal to the shell vertical centerline and below the top of the bundle, (see Figure 7) may require impingement protection.

7.5.4.11 For kettles that have a 2-phase inlet above the liquid level a deflector box shall be provided and shall meet the following requirements:

- the length and width of the box shall extend at least 25mm (1 in.) beyond the shell opening;
- the box shall have the opening located such that the inlet flow is directed downward;
- the depth of the box shall be such that the cross-sectional area of the box opening shall be at least equal to the nozzle's cross-sectional area.

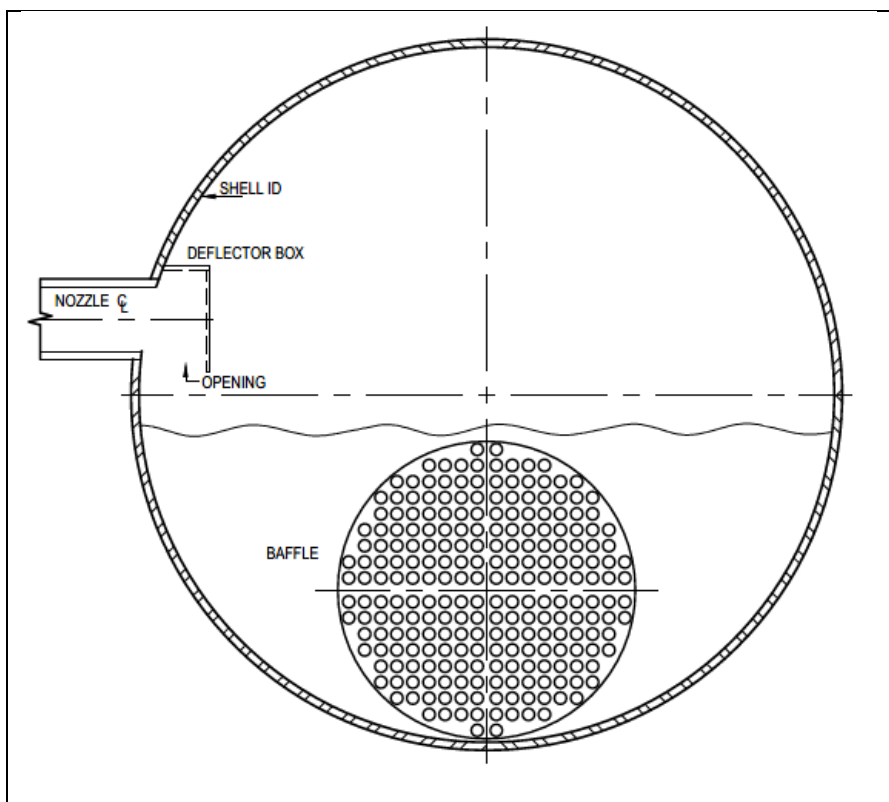


Figure 7 - Deflector Box Located in TEMA Type K Shells

7.5.5 Bypass Sealing Devices

7.5.5.1 Bypass sealing devices (such as seal bars, dummy tubes or tie-rods) as shown in Figure 8 shall be used for non-isothermal shell side services if bypass clearances exceed 16 mm (⁵/₈ in.) and shall be located as follows:

- If the distance between baffle cut edges is six tube pitches or less, a single seal located approximately halfway between the baffle cut shall be provided.
- If the distance between baffle cut edges exceeds six tube pitches, multiple seals shall be provided. A seal shall be located every five to seven tube pitches between the baffle cuts, with the outermost seals not more than 75 mm (3 in.) from each baffle cut edge.

7.5.5.2 Peripheral bypass seals shall extend from the peripheral edge of the transverse baffle into the tube bundle so that the clearance to the nearest tube does not exceed the nominal clearance between tubes.

7.5.5.3 Internal bypass seals shall be installed so that the clearance to the nearest tube does not exceed the nominal clearance between tubes.

7.5.5.4 Bypass sealing devices shall either be located to minimize obstruction of mechanical cleaning lanes or shall be readily removable. Continuous cleaning lanes shall be maintained for square (90°) and rotated-square (45°) pitch.

7.5.5.5 The nominal thickness of seal strips shall be the nominal thickness of the transverse baffles or 6 mm ($\frac{1}{4}$ in.), whichever is less.

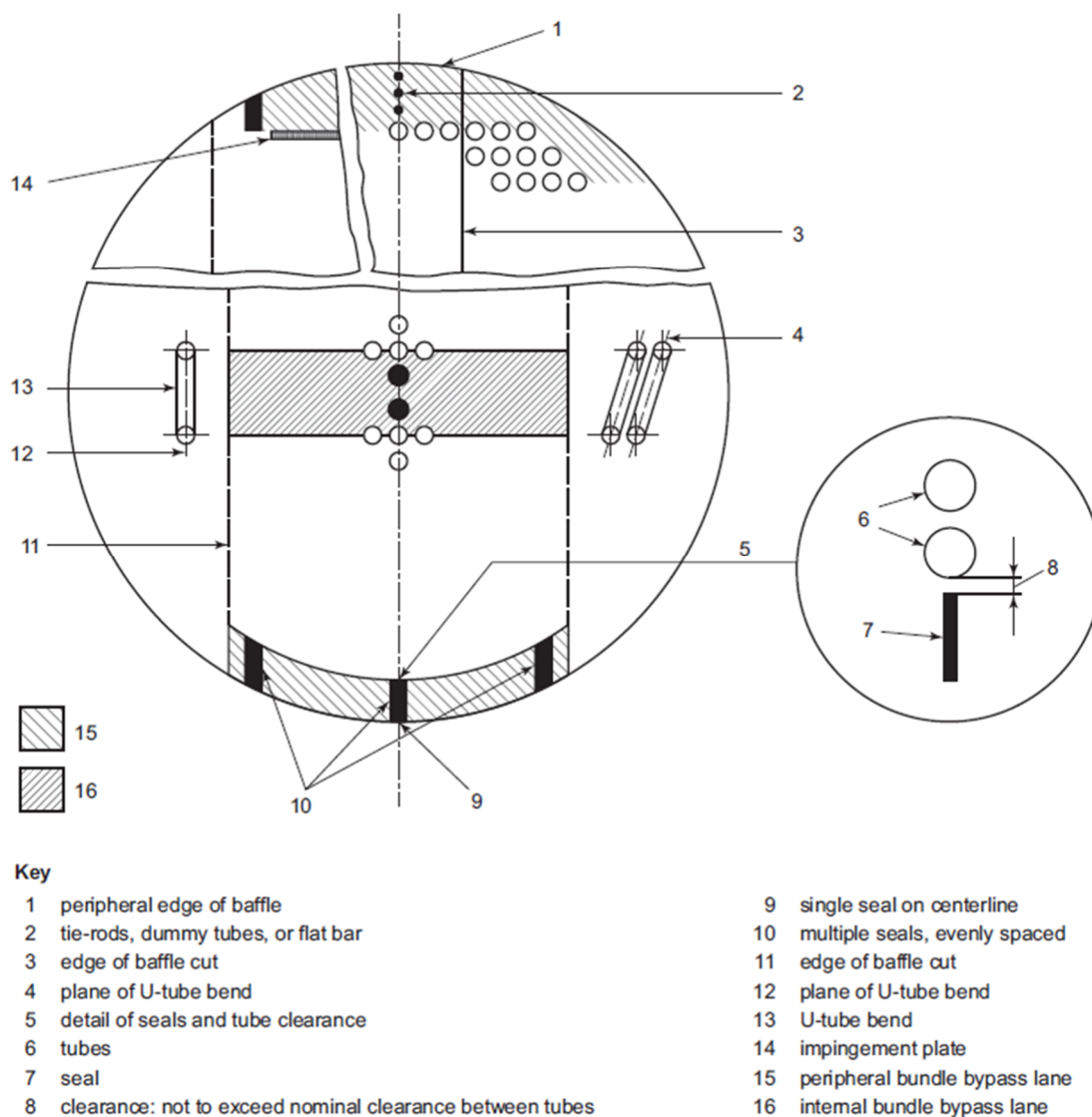


Figure 8 —Typical Cross-sections of Tube Bundle Showing Locations of Bypass Sealing Devices

7.5.5.6 Bypass seal strips shall be attached to the transverse baffles by continuous welds on one side of each baffle.

7.5.5.7 The leading and trailing edges of seal strips shall be provided with a radius or a bevel to prevent damage to the shell when inserting or removing the tube bundle.

7.5.5.8 Peripheral bypass seal strips shall not restrict the tube bundle inlet or outlet connections.

7.5.6 Tube Bundle Skid Bars

7.5.6.1 For all removable tube bundles with a mass of more than 5500 kg (12,000 lb), continuous sliding surfaces shall be provided to facilitate bundle removal.

7.5.6.2 If skid bars are used, they shall comply with the following:

- A minimum of two shall be provided.
- They shall be welded to the transverse baffles and support plates.

- c) They shall not restrict the tube bundle inlet or outlet connections.
- d) They shall protrude 1 mm (1/32 in.) beyond the outside diameter of baffle and support plates.
- e) The leading and trailing edges of skid bars shall be provided with a radius or a bevel to prevent damage to the shell when inserting or removing the tube bundle.
- f) The skid bars shall be radially orientated and located between 15 ° to 20 ° from the vertical centerline.
- g) The skid bar dimensions shall be in accordance with Table 2. When the tube layout does not allow for the dimensions to be met, the design shall be subject to agreement between the purchaser and the vendor.
- h) The skid design for bundles over 18,150 kg (40,000 lb) shall be subject to approval by the purchaser. See A.4.6 for additional guidance.

Table 2 — Skid Bar Dimensions

	Skid bar dimensions (minimum) mm (in.)	
< 760 (30)	13 (0.5)	38 (1.5)
761 to 1067 (30 to 42)	19 (0.75)	38 (1.5)
<1088 (42)	25 (1.0)	50 (2.0)

7.5.6.3 If a tube or pipe is used as a skid or alignment device, the tube or pipe shall be either sealed on both ends with a single 12.5 mm (0.5 in.) hole in the tube or pipe at the stationary tubesheet or sealed at the inlet end only and be self draining.

7.5.7 Tube-to-Tubesheet Joint

See A.4.7 for additional guidance on tube-to-tubesheet joints.

7.5.7.1 Expanded only tube-to-tubesheet joints shall be expanded with a minimum of two ring grooves.

7.5.7.2 Strength-welded tube-to-tubesheet joints shall be used for the following conditions:

- a) If the design pressure on either side exceeds 7000 kPa (ga) (1000 psig).
- b) Operating temperatures greater than those defined in Table 3 for similar tube and tubesheet materials. See A.4.7 for guidance on dissimilar tube and tubesheet materials and materials not shown in Table 3.
- c) Cyclic service.
- d) When the tubesheet thickness is less than the TEMA minimum thickness or when the tubesheet thickness is less than the tube nominal OD, e.g., flexible tubesheet designs such those used in heat recovery steam generators and sulfur plant applications.
- e) Where tube pitch is less than 1.25 times the tube OD for tubes up to and including 25 mm (1 in.) to prevent ligament interaction.
- f) For tubes with diameter to nominal thickness ratios less than 6 or for tubes of nominal thickness greater than or equal to 3.404 mm (0.134 in.).
- g) When f_y (defined as the minimum of $[S_y/S_{y,t}, 1.0]$ in ASME BPVC Section VIII, where S_y is the tubesheet yield strength and $S_{y,t}$ is the tube yield strength) is less than 0.6. For clad tubesheets, the tubesheet yield stress used in the calculation of f_y should be that of the base material.
- h) Other services when specified by the purchaser.

Table 3 —Temperature Limits Used to Define the Creep Range for Tube-to-Tubesheet Joints with Similar Tube and Tubesheet Materials

Material	Operating Temperature Limit ^a
Carbon Steel	343°C (650°F)
1.25Cr to 9Cr (not V modified)	427°C (800°F)
2.25Cr to 3Cr (V modified)	441°C (825°F)
12Cr	482°C (900°F)
Type 304/304H	510°C (950°F)
Type 316/316H	538°C (1000°F)
Type 321/321H	538°C (1000°F)
Type 347/347H	538°C (1000°F)
Alloy 400	399°C (750°F)
Alloy 600	399°C (750°F)
Alloy 800/800H/800HT	565°C (1050°F)
a These suggested temperature limits are only applicable for expanded joints with similar tube and tubesheet thermal coefficients of expansion. See A.4.7 for additional guidance.	

7.5.7.3 If welded tube-to-tubesheet joints are specified, the joint shall be welded by one of the following methods:

- a) strength-welded only;
- b) strength-welded and expanded;
- c) seal-welded and expanded.

- **7.5.7.4** When strength welds are applied, the degree of expansion and the use of ring grooves shall be specified or agreed to by the purchaser. See A.4.7 for additional guidance.

7.5.7.5 Where expanded tube-to-tubesheet joints are used with duplex stainless steel in wet H₂S or sour service and the tube wall reduction exceeds 3 %, hardening of duplex stainless steel tubes in excess of NACE MR0175 or MR0103 hardness limits can occur. Qualification of the tube-to-tubesheet expansion procedure at the maximum proposed % tube wall reduction shall include hardness testing when in wet H₂S or sour service and the degree of expansion exceeds 3%. Alternately, strength-welded tube-tubesheet joints may be used, without expansion, or with light expansion (up to a maximum of 3 %).

7.5.8 Bundle Handling Devices

- The purchaser shall specify if additional methods of support to aid in job site handling of the bundle are required for the following removable bundle designs:
 - a) Bundles with a diameter greater than mm (48 in.).
 - b) A bundle mass above 40,000 lbs, (18,000 kg).

- c) A tube lengths exceeding 7.3 m (24 ft).
- d) Bundles with vertical cut baffles.
- e) Bundles with longitudinal baffle seals.
- f) Bundles that contain unusual transverse baffle and support plate arrangements.
- g) Bundles with NTIW baffles.

See A.4.8 for additional guidance.

7.6 Nozzles and Other Connections

- **7.6.1** The purchaser shall specify the required flange design code (e.g. ASME B16.47 A or B, ASME B16.5).

7.6.2 Connection sizes of DN 40 (NPS 1-1/2) and larger shall be flanged unless otherwise specified by the purchaser.

7.6.3 When butt-welded connections to the piping are to be used, the purchaser shall specify the corresponding piping wall thickness. The wall thickness at the prepared end shall be equal to that of the connecting piping. The weld ends shall be beveled by others in accordance with ASME B16.25.

7.6.4 Non-flanged connections smaller than DN 40 (NPS 1¹/₂) shall be forged couplings with an appropriate rating, for example equivalent to ASME B16.11 class 6000, or shall be integrally reinforced welding fittings with appropriate tapered threads, for example, equivalent to ASME B1.20.1 and shall comply with the pressure design code. Threaded connections shall not be used in hydrogen, sour, or wet hydrogen sulfide service. This includes auxiliary connections such as vents, drains, instrument connections, and chemical cleaning connections.

7.6.5 Flanged connections shall be of one of the following types:

- a) Forged integrally flanged.
- b) Pipe or forged cylinder welded to forged welding-neck flange.
- c) Rolled plate welded to forged welding-neck flange (if approved by the purchaser).
- d) Pipe welded to a forged slip-on flange, except as noted in 7.6.6.

7.6.6 Slip-on flanges shall not be used in any of the following conditions:

- a) for design pressure greater than 2100 kPa (ga) (300 psig);
- b) for design temperature greater than 400 °C (750 °F);
- c) for corrosion allowance greater than 3 mm (¹/₈ in.);
- d) in hydrogen service;
- e) in sour or wet hydrogen sulfide service or other hydrogen charging services;
- f) when PWHT is required;
- g) in services that normally operate in vacuum conditions;
- h) in service with operating temperature below -50 °C (-58 °F);
- i) in cyclic service.

- **7.6.7** The projection of flanged connections shall allow through-bolting to be removed from either side of the flange without removing the insulation. For stacked units, this requirement need only be applied to one side of directly coupled connections. This requirement does not apply to auxiliary connections. The insulation thickness shall be specified by the purchaser.

7.6.8 Integrally reinforced nozzles shall be designed so that standard spanners (wrenches) fit the nuts without interference from nozzle neck reinforcement.

- **7.6.9** The purchaser shall specify if chemical cleaning connections are required. Their nominal size shall be not less than DN 50 (NPS 2).

7.6.10 Nozzles shall be designed to withstand the simultaneous application of forces and moments in the corroded condition, as defined in Figure 9 and listed in Table 5 or Table 6, unless otherwise specified by the purchaser. Non-piped auxiliary connections, such as vents, drains, and cleaning connections, are excluded from this requirement. Evaluation shall be performed in accordance with WRC 297, WRC 537 or finite element analysis. Any other standard or numerical methods may be used subject to approval by the purchaser. All geometrical limits specified in methods (such as WRC etc.) used for local load analysis shall be followed. Extrapolation beyond stated limits is not allowed. When the application of these nozzle loads would require an increase in the local cylinder or head thickness, acceptable loads shall be agreed mutually with the purchaser.

- **7.6.11** For nozzle sizes larger than those listed in Table 5 and Table 6, the purchaser shall specify the moments and forces.

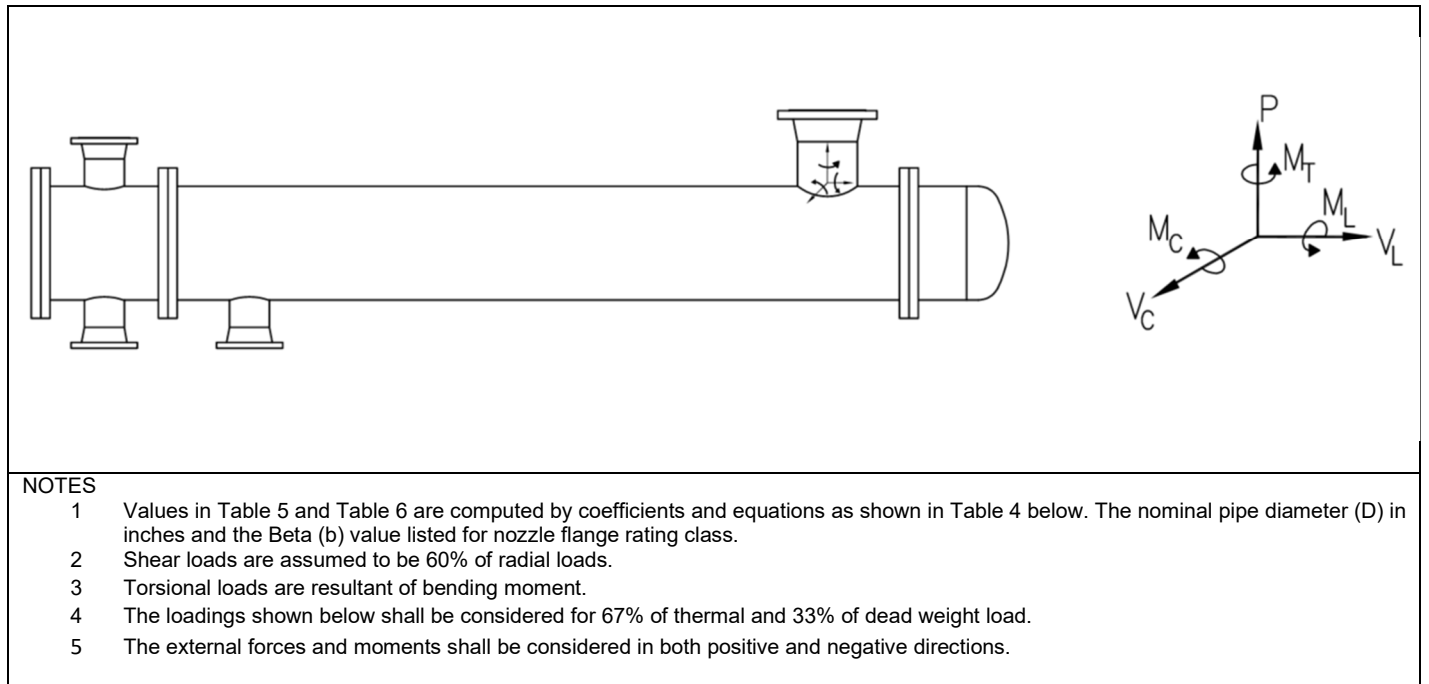


Figure 9 - Directions of Moments and Forces on Nozzles

Table 4 - Basis of Nozzle Forces and Moments

Flange Rating Class	150#	300#	600#	900#	1500#	2500#
β Value	0.6	0.7	0.8	0.9	1.0	1.1

Longitudinal Bending Moment M_L (lbf - ft)	$\beta \times 110 \times D^2$
Circumferential Bending Moment M_C (lbf - ft)	$\beta \times 85 \times D^2$
Radial Load (Tension or Compression) P (lbf)	$\beta \times 500 \times D$
Torsional Moment, M_T (lbf - ft)	$\beta \times 140 \times D^2$
Shear Load V_C or Shear Load V_L (lbf)	$\beta \times 300 \times D$

Table 5 —Allowable Forces and Moments at the Nozzle Neck to Shell/Channel Interface – SI units

Nozzle Forces and Moments for 150# Flange Rating

Nominal Diameter (DN)	Circumferential Moment (M_c) (N-m)	Longitudinal Moment (M_L) (N-m)	Torsional Moment (M_T) (N-m)	Shear Load (V_c) (N)	Radial Load (P) (N)	Shear Load (V_L) (N)
50	277	358	452	2669	1601	2669
80	622	805	1018	4003	2402	4003
100	1106	1432	1810	5338	3203	5338
150	2490	3222	4072	8007	4804	8007
200	4426	5728	7239	10676	6405	10676
250	6916	8950	11310	13345	8007	13345
300	9958	12887	16287	16014	9608	16014
350	13555	17541	22168	18683	11210	18683
400	17704	22911	28954	21351	12811	21351
450	22407	28997	36645	24020	14412	24020
500	27662	35798	45241	26689	16014	26689
600	39834	51550	65147	32027	19216	32027
Nozzle Forces and Moments for 300# Flange Rating						
50	323	418	528	3114	1868	3114
80	536	940	1188	4671	2802	4671
100	952	1671	2111	6227	3736	6227
150	2142	3759	4750	9341	5605	9341
200	3808	6682	8445	12455	7473	12455
250	5950	10441	13195	15569	9341	15569
300	8568	15035	19001	18682	11209	18682
350	11662	20465	25863	21796	13078	21796
400	15232	26729	33780	24910	14946	24910
450	19278	33829	42753	28024	16814	28024
500	23800	41765	52781	31137	18682	31137
600	34272	60141	76005	37365	22419	37365

Nozzle Forces and Moments for 600# Flange Rating						
Nominal Diameter (DN)	Circumferential Moment (M_c) (N-m)	Longitudinal Moment (M_L) (N-m)	Torsional Moment (M_T) (N-m)	Shear Load (V_c) (N)	Radial Load (P) (N)	Shear Load (V_L) (N)
50	369	477	603	3559	2135	3559
80	830	1074	1357	5338	3203	5338
100	1475	1909	2413	7117	4270	7117

150	3319	4296	5429	10676	6405	10676
200	5901	7637	9651	14234	8541	14234
250	9221	11933	15080	17793	10676	17793
300	13278	17183	21716	21351	12811	21351
350	18073	23388	29557	24910	14946	24910
400	23605	30548	38606	28469	17081	28469
450	29875	38662	48860	32027	19216	32027
500	36883	47731	60321	35586	21351	35586
600	53112	68733	86862	42703	25622	42703
Nozzle Forces and Moments for 900# Flange Rating						
50	415	537	679	4003	2402	4003
80	934	1208	1527	6005	3603	6005
100	1660	2148	2714	8007	4804	8007
150	3734	4833	6108	12010	7206	12010
200	6639	8592	10858	16014	9608	16014
250	10373	13424	16965	20017	12010	20017
300	14938	19331	24430	24020	14412	24020
350	20332	26312	33252	28024	16814	28024
400	26556	34366	43431	32027	19216	32027
450	33610	43495	54968	36031	21618	36031
500	41494	53698	67861	40034	24020	40034
600	59751	77325	97720	48041	28824	48041

Nozzle Forces and Moments for 1500# Flange Rating						
Nominal Diameter (DN)	Circumferential Moment (M _c) (N-m)	Longitudinal Moment (M _L) (N-m)	Torsional Moment (M _T) (N-m)	Shear Load (V _c) (N)	Radial Load (P) (N)	Shear Load (V _L) (N)
50	461	597	754	4448	2669	4448
80	1037	1342	1697	6672	4003	6672
100	1844	2387	3016	8896	5338	8896
150	4149	5370	6786	13345	8007	13345
200	7377	9546	12064	17793	10676	17793
250	11526	14916	18850	22241	13345	22241
300	16597	21479	27145	26689	16014	26689
350	22591	29235	36947	31138	18683	31138

400	29507	38185	48257	35586	21351	35586
450	37344	48328	61075	40034	24020	40034
500	46104	59664	75401	44482	26689	44482
600	66390	85916	108578	53379	32027	53379
Nozzle Forces and Moments for 2500# Flange Rating						
50	507	656	829	4893	2936	4893
80	1141	1477	1866	7340	4404	7340
100	2029	2625	3318	9786	5872	9786
150	4564	5907	7465	14679	8807	14679
200	8114	10501	13271	19572	11743	19572
250	12679	16408	20735	24465	14679	24465
300	18257	23627	29859	29358	17615	29358
350	24850	32159	40641	34251	20551	34251
400	32457	42003	53083	39144	23487	39144
450	41079	53161	67183	44037	26422	44037
500	50714	65630	82942	48930	29358	48930
600	73029	94508	119436	58717	35230	58717

Table 6 —Nozzle Allowable Forces and Moments at the Nozzle Neck to Shell/Channel Interface – US customary units

Nozzle Forces and Moments for 150# Flange Rating						
Nominal Diameter (NPS)	Circumferential Moment (M_c) (lbf-ft)	Longitudinal Moment (M_L) (lbf-ft)	Torsional Moment (M_T) (lbf-ft)	Shear Load (V_c) (lbf)	Radial Load (P) (lbf)	Shear Load (V_L) (lbf)
2	204	264	334	360	600	360
3	459	594	751	540	900	540
4	816	1,056	1,335	720	1200	720
6	1,836	2,376	3,003	1080	1800	1080
8	3,264	4,224	5,338	1440	2400	1440
10	5,100	6,600	8,341	1800	3000	1800
12	7,344	9,504	12,011	2160	3600	2160
14	9,996	12,936	16,348	2520	4200	2520
16	13,056	16,896	21,353	2880	4800	2880
18	16,524	21,384	27,024	3240	5400	3240
20	20,400	26,400	33,363	3600	6000	3600
24	29,376	38,016	48,043	4320	7200	4320

Nozzle Forces and Moments for 300# Flange Rating						
2	238	308	389	420	700	420
3	536	693	876	630	1,050	630
4	952	1,232	1,557	840	1,400	840
6	2,142	2,772	3,503	1,260	2,100	1,260
8	3,808	4,928	6,228	1,680	2,800	1,680
10	5,950	7,700	9,731	2,100	3,500	2,100
12	8,568	11,088	14,013	2,520	4,200	2,520
14	11,662	15,092	19,073	2,940	4,900	2,940
16	15,232	19,712	24,911	3,360	5,600	3,360
18	19,278	24,948	31,528	3,780	6,300	3,780
20	23,800	30,800	38,924	4,200	7,000	4,200
24	34,272	44,352	56,051	5,040	8,400	5,040

Nozzle Forces and Moments for 600# Flange Rating						
Nominal Diameter (NPS)	Circumferential Moment (M_c) (lbf-ft)	Longitudinal Moment (M_L) (lbf-ft)	Torsional Moment (M_T) (lbf-ft)	Shear Load (V_c) (lbf)	Radial Load (P) (lbf)	Shear Load (V_L) (lbf)
2	272	352	445	480	800	480
3	612	792	1,001	720	1,200	720
4	1,088	1,408	1,779	960	1,600	960
6	2,448	3,168	4,004	1,440	2,400	1,440
8	4,352	5,632	7,118	1,920	3,200	1,920
10	6,800	8,800	11,121	2,400	4,000	2,400
12	9,792	12,672	16,014	2,880	4,800	2,880
14	13,328	17,248	21,797	3,360	5,600	3,360
16	17,408	22,528	28,470	3,840	6,400	3,840
18	22,032	28,512	36,033	4,320	7,200	4,320
20	27,200	35,200	44,485	4,800	8,000	4,800
24	39,168	50,688	64,058	5,760	9,600	5,760

Nozzle Forces and Moments for 900# Flange Rating						
2	306	396	500	540	900	540
3	689	891	1,126	810	1,350	810
4	1,224	1,584	2,002	1,080	1,800	1,080
6	2,754	3,564	4,504	1,620	2,700	1,620

8	4,896	6,336	8,007	2,160	3,600	2,160
10	7,650	9,900	12,511	2,700	4,500	2,700
12	11,016	14,256	18,016	3,240	5,400	3,240
14	14,994	19,404	24,522	3,780	6,300	3,780
16	19,584	25,344	32,029	4,320	7,200	4,320
18	24,786	32,076	40,537	4,860	8,100	4,860
20	30,600	39,600	50,045	5,400	9,000	5,400
24	44,064	57,024	72,065	6,480	10,800	6,480

Nozzle Forces and Moments for 1500# Flange Rating

Nominal Diameter (NPS)	Circumferential Moment (M _c) (lbf-ft)	Longitudinal Moment (M _L) (lbf-ft)	Torsional Moment (M _T) (lbf-ft)	Shear Load (V _c) (lbf)	Radial Load (P) (lbf)	Shear Load (V _L) (lbf)
2	340	440	556	600	1,000	600
3	765	990	1,251	900	1,500	900
4	1,360	1,760	2,224	1,200	2,000	1,200
6	3,060	3,960	5,005	1,800	3,000	1,800
8	5,440	7,040	8,897	2,400	4,000	2,400
10	8,500	11,000	13,901	3,000	5,000	3,000
12	12,240	15,840	20,018	3,600	6,000	3,600
14	16,660	21,560	27,247	4,200	7,000	4,200
16	21,760	28,160	35,588	4,800	8,000	4,800
18	27,540	35,640	45,041	5,400	9,000	5,400
20	34,000	44,000	55,606	6,000	10,000	6,000
24	48,960	63,360	80,072	7,200	12,000	7,200

Nozzle Forces and Moments for 2500# Flange Rating

2	374	484	612	660	1,100	660
3	842	1,089	1,376	990	1,650	990
4	1,496	1,936	2,447	1,320	2,200	1,320
6	3,366	4,356	5,505	1,980	3,300	1,980
8	5,984	7,744	9,787	2,640	4,400	2,640
10	9,350	12,100	15,292	3,300	5,500	3,300
12	13,464	17,424	22,020	3,960	6,600	3,960
14	18,326	23,716	29,972	4,620	7,700	4,620
16	23,936	30,976	39,146	5,280	8,800	5,280

18	30,294	39,204	49,545	5,940	9,900	5,940
20	37,400	48,400	61,166	6,600	11,000	6,600
24	53,856	69,696	88,080	7,920	13,200	7,920

7.6.12 Reinforcement pads shall not be used for nozzles in cyclic service, HTHS, or other services with operating temperatures above 400 °C (750 °F).

- **7.6.13** The purchaser shall specify whether reinforcing pads shall not be used in sour or wet hydrogen sulfide services. For additional information A.2.2.

7.6.14 The maximum allowable working pressure shall not be limited by nozzle reinforcement.

7.7 Flanged External Girth Joints

7.7.1 Flange design shall be in accordance with the pressure design code unless otherwise specified by the purchaser.

Note: See A.4.9 and A.4.10 for guidance on girth flange.

7.7.2 Channel and shell external girth joints shall be of through-bolted construction. Studded-in bolts may be used when specified or approved by the purchaser.

7.7.3 Stud length for blind holes shall be sufficient to accommodate two heavy hex nuts beyond the made-up flange assembly.

7.7.4 For studs in blind holes, the bottom of the hole shall be relieved to prevent galling. Thread relief shall be per ASME PCC-2, Article 303, Mandatory Appendix 1c.

7.7.5 Flanges for external girth joints shall be of the forged welding-neck type, unless otherwise specified or approved by the purchaser.

7.7.6 Nubbins shall not be used.

7.7.7 The design clearance between mating flanges shall not be less than 3 mm (¹/₈ in.) at the periphery of the flanged joint. After assembly of the girth flange joint, the measured clearance between the mating flanges shall not be less than 1.5 mm (¹/₁₆ in.). The clearance between flanges shall extend within the bolt circle to allow flanges to be checked for radial distortion caused by an excessive bolt load.

7.7.8 Spot facing or back facing of the bolting bearing surfaces shall be in accordance with ASME B16.5 and the pressure design code.

7.7.9 Hydraulic bolt tensioning shall be applied for all bolt diameters equal to or greater than 50 mm (2 in.), or in hydrogen service where bolt diameters are equal to or greater than 40 mm (1¹/₂ in.), or when specified by the purchaser.

7.7.10 When the effective bolt length to diameter ratio (L/D) is less than 6, alternate forms of bolt tightening shall be applied in lieu of tensioning; the method to be applied is subject to the agreement of the purchaser. The effective bolt length is as defined by ASME PCC-1, appendix J.

7.7.11 When bolt tensioning is used, the bolting shall have additional thread length equivalent to one bolt diameter, extending from the nut at one end, to allow attachment of the bolt-tightening device.

7.7.12 When either manual or hydraulic torquing is applied, through-hardened washers shall be provided under nuts for all bolts of external girth flanges. When normal process operating temperatures are below -50C (-58F), the use and type of washers shall be per the agreement between the purchaser and manufacturer. For additional information see A.4.11.

7.7.13 Washers shall conform to the requirements of ASME PCC-1, Appendix M. The washers shall be suitable for re-use as defined in ASME PCC-1, Appendix M, unless otherwise specified by the purchaser.

7.7.14 Stationary tubesheet flange assembly shall be designed with the same type of gasket on the tube side and shell side.

7.7.15 Allowable stresses that have been established on the basis of short-time tensile strength shall not be used for the design of girth flanges, gasketed tubesheets, and gasketed flat covers.

NOTE In some pressure design codes the allowable stresses of some alloys have been established in this way. These stress values may result in dimensional changes due to permanent strain where slight distortion can cause leakage.

7.7.16 Custom designed external girth flanges shall be provided with 3 mm ($1/8$ in.) future machining allowance on the gasket contact seating surface (including pass-partition surfaces). The additional thickness shall not be used in the calculation of maximum allowable working pressure.

NOTE: This requirement does not apply to clad or weld overlay construction.

7.7.17 Tongue and groove type configuration shall not be used, unless specified or agreed by the purchaser.

7.7.18 Maximum spacing between bolt centers shall be as calculated by TEMA.

7.7.19 The joint component approach, as defined in ASME PCC-1, Appendix O, and the requirements contained within this section, shall be applied to each external girth flange of the heat exchanger within the following: See A.4.9 and A.4.10 for additional guidance.

NOTE: Definition of terms and symbology contained within this section are consistent with ASME PCC-1, except for pressure terms as modified for vacuum pressure conditions.

7.7.19.1 The minimum permissible bolt stress, Sb_{min} , and the maximum permissible bolt stress, Sb_{max} , shall be 40 % and 70 %, respectively, of the listed yield values in the pressure design code at assembly conditions (i.e. ambient temperature).

7.7.19.2 At least one-half of the gasket pass-partition rib area shall be added to the peripheral ring gasket area, Ag , unless otherwise agreed with the purchaser.

7.7.19.3 The selected assembly bolt stress, Sb_{sel} , shall be established so as to ensure that the gasket stresses comply with Table 7. For gasket types not listed, the minimum and maximum gasket stresses shall be agreed between the purchaser and vendor.

7.7.19.4 The fraction of a single gasket load remaining after relaxation, ϕ_g , used in the calculations shall be 0.7, unless otherwise specified by the purchaser. Stationary tubesheets with gaskets on both the shell and tube sides of the tubesheet shall consider a value of based on the relaxation of both gaskets combined.

7.7.19.5 To minimize potential for flange rotation, the rigidity index as calculated in ASME BPVC, Section VIII Division 1, Mandatory Appendix 2, shall not exceed 1.0.

Table 7- Assembly Gasket Stress

Peripheral Gasket Type ^a	Maximum Permissible Gasket Stress, $S_{g_{max}}$ MPa (psi)	Minimum Gasket Seating Stress, $S_{g_{min-S}}$ MPa (psi)	Minimum Gasket Operating Stress, $S_{g_{min-O}}$ MPa (psi)
Grooved metal with soft facing	380 (55,000)	140 (20,000)	97 (14,000)
Corrugated metal with soft facing	275 (40,000)	140 (20,000)	97(14,000)
Spiral-wound	b	140 (20,000)	97 (14,000)

- ^a These stresses are based on the use of facing layers or filler materials, such as flexible graphite, PTFE, or other conformable materials, and excludes core materials of carbon steel, brass, copper, or aluminum alloys.
- ^b The maximum permissible gasket stress does not apply when a means to prevent over-compression of the gasket is employed (e.g. centering rings with spiral-wound gaskets).

7.7.19.6 Stationary tubesheets with gaskets on both the shell and tube sides of the tubesheet shall incorporate the following into the design:

- a) A common selected bolt load shall be applied for the joint.
- b) Gasket operating stress determined by PCC-1 Step 6 Equation O-8 shall be calculated for the shell side and tube side independently based on their respective design pressures.
- c) The following additional shell side checks, using Equation O-8 as the basis for these checks:
 - 1) The gasket stress on the shell side shall be checked for the conditions where the tube side is pressured with no pressure on the shell side (P_{max} taken to be $(-P_t)$). For this condition, the gasket stress is evaluated to the minimum and maximum permissible gasket stress as defined in PCC-1, Appendix O.
 - 2) The gasket stress on the shell side shall be checked for the conditions where the shell side is pressured with no pressure on the tube side (P_{max} taken to be (P_s)). For this condition, the gasket stress is evaluated to the minimum and maximum permissible gasket stress as defined in PCC-1, Appendix O.
 - 3) The gasket operating stress shall be checked for the conditions where the tube side is pressured with both sides in operation (P_{max} taken to be $(P_s - P_t)$). For this condition, the gasket stress is evaluated to the minimum gasket operating stress as defined in PCC-1, Appendix O.
- d) The following additional tube side checks, using Equation O-8 as the basis for these checks:
 - 1) The gasket stress on the tube side shall be checked for the conditions where the shell side is pressured with no pressure on the tube side (P_{max} taken to be $(-P_s)$). For this condition, the gasket stress is evaluated to the minimum and maximum permissible gasket stress as defined in PCC-1, Appendix O.
 - 2) The gasket stress on the tube side shall be checked for the conditions where the tube side is pressured with no pressure on the shell side (P_{max} taken to be (P_t)). For this condition, the gasket stress is evaluated to the minimum and maximum permissible gasket stress as defined in PCC-1, Appendix O.
 - 3) The gasket operating stress shall be checked for the conditions where the shell side is pressured with both sides in operation (P_{max} taken to be $(P_t - P_s)$). For this condition, the gasket stress is evaluated to the minimum gasket operating stress as defined in PCC-1, Appendix O.
- e) P_s and P_t shall be adjusted to account for a coincidental vacuum condition on the other side when applicable.

7.7.19.7 If the requirements contained within this section require the bolt area to be increased or gasket geometry revised, then the pressure design code calculations shall be updated, and the required flange thickness re-calculated.

7.7.19.8 Controlled bolt-tightening procedures shall be applied, and records provided, to demonstrate that the selected assembly bolt stress, $S_{b_{sel}}$, has been achieved.

- **7.7.19.9** The purchaser shall specify the joint component method in 7.7.19 shall be applied for floating-head joints.

7.8 Expansion Joints

7.8.1 Expansion joints in shells shall be thin-wall bellows or thick-walled (e.g. flanged, or flanged-and-flued). The type of expansion joint to be approved by the purchaser. Floating head tail-pipe expansion joints shall be thin-wall bellows type. See A.2.3.

7.8.2 External expansion joints made of thin-walled bellows shall:

- a) Be protected from internal damage with a metallic liner mounted flush with the inside of the shell.
- b) The liner material shall be compatible with the base material to which it is attached.
- c) Carbon steel liners shall be at least 6 mm (1/4 in.) thick.
- d) Liners of stainless steel or other alloys shall be at least 3 mm (1/8 in.) thick.
- e) The liner should be attached to the upstream side to protect the bellows from the flow direction.
- f) For vertically mounted bellows, adequate means of drainage and particulate build up prevention shall be provided.
- g) Be protected from external damage with a metallic cover that is designed to remain intact during shipping, handling, and operation.
- h) Be designed to meet the requirements of the pressure design code. In no case shall the cycle life as calculated by EJMA be less than 1000 normal operating cycles.
- i) Be designed for any torsional, lateral, or angular deflection allowance needed for the initial fit-up and consider this in the design life of axial extension/compression cycles.
- j) Have the bellows formed from a cylinder having only a single longitudinal weld that shall be 100 % radiographed before forming. The acceptance criteria shall comply with the pressure design code.
- k) Have bellows welds that are 100 % examined by the liquid-penetrant method including attachment welds. The acceptance criteria shall comply with the pressure design code.
- l) When PWHT is required after installation see 9.6.13 and A.2.3.
- m) Be designed with stays to prevent damage during shipping and handling. Stays that require removal or adjustment prior to operation must be clearly identified.

7.8.3 Internal tail-pipe expansion joints made of thin-walled bellows shall:

- a) Be thin wall bellow construction for internal floating-head tail-pipe expansion joints and be fitted with permanent stays to prevent damage due to over extension or lack of weight bearing support during maintenance and hydrostatic testing with the shell cover removed. The stays permit the expansion joint's full design movements.
- b) Fitted with welded ends unless otherwise specified by the purchaser.
- c) Be protected from internal damage with a metallic liner mounted flush with the inside of the shell (or tail-pipe spool).
- d) The liner material shall be compatible with the base material to which it is attached.
- e) Carbon steel liners shall be at least 6 mm (1/4 in.) thick.
- f) Liners of stainless steel or other alloys shall be at least 3 mm (1/8 in.) thick.
- g) The liner should be attached to the upstream side to protect the bellows from the flow direction.
- h) For vertically mounted bellows, adequate means of drainage and particulate build up prevention shall be provided.
- i) When PWHT is required after installation see 9.6.13 and A.2.3.
- j) Be designed to meet the requirements of the pressure design code. In no case shall the cycle life as calculated by EJMA be less than 1000 normal operating cycles.

- k) Be designed for any torsional, lateral, or angular deflection allowance needed for the initial fit-up, and consider this in the design life of axial extension/compression cycles.
- l) Have the bellows formed from a cylinder having only a single longitudinal weld that shall be 100 % radiographed before forming. The acceptance criteria shall comply with the pressure design code.
- m) Have bellows welds that are 100 % examined by the liquid-penetrant method including attachment welds. The acceptance criteria shall comply with the pressure design code.
- n) Be designed with stays to prevent damage during shipping and handling. Stays that require removal or adjustment prior to operation must be clearly identified.

NOTE: See A.2.3 for additional information.

7.8.4 Expansion joints made of thick-walled flexible shell elements shall:

- a) Be constructed with either flanged only or flanged-and-flued flexible shell elements.
- b) Have expansion joint material of the same P Number as the shell material.
- c) Be supplied either with an internal sleeve or have transverse baffles located between convolutions to eliminate bypass in the expansion joint. If a sleeve is used, it shall have an opening to allow shell side fluid to heat the expansion joint.
- d) Consider if multiple flexible shell elements located in series may be necessary to accommodate the differential movement between the shell and tubes.
- e) A means of venting and draining the high and low points shall be provided. The method of venting and draining shall be subject to the agreement between the purchaser and vendor.

NOTE: See A.2.3 for additional information.

7.9 Gaskets

7.9.1 Flanged external girth joint gaskets and floating-head gaskets in hydrocarbon or steam service shall be double-jacketed metal with soft gasket filler, spiral-wound, grooved metal with soft gasket-seal facing, or corrugated metal with soft gasket-seal facing. Solid metal gaskets shall only be used with welded closures or self-energizing closures. See A.4.12 for additional guidance.

NOTE: The term grooved is synonymous with the term serrated as applied to gasket type.

7.9.2 Double-jacketed gaskets shall not be used for the peripheral portion of the gasket in the following services:

- a) Hydrogen service.
- b) Operating temperatures above 204 °C (400 °F).
- c) Cyclic service.
- d) Sour or wet hydrogen sulfide service.
- e) Design pressure equal to or greater than 2050 kPa (ga) (300 psig).
- f) Shell diameter greater than 1200 mm (48 in.).

7.9.3 Pass-partition rib(s) shall be of the same material grade as the peripheral portion of the gasket.

7.9.4 Pass-partition ribs shall not be made from flat solid metal.

7.9.5 Pass-partition ribs shall be designed and constructed such that they do not inhibit the compression or seating of the peripheral portion of the gasket.

7.9.6 If the gasket OD is less than or equal to 610 mm (24 in.), the peripheral portion shall be of one piece (non-welded) construction. If the gasket OD exceeds 610 mm (24 in.), the maximum quantity of welds in the peripheral portion shall not exceed the following:

- a) for double jacketed gaskets, one;
- b) for other gasket types, the greater of either two, or the ratio of gasket OD/610 mm (gasket OD/24 in.).

7.9.7 The gasket welds shall be as follows:

- a) The welds in the peripheral portion of the gasket shall be continuous and full-penetration. The cross-section, finish, and flatness of these welded areas must match the remainder of the peripheral gasket.
- b) The welds shall not inhibit the sealing or compression of the peripheral gasket or pass ribs.

7.9.8 When adhesives are used to bond facing materials (e.g. graphite) to grooved and corrugated metal gaskets, they shall be compatible with the metal core (e.g. chloride content).

7.9.9 Double jacketed gaskets shall be as follows:

- a) They shall be manufactured with the filler material completely enclosed in metal.
- b) Filler materials shall be selected to ensure no degradation occurs during manufacturing (e.g. welding).

7.9.10 Grooved metal gaskets shall be as follows:

- a) The top of the serrations at the weld shall be flat in comparison with the rest of the gasket. The grooves shall be dressed to match the standard profile.
- b) Thickness variations in the metallic core, after completion of gasket fabrication, shall not exceed 0.13 mm (0.005 in.).
- c) Where the outside diameter is less than 1 m (40 in.), they shall be manufactured with one-piece continuous facing materials (no joints). Facing joints in larger-diameter gaskets shall include a 1.5 mm (¹/₁₆ in.) to 3 mm (¹/₈ in.) overlap of the facing material; however, butt joints (gap free) are allowed for flexible graphite facings. Facing joints shall not be coincident with a facing joint on the opposite sealing face, or with any welds in the core.
- d) Graphite shall be affixed to the metal core with a thin coat of suitable adhesive that firmly holds the graphite to the ring. Prefabricated, adhesive-backed graphite shall not be used.

7.9.11 Corrugated metal gaskets shall be as follows:

- a) minimum core thickness of 0.46 mm (0.018 in.);
- b) minimum width of 12 mm (¹/₂ in.);
- c) corrugations in the weld area that match the gasket profile;
- d) where the gasket outside diameter is less than 1000 mm (40 in.), they shall be manufactured with one-piece continuous facing materials (no joints). Facing joints in larger-diameter gaskets shall include a 1.5 mm (¹/₁₆ in.) to 3 mm (¹/₈ in.) overlap of the facing material; however, butt joints (gap free) are allowed for flexible graphite facings. Facing joints shall not be coincident with a facing joint on the opposite sealing face or with any welds in the core.

7.9.12 Spiral-wound gaskets shall be provided with a flat inner ring to prevent over-compression and inward radial buckling. Spiral wound gasket for shell inside diameters less than or equal to 1000 mm (40 in.) shall not have an outer ring. For larger diameters an outer ring shall be used for handling purposes.

7.9.13 The maximum gasket width shall be 50 mm (2 in.), except for non-metallic sheet type gaskets.

7.9.14 Where serrated, corrugated metal or grooved metal gaskets with graphite, polytetrafluoroethylene (PTFE), or other similar facing are used, the values of m and y used in the flange design and analysis shall be greater than or equal to 3.75 and 52.4 MPa (7600 psi) respectively.

7.9.15 When welded lip seal or welded diaphragm joints are specified, they shall be designed to accommodate differential thermal growth. The design and method of analysis shall be subject to approval of the purchaser.

7.10 Handling Devices

7.10.1 The lifting device shall be a weld-on lug or ring provided with a hole not less than 38 mm (1½ in.) in diameter. Wherever possible, the lug or ring shall be located at the top of the component, above its center of gravity; otherwise, two suitably located lugs or rings shall be provided. The lifting device shall be designed to support at least twice the mass of the component.

7.10.2 For stacked heat exchangers, two lifting lugs or rings shall be provided on all covers, located at the top, approximately 45° from the vertical centerline.

7.10.3 Stationary tubesheets shall be provided with pulling lugs or tapped holes on their tubeside face for the insertion of eyebolts to aid removal of the tube bundle from the shell. Pulling lugs and tapped holes shall be designed for a pulling force equal to at least 150 % of the tube bundle mass. Each tapped hole shall be fitted with a threaded plug of the same material as the tubesheet face. The exposed section of the plug shall be at least 50 mm (2 in.) long.

7.10.4 All vertical heat exchangers shall be provided with lifting devices for the entire heat exchanger. The lifting devices shall be provided above the center of gravity of the heat exchanger and shall not transmit the load through a bolted joint unless approved by the purchaser.

7.10.5 Localized stresses caused by concentrated loads due to lifting of the exchanger shall be evaluated. Evaluation shall be performed in accordance with WRC 537 or finite element analysis. Any other standard or numerical methods may be used subject to approval by the purchaser. All geometrical limits specified in methods (such as WRC, etc.) used for local load analysis shall be followed. Extrapolation beyond stated limits is not allowed.

7.10.6 For bundle handling devices see 7.5.8.

7.11 Requirements for Hydrogen Service

7.11.1 All external attachment welds to the pressure boundary shall be full penetration. If it is not possible to avoid enclosed spaces between welds, then the space shall be vented with a 6 mm (¼ in.) diameter hole.

- **7.11.2** The purchaser shall specify any supplemental requirements for low chrome steels in high temperature or high pressure hydrogen service (e.g. API Recommended Practice 934-A, API Recommended Practice 934-C). See section 13.

8 Materials

8.1 General

8.1.1 Castings shall not be used, unless approved by the purchaser.

8.1.2 Material for external parts that are welded directly to the heat exchanger, such as pads, brackets, and lugs, shall be of the same nominal composition as the material to which they are welded.

8.1.3 If alloy linings are specified by the purchaser, they shall be weld overlay, integrally-clad, or explosion-bonded. Loose liners or sleeves shall not be used without the approval of the purchaser.

- **8.1.4** The purchaser shall specify materials for all components, including bolting and gaskets.

8.1.5 Tubesheets that are welded to the adjacent shell or channel cylinder by use of a corner joint can be provided as forged or plate materials. If plate material is used, it shall be 100 % ultrasonically examined for laminations and other defects prior to attachment to the cylinder.

8.1.6 Baffles, tie rods, spacers, impingement devices, detuning devices, and sealing devices shall be manufactured from material with equivalent chemical composition and corrosion resistance to those for the tubes unless otherwise specified by the purchaser.

- **8.1.7** The purchaser shall specify if stainless steel grades 321 and 347 shall be stabilized after solution annealing heat treatment.

8.1.8 Duplex stainless steel materials shall meet the requirements specified in API TR 938-C, Annex A (Plate, Pipe, Forgings, Fittings, Bar, and Tubing) and Annex D (Castings).

8.1.9 For guidance on low temperature structural support requirements to ensure adequate toughness see A.5.1.

8.1.10 Use of non-impact tested materials as allowed by ASME Section VIII, Division 1, clause UG-20 (f), or equivalent exemptions from other pressure design codes, shall not be permitted.

8.1.11 The maximum allowable carbon equivalent (CE) for carbon steel plate shall be in accordance with Table 8.

Table 8— Maximum Allowable Carbon Equivalent in Carbon Steel

Plate thickness	Maximum allowable CE
≤ 50 mm (2 in.)	0.43
> 50 mm (2 in.) ≤ 100 mm (4 in.)	0.45
> 100 mm (4 in.)	0.48

8.2 Gaskets

8.2.1 Gaskets shall not contain asbestos.

8.2.2 Material for metal-jacketed, grooved-metal, corrugated-metal, or solid-metal gaskets shall have a corrosion resistance at least equal to that of the gasket contact surface material.

8.2.3 Metal windings and inner rings (when used) of spiral-wound gaskets shall be of austenitic stainless steel, unless otherwise specified or approved by the purchaser. Where materials of construction are a higher alloy, then the metal windings shall be the same alloy material.

8.2.4 Grooved, or metal-jacketed gaskets, shall be softer than the gasket contact surface.

8.2.5 Gasket material, including filler and facing material, shall be selected to withstand the maximum design temperature. See A.5.2 for additional information on gasket material.

8.2.6 Compressed sheet type gaskets shall not be used in hydrocarbon, steam, hydrogen, sour or wet hydrogen sulfide service.

8.3 Tubes

8.3.1 Integrally finned tubes shall comply with the following:

- ASTM A498, seamless and welded carbon steel heat-exchanger tubes with integral fins.
- ASTM A1012, standard specification for seamless and welded ferritic, austenitic and duplex steel condenser and heat exchanger tubes with integral fins.

- c) ASTM B891, standard specification for seamless and welded titanium and titanium alloy condenser and heat exchanger tubes with enhanced surface for improved heat transfer.
- d) ASTM B924, standard specification for seamless and welded nickel alloy condenser and heat exchanger tubes with integral fins.
- e) ASTM B359 / ASME SB-359, standard specification for copper and copper-alloy seamless condenser and heat exchanger tubes with integral fins.

8.3.2 In addition to the requirements by the tube material specification, integrally finned tubes and other types of enhanced-shape tubes, shall be individually hydrotested or pneumatically tested after cold forming (or after heat treatment, when applicable), in accordance with the applicable pressure design code.

8.3.3 Materials and services where integrally-finned tubes require heat-treatment after cold forming shall include, but are not necessarily limited to, the following (for additional guidance for finned tubes A.5.3):

- a) Copper alloy UNS Nos. C23000, C44300, C44400, C44500, C60800, and C68700 shall be annealed (O61) after finning.
- b) Ferrous tube materials, as listed in NACE MR0103 or MR0175, when wet H₂S or sour service is specified for the shell side by the purchaser.

Note: NACE MR0103 requires heat treatment for ferrous materials if the cold forming process results in deformation in excess of 5 %. Refer to NACE MR0103 and MR0175 as applicable, for a complete list of requirements for cold-formed shapes.

- c) Duplex stainless steel tubes, when wet H₂S or sour service is specified for the shell side by the purchaser (solution-annealing type heat treatment is required after cold forming of fins for duplex stainless steels).

Note: Refer to API Technical Report 938-C for more information and experience with integrally-finned duplex stainless steel tubes.

- d) Carbon steel tubes in amine solvent service on the shell side.

Note: Refer to API Recommended Practice 945 for more information.

- e) Carbon steel tubes in caustic service on the shell side, when the maximum service temperature exceeds 49 °C (120 °F).

Note: The maximum allowable service temperatures for non-stress relieved CS and stress-relieved carbon steel in caustic service depends on caustic concentration and temperature. These limits can be found in the caustic soda service chart in NACE SP0403.

- f) Austenitic stainless steel and high nickel alloy tubes, when wet H₂S or sour service is specified for the shell side by the purchaser, if the cold forming process results in hardness values exceeding NACE MR0103 or MR0175, as applicable. Compliance with the maximum allowable hardness in the cold-formed condition must be demonstrated by the manufacturer, otherwise a solution-annealing heat treatment is required.

- **8.3.4** The purchaser shall specify if heat treatment is required for integrally finned-tubes and other types of enhanced shape tubes if other types of environmental cracking mechanisms are applicable, considering the tube material and service conditions (e.g. chloride stress corrosion cracking (SCC), alkaline SCC, including caustic SCC for materials other than carbon steel, carbonate SCC, ethanol SCC, etc.). For additional guidance for finned tubes A.5.3.

8.3.5 The purchaser and manufacturer shall agree on the heat-treatment method and procedure for low-fin tubes and other types of enhanced shape tubes.

8.3.6 All welded tubes shall be non-destructive electric tested in the finished condition (prior to any tube enhancement or bending) over their full length.

8.3.7 For welded duplex stainless steel heat exchanger tubing:

- a) The seam weld shall be provided with corrosion testing in accordance with ASTM A923 Method C for standard and super duplex stainless steel grades, ASTM A1084 Method C for lean duplex stainless steel grades, or ISO 1778.
- b) For tubing which will be expanded only into tubesheets or integrally-finned, seam-weld bead working or tube redrawing shall be completed.
- c) Electric test (eddy current) and hydrotest or air underwater test shall be completed.

9 Fabrication

9.1 Shells

9.1.1 All longitudinal and circumferential welds of shells shall be finished flush with the inner contour. This allows for easier tube bundle insertion and withdrawal, and flow pattern disruptions. For kettle-type heat exchangers, this requirement shall not apply to welds in the enlarged section, unless they are in the bottom quadrant of the shell.

9.1.2 The main cylinders (shell and channel) seams, connections, supports, and other external attachments (except for insulation support rings) shall be laid out so that connections or reinforcement pads do not intersect the longitudinal or circumferential weld seams. Openings and attachments should clear the toe of weld seams by at least 50 mm (2 in.). If this is not possible, unless otherwise approved by the purchaser, the seam weld should be ground flush and inspected for a distance of 100 mm (4 in.) on either side of the opening, or for the full length of the seam covered by an attachment. Inspection of the seam welds shall be by full radiography, or 100% volumetric and magnetic particle examination.

9.1.3 Longitudinal weld seams on horizontal exchangers shall be located above the horizontal centerline. When this is not possible, longitudinal seams shall not be located more than 30° below the horizontal centerline unless approved by the purchaser. Where possible weld seams should not be located under saddle bearing plates.

9.1.4 Where attachments will cover main seam welds, the length of the main seam weld covered by the attachment and projecting at least 50 mm (2 in.) beyond each side of the attachment shall be ground flush and inspected with 100 % volumetric examination and magnetic particle or liquid penetrant examination.

9.1.5 For removable-bundle heat exchangers, the permissible out-of-roundness of a completed shell, after all welding and any required heat treatment, shall allow a metal template to pass through the entire shell length without binding. The template shall consist of two rigid disks (each with a diameter equal to the diameter of the transverse baffle or support plate) rigidly mounted perpendicularly on a shaft and spaced not less than 300 mm (12 in.) apart.

9.1.6 Transverse baffle-to-shell clearances greater than those indicated in TEMA, Table RCB-4.3, shall not be used, unless approved by the purchaser.

9.2 Pass-partition Plates and Longitudinal Baffles

Longitudinal baffles which are welded to the shell and pass-partition plates for channels and floating-heads shall be welded full length, either from both sides or with full-penetration welds, except for special designs approved by the purchaser. If welded from both sides, the first 50 mm (2 in.) from the gasket face shall be a full-penetration weld.

9.3 Connection Junctions

9.3.1 All inside edges of nozzles and couplings without clad or weld overlay shall be rounded off to a radius as follows:

- a) In cyclic service, 25 % of thickness of penetrated shell or head, but not less than 6 mm (1/4 in.) or more than 19 mm (3/4 in.).
- b) All other services to a minimum of 3 mm (1/8 in.).

9.3.2 All inside edges of nozzles with clad or weld overlay shall be rounded off to a radius as specified in Table 9 unless otherwise specified by the purchaser.

Table 9 — Nozzle Junction Radius Requirements for Clad or Weld Overlay Construction

	Base Metal	Clad or Weld Overlay (if applicable)
Cyclic Service	25% of thickness of penetrated shell/head but not less than 8 mm (5/16") or more than 19 mm (3/4")	Radius shall not exceed 1.5 mm (1/16") less than that of the base metal. The minimum required weld overlay thickness shall be maintained.
All Other Services	8 mm (5/16") minimum	6 mm (1/4") minimum. The minimum required weld overlay thickness shall be maintained.

9.4 Tubes

9.4.1 All tubes, including U-tubes, shall be formed from a single length and shall have no circumferential welds, unless approved by the purchaser.

9.4.2 No tubes or tube holes shall be plugged without notifying the purchaser. The method and procedure of plugging shall be subject to approval of the purchaser.

9.5 Welding

9.5.1 Welding and associated weld testing shall comply with the welding requirement of API Recommended Practice 582, or as specified by the purchaser.

9.5.2 Category A and B welded joints shall be full-penetration welds.

9.5.3 All welds attaching connections to cylinders or to heads shall fully penetrate the total thickness of the component wall or the connection wall forming the attachment.

9.5.4 Backing strips that remain in place on the inside of a component after completion of welding shall not be used, unless approved by the purchaser.

9.5.5 The method used to position a tube in advance of tube-end welding shall be done in such a way as to prevent the trapping of gases during the welding process.

9.5.6 Strength-welded tube-to-tubesheet welds shall have a minimum of two weld passes and be produced by gas tungsten arc welding (GTAW) using filler material, unless otherwise specified or approved by the purchaser.

9.5.7 Autogenous tube-to-tubesheet seal welded joints are not permitted.

9.5.8 The qualification of the weld procedure for a strength-welded tube-to-tubesheet joint shall be in accordance with ASME BPVC Section IX QW-193.1 or equivalent, and the following additional requirements:

- The minimum leak path (i.e. the distance from the root of the weld to the outer surface nearest to the root) shall not be less than two-thirds of the nominal tube wall thickness.
- A minimum of three tensile pull-tests shall be performed on the qualification test coupon whenever required to support the calculations of ASME BPVC Section VIII, Division 1, Joint Strength by Calculation, or equivalent, or when requested by the purchaser.

9.5.9 Welds attaching nonpressure external attachments (such as lugs or structural steel supports) shall be continuous, except for insulation support rings.

9.5.10 Repair-associated welding procedures, including heat treatment and non-destructive examination procedures, shall be submitted to the purchaser for review prior to the start of the repair.

9.5.11 Full-penetration welds shall be used for all internal attachments to the pressure boundary components that will be exposed to hydrogen service.

9.5.12 The performance qualification of the weld operators for a strength-welded tube-to-tubesheet joint shall be in accordance with ASME BPVC Section IX QW-193.2 or equivalent.

9.5.13 A minimum of two layers shall be applied for all weld overlays. Electroslag strip cladding may be performed with a single pass, subject to the purchaser's approval.

9.5.14 Dissimilar metal welds shall not be used for pressure-retaining joints unless approved by the purchaser.

9.5.15 Temporary attachment welds shall be removed and the affected area examined by either liquid-penetrant or magnetic particle examination. If necessary, the locations may be repaired by fill-welding and ground smooth prior to the examination.

9.5.16 Production test plates, if required per the design code or specified by the purchaser, shall be welded and heat treated in accordance with the procedures used for production welds.

9.6 Heat Treatment

9.6.1 Machined contact surfaces, including any threaded connections, shall be suitably protected to prevent scaling or loss of finish during heat treatment.

9.6.2 Unless otherwise specified by the purchaser, the following shall be subject to heat treatment after tube bending:

- a) Carbon steel, carbon-molybdenum, and low chromium steel U-bends (C - 0.5 % Mo through 5 % Cr steels) whenever the purchaser specifies that the equipment is in sour or wet hydrogen sulfide, amine, caustic, carbonate/bi-carbonate, HF acid service, or hydrogen service.
 - b) Ni-Cu (e.g., Alloy 400) and Cu-Ni alloy U-bends in HF acid service.
 - c) U-bends in austenitic stainless steels.
 - d) U-bends in standard grades of duplex stainless steel (up to 22 % Cr) with bend radius less than 3.3 times the tube outside diameter (see 9.6.5 for more information).
- **9.6.3** For ferritic and martensitic stainless steels and high nickel alloys (Ni > 30 %), heat treatment of U-tubes shall be applied if cold working can induce susceptibility to stress corrosion. The purchaser shall specify when heat treatment is required in such cases.
 - **9.6.4** For chemically stabilized grades of austenitic stainless steel (e.g., 347, 321 grades), the purchaser shall specify if a stabilization heat treatment, between 870 °C and 900 °C (1600 °F and 1650 °F, is required for these materials prior to being used in fabrication, including tubes and U-bends, following the application of solution-annealing heat treatment. Soak time and temperature for this type of heat treatment shall be agreed between the purchaser and vendor.

9.6.5 Duplex stainless steel tubes and U-bends shall not be heat-treated after bending or straightening, unless otherwise approved or specified by the purchaser. For standard DSS grades (up to 22 % Cr), cold work shall be limited to 15 % maximum (equivalent to limiting U-bend radii to a minimum of 3.3 times the tube diameter) without heat treatment required. For 25 % Cr and 27 % Cr DSS grades, U-bend radii may be a minimum of 2 times the tube diameter without heat treatment required. When these minimum values cannot be practically met or when heat treating is deemed necessary by the purchaser due to the service environment, the following specifications shall be followed:

- a) Prior to heat treatment the tubes shall be cleaned with a solvent to remove lubricants and markings in the bent section.
- b) The tube ID shall be purged with argon gas before and maintained in the tube during the heat treatment to prevent oxidation of the ID surface.

- c) Electric-resistance method shall be used. Soak temperature shall be consistent with the governing material specification (e.g., ASTM A789) and not exceed a hold time of 30 seconds. This shall be followed by rapid cooling to below 315 °C (600 °F) within 2 minutes. The total time above 315 °C (600 °F) during heat up and quenching shall not exceed 5 minutes.
- d) The U-tube outside surface shall be free of heavy scaling. This may be accomplished by bright annealing, white pickling or polishing with mineral belts. A light oxide on the outside surface of the heat-treated area is acceptable.
- e) Two tube cross-sections, each 25 mm (1 in.) in length, shall be qualification tested via mock-up: one from the U-bend section and one from the straight tube leg adjacent to the ER clamp location, for each heat and lot produced with the same bending and heat treatment procedure, and subjected to corrosion testing as per ASTM A923 Method C or ISO 17781, as defined by the purchaser.

NOTE Refer to API Technical Report 938-C for additional information and supporting data on duplex stainless steel U-tube bend heat treatment. Heat treatment of DSS U-tube bends can result in a loss of corrosion resistance due to formation of harmful inter-metallic phases

9.6.6 When applied, U-bend heat treatment shall be completed by electric resistance or furnace methods. Open flame heat treatment is not permitted.

9.6.7 For electric resistance or induction heating of U-bends:

- a) The heat-treated portion of the U-bend shall extend at least 300 mm (12 in.) beyond the tangent point.
- b) The soak temperature should be higher than furnace heat treatment. For carbon steel U-bends, the soak temperature should be in the range 663 °C to 691 °C (1225 °F to 1275 °F). For stainless steel U-bends the minimum soak temperature shall be 1040 °C (1900 °F) and meet the governing material specification annealing temperature.
- c) Time during which the lowest measured metal temperature exceeds the minimum soak temperature shall be between 5 and 15 seconds. Tube metal temperature shall be controlled by calibrated optical or emission pyrometer.
- d) Clamp areas and adjacent regions of the tubes shall be inspected for damage from arc burns prior to bundle cage tube insertion. Tubes with any indication of arc burns shall be rejected.
- e) ER or induction heat treatment procedure shall be submitted for review by the purchaser.

9.6.8 For furnace heating of U-bends:

- a) The heat treatment shall extend at least 600 mm (24 in.) beyond the tangent point. The portion of tubes located outside of furnace shall use insulation to reduce thermal gradients.
 - b) The tubes (or tube bundle) shall be adequately supported during heat treatment.
 - c) The soak temperatures should be consistent with the PWHT ranges specified in API Recommended Practice 582.
 - d) Holding time at the soak temperature shall be at least 15 minutes.
 - e) Thermocouples shall be provided to ensure that all portions of the U-bends, and at least 300 mm (12 in.) of the straight portion of the tube measured from the tangent line achieve the required soak temperature for the specified holding time.
- **9.6.9** The purchaser shall specify if the heat treated portion of U-tubes shall receive a de-scaling treatment. Slight discoloration of the surface (such as level 4 of AWS D18.2) due to oxidation is acceptable.

9.6.10 Carbon steel and low-alloy (maximum 9 % nominal chromium) steel channels and bonnets shall have the base metal welds post weld heat treated for the following:

- a) channels and bonnets with six or more tube passes in which pass partition plates are welded directly to the base metal;

- b) channels and bonnets whose nozzle-to-cylinder internal diameter ratios are 0.5 or greater, except where a conical reducer is used in place of the channel or bonnet.

9.6.11 Postweld heat treatment of base metal welds shall be performed for all carbon steel and low-alloy (maximum 9 % chromium) steel floating-head covers that are fabricated by welding a dished-only head into a ring flange.

- **9.6.12** The purchaser shall specify if postweld heat treatment of shell side or tube side components is required for process reasons.

9.6.13 Thin-wall bellows type (internal & external) expansion joints shall not be subjected to any heat treatment other than that performed by the expansion joint manufacturer, or agreed to by the expansion joint manufacturer when further heat treatment is to be performed after its installation. See A.2.3.

9.6.14 Carbon steel heads shall be normalized after cold-forming. Hot-formed carbon steel heads which were not formed in the normalizing range shall be normalized.

9.6.15 Solid or clad duplex stainless steel materials, including weld deposit overlay, shall not be post weld heat treated.

9.7 Dimensional Tolerances

9.7.1 Manufacturing tolerances shall be such that nominally identical parts are interchangeable.

9.7.2 Heat exchangers that will be stacked in series shall be stacked in the shop to check connection alignment.

9.7.3 For stacked heat exchangers, mating nozzle flanges shall not be out-of-parallel with each other by more than 1 mm ($1/32$ in.), measured across any diameter. Separation of mating nozzle flanges shall not exceed 3 mm ($1/8$ in.) after installation of the gasket. Bolts shall be capable of being inserted and removed freely without binding. Shims shall be installed as required between the supports and shall be tack-welded in place.

9.7.4 Dimensional tolerances on U-bend forming to be as follows:

- a) Flattening after bending shall not exceeding 10 % of the tube outside diameter.
- b) Tube wall thinning shall be in accordance with TEMA.
- c) The bend diameter, measured at the point of tangency, shall be ± 1.5 mm ($1/16$ in.).
- d) The bend portion of the U-tube shall be uniform in curvature and not exceed ± 1.5 mm ($1/16$ in.) of the nominal center-line radius.
- e) The bend deviation from the plane shall be ± 1.5 mm ($1/16$ in.).

9.8 Gasket Contact Surfaces Other Than Nozzle Flange Facings

9.8.1 Gasket contact surfaces shall have finishes as shown in Table 10Table 10.

9.8.2 The flange seating face flatness tolerance shall be in accordance with PCC-1, Appendix D. The definitions for hard gaskets and soft gaskets used for the tolerance criteria in PCC-1, Appendix D shall be as follows:

- a) For hard gaskets: grooved metal gaskets, corrugated metal gaskets, metal jacketed gaskets, flat solid gaskets, and 1.5 mm ($1/16$ in.) or lesser thickness sheet gaskets. Hard gaskets may or may not have soft facings applied.
- b) For soft gaskets: spiral wound and sheet gaskets of thickness > 1.5 mm ($1/16$ in.).

Table 10 — Gasket Contact Surface Finishes

Type	Surface Roughness R_a^a
------	------------------------------

Solid flat metal gaskets	1.6 (63) ^b maximum
Double jacketed gaskets	1.6 to 3.2 (63 to 125)
Spiral-wound gaskets	3.2 to 6.3 (125 to 250)
Grooved or corrugated metal gaskets with soft gasket-seal facing	
Non-metallic soft cut sheet gaskets ≥ 1.5 mm (1/16 in.) thickness	3.2 to 6.3 (125 to 250)
Non-metallic soft cut sheet gaskets ≤ 1.5 mm (1/16 in.) thickness	3.2 to 12.7 (125 to 500)
^a R_a is roughness average.	
^b Dimensions are in micrometers (micro-inches).	

9.8.3 The flatness of gasket contact surfaces shall be measured with a dial gauge. However, the flatness of the pass-partition grooves and mating pass-partition plate edges can be measured with a straight edge.

9.8.4 Flange flatness tolerance and surface finish shall be measured after the flange has been welded to the component cylinder or the cover and after any post weld heat treatment.

9.8.5 The flatness of tubesheet gasket-contact surfaces shall be measured after the tube-to-tubesheet joints have been completed, including any heat treatment.

9.9 Tube Holes

9.9.1 Tube hole grooves shall be square-edged, concentric, and free from burrs.

9.9.2 If austenitic stainless steel, duplex stainless steel, titanium, cupronickel, or nickel alloy tubes are specified, the tube holes shall be machined in accordance with TEMA, Table RCB-7.21, column (b) (Special Close Fit).

9.10 Tube-to-Tubesheet Joints

Note: See A.4.7 for additional information.

9.10.1 If roller-expanded joints are utilized, the tube wall thickness reduction shall be in accordance with TEMA. However, these can be increased by a further 2 %, if approved by the purchaser.

9.10.2 If welded-and-expanded joints are specified, tube wall thickness reduction due to expansion shall begin at least 6 mm (1/4 in.) away from welds.

9.10.3 When tubes are to be expanded into the tubesheet, they shall be expanded through the full thickness of the tubesheet. However, in no case shall the expansion extend within 3 mm (1/8 in.) of the shell side face of the tubesheet.

9.10.4 When a strength welded tube-to-tubesheet joint is applied, the purchaser should consider a tube end expansion in addition to the strength weld, where it is desirable to seal the crevice between the tube and tubesheet hole on the shell side. It is not necessary to provide a tube hole ring groove for such services. The expanded depth should extend to within 3 mm (1/8 in.) of the shell side face of the tubesheet to minimize the crevice. For services where work hardening results in potential for environmental cracking or corrosion, a light tube expansion (e.g., 3 % wall reduction) should be used.

9.10.5 For shell side clad (or weld overlay) tubesheets, the tube shall be expanded to seal against the cladding material for a minimum distance of 6 mm (1/4 in.).

9.10.6 A ring groove shall not be used within shell side cladding, unless specified by the purchaser.

9.10.7 For tube side clad (or weld overlay) tubesheets with either strength-welded or seal-welded and expanded tube-to-tubesheet joints, tube hole ring grooves, when provided, shall be located in the base material and not in the applied cladding or weld overlay.

9.10.8 For welded-and-expanded tube-to-tubesheet joints requiring postweld heat treatment, the tubes shall be expanded after post weld heat treatment.

9.10.9 Tubes shall be flush with or extended by no more than 5 mm ($3/16$ in.) beyond the face of the tubesheet, except in vertical exchangers when tubes shall be flush with the top tubesheet, unless otherwise specified by the purchaser.

9.10.10 When not specified by the pressure design code the joint strength calculation of the tube-to-tubesheet expanded joint shall be in accordance with ASME BPVC Section VIII Div 1, UW-20.1.

9.11 Assembly

9.11.1 Match marks shall be provided to prevent mis-assembly of the following bolted joints:

- a) floating-head cover to tubesheet;
- b) channel to tubesheet;
- c) grooved channel cover to channel;
- d) stationary tubesheet to shell.

9.11.2 The threads of external studs and nuts shall be coated with a suitable anti-seize compound to prevent galling.

9.11.3 Unless otherwise specified by the purchaser, a thin film of adhesive may be applied on the gasket seating surface to hold the gaskets in place during assembly. The adhesive shall:

- a) Not contain zinc, lead, or copper.
- b) Not be comprised of tape, shellac, or grease.
- c) Be compatible with the heat exchanger process fluid and gasket materials.
- d) Be resilient and non-hardenable to ensure it does not interfere with gasket seating or otherwise alter sealing surface profile.
- e) Be low chloride or chloride free when used with stainless steel.

10 Inspection and Testing

10.1 Quality Control

10.1.1 Pressure containing welds inspection and testing shall be performed in accordance with the pressure design code. The following additional requirements apply:

- a) At least one spot-radiograph shall be made of each accessible butt-welded joint. Nozzle-to-shell welds are exempt from this requirement, unless required by the pressure design code.
- b) Spot radiographs shall include each start and stop of welds made by the automatic submerged-arc welding process.
- c) Spot radiographs shall be at least 250 mm (10 in.) long, or shall be full length if the weld is less than 250 mm (10 in.) long.
- d) Weld porosity limits for spot radiographs shall be as stated in the pressure design code for fully radiographed joints.
- e) 100 % radiographic examination shall be performed on any welded seam in any component that has been subjected to severe working (e.g. a ratio of thickness to local radius greater than 5 %) after welding.

- f) If set-on connections are used and the pressure envelope is plate, after welding of the connections, 100 % ultrasonic examination on the attachment welds and the plate shall be completed for at least 100 mm (4 in.) from the connection.
- g) Unless otherwise specified by the purchaser all butt welds for sour or wet hydrogen sulfide service shall be subject to 100 % volumetric examination. Nozzle to shell, nozzle to channel wall, and tubesheet to cylinder joints shall be 100 % ultrasonic tested.

10.1.2 If set-on connections are used, the edge of the hole in the plate to which the connections are attached shall also be examined by the magnetic-particle or liquid penetrant method prior to welding. Subject to agreement with the purchaser, indications found shall be cleared to sound metal and then repair-welded.

10.1.3 The magnetic-particle examination method, extent, and acceptance criteria shall comply with the pressure design code. If not specified by the design code, the minimum extent of magnetic particle examination shall be as per ASME BPVC, Section VIII, Division 2, Table 7.2.

10.1.4 For nonmagnetic materials, a liquid-penetrant examination shall be used in place of any required magnetic-particle examination.

10.1.5 The liquid-penetrant examination method, extent, and acceptance criteria shall comply with the pressure design code. If not specified by the design code, the minimum extent of liquid penetrant test examination shall be as per ASME BPVC, Section VIII, Division 2, Table 7.2.

10.1.6 Production weld-hardness testing of pressure-retaining welds shall be in accordance with the pressure design code, or the following requirements, whichever is the more stringent.

- a) Welds in components made of carbon, Cr-Mo, 11/13/17 % chromium steels, and duplex stainless steels shall be hardness tested. Hardness testing of the heat affected zone shall be conducted if required by the pressure design code, or when specified by the purchaser.
- b) If postweld heat treatment is required, examination shall be made after the post-weld heat treatment is completed.
- c) Unless otherwise agreed between the vendor and purchaser, the weld hardness shall not exceed the values listed in Table 11.

Table 11 — Hardness Limits

Material	Maximum Weld Hardness
Carbon Steel	225 HBW
Chromium Steel (up to 3 % Cr)	225 HBW
Chromium Steel (5 % Cr to 17 % Cr)	241 HBW
Duplex Stainless Steel (22 % Cr)	to be agreed with purchaser
Super Duplex Stainless Steel (25 % Cr)	to be agreed with purchaser
Carbon steel and chromium steels (up to 17%) in sour or wet hydrogen sulfide services	To be specified by purchaser

- d) Hardness readings shall be taken with a portable Brinell hardness tester. Other hardness testing techniques can be employed if approved by the purchaser. When access is available, tests shall be performed on the side of the weld in contact with the process fluid.
- e) One longitudinal weld, one circumferential weld, and each connection-to-component weld if the connection is DN 50 (NPS 2) or larger, shall be tested.

- f) If more than one welding procedure is used to fabricate longitudinal or circumferential welds, hardness readings shall be made on welds deposited by each procedure.
- g) Hardness test results and locations shall be recorded.

10.1.7 At welded joints in alloy-clad construction, the weld in the base metal and the area adjacent to the weld where the cladding has been stripped back shall be examined by magnetic-particle or liquid-penetrant inspection before weld overlay of the joint.

10.1.8 All finished carbon steel welds where postweld heat treatment is specified for process reasons (e.g. to prevent environmental cracking), and low-alloy steel welds in all services, shall be examined after postweld heat treatment (unless the pressure design code specifies examination after hydrostatic testing) by the magnetic-particle method. The wet-fluorescent method shall be used to inspect accessible process-wetted surfaces of welds and heat affected zones in sour or wet hydrogen sulfide service.

10.1.9 Final welds in all nonmagnetic materials, whether of solid alloy or alloy-clad plate and all bimetallic welds between nonmagnetic and magnetic materials shall be examined by the liquid-penetrant method after any required postweld heat treatment.

10.1.10 A final visual weld inspection shall be performed after any postweld heat treatment.

10.1.11 After cladding, but prior to fabrication, integrally-clad material shall be subjected to an ultrasonic examination from the clad side in accordance with the pressure design code.

10.1.12 Overlay weldments, back-cladding, and attachment welds to overlay weldments shall be liquid-penetrant examined after any postweld heat treatment.

10.1.13 When ferrite testing is applied, a minimum of three tests shall be made on each 1.5 m (5 ft) of weld, with at least three tests made on each circumferential seam, three tests on each longitudinal weld seam, and three tests on each nozzle weld. The acceptance criteria for the minimum and maximum ferrite content shall be agreed between the purchaser and vendor.

- **10.1.14** The purchaser shall specify whether all carbon steel plate in sour or wet hydrogen sulfide service shall be subjected to an ultrasonic lamination check (e.g. to EN 10160 grade S2E2 or ASME SA-578, acceptance level A supplementary requirement S1).

10.1.15 If tube-to-tubesheet joints are roller expanded (not welded), the tube wall thickness reduction shall be randomly checked and recorded. A minimum of 2 % of the joints per tube bundle, with a minimum of 5 joints, shall be checked (see 9.10.1).

10.1.16 When weld deposit overlay is applied on pressure retaining components, production weld chemistry tests shall be performed at the specified depth as follows.

- a) Weld overlay test samples (including weld overlay restoration) shall be taken at a depth of at least 1.5 mm (¹/₁₆ in.) from the finished surface, unless otherwise specified by the purchaser.
- b) When weld overlay is applied to the surface of exchanger shell or channel cylinders, at least two deposit analyses shall be made for every 10 m² (100 ft²) of overlay surface, or fraction thereof, for each welding procedure applied. The analysis shall be taken from each end of the cylinder at locations that are diametrically opposed.
- c) When weld overlay is applied to the surface of formed heads, channel covers, girth and floating-head flanges and tubesheets, at least one deposit analysis shall be made for each welding procedure used and for each component manufactured.
- d) For tubesheets that are weld overlaid on both sides, at least one deposit analysis is required on each side of the tubesheet.

- e) When weld overlay is applied on nozzle assemblies, at least one deposit analysis shall be made for each welding procedure used. Where multiple nozzle assemblies are manufactured with the same welding procedure, the deposit analysis shall be taken from the smallest diameter nozzle manufactured for each welding procedure.
- f) When weld overlay is applied to restore the clad area of weld joints between integrally clad components, at least one deposit analysis shall be taken per welding procedure for each main longitudinal or circumferential weld seam.
- g) When weld overlay is applied for the surface repair of clad components or for the restoration of surfaces (other than weld seams) where the cladding has been removed, at least one deposit analysis shall be taken for each welding procedure used.
- h) Production test locations shall be restored after testing.

10.1.17 For quality requirements of integrally finned tubes, see 8.3.

- **10.1.18** The purchase shall specify if U-bends shall be metallurgically examined and corrosion tested prior to the start of production bending. One tube of each heat shall be bent to the smallest radius, and heat treated (if applicable) prior to the test. Test procedures and acceptance criteria shall be agreed to by the purchaser and the manufacturer.

10.1.19 Integrally clad formed heads shall be ultrasonically tested after forming in accordance with the requirements of ASTM A263, ASTM A264 or ASTM A265 including supplementary requirement S12 with a bond quality level of Class 1.

10.1.20 Cut areas of integrally clad material shall be ultrasonically examined within 50 mm (2 in.) from the edge of the opening.

10.1.21 The test acceptance criteria for liquid penetrant inspection of weld overlay shall be in accordance with the requirements of API Recommended Practice 582. Any linear indication in the weld overlay as identified by the liquid penetrant test shall be repaired.

10.1.22 Weld overlaid surfaces shall be examined with the liquid penetrant method after heat treatment and/or final machining.

- **10.1.23** The purchaser shall specify if clad and weld overlay construction shall be tested for susceptibility to hydrogen disbonding. The testing requirements and acceptance criteria shall be agreed between the purchaser and vendor.
- **10.1.24** The purchaser shall specify if the material composition and grade for pressure part components including cladding, overlay, weld consumables, heat transfer tube(s), and completed welds shall be verified by positive material identification. The extent required shall be agreed between the purchaser and the vendor.

10.2 Pressure Testing

10.2.1 In the case of welded-and-expanded tube-to-tubesheet joints, the tube weld integrity shall be verified before final expansion of the tubes by a pneumatic test from the shell side at a gauge pressure of between 50 kPa (7.5 psi) and 100 kPa (15 psi), using a soap-water solution to identify leaks.

10.2.2 When pneumatic testing of the tube-to-tubesheet joints from the shell side is not practical (i.e. replacement tube bundles without shells, floating-head exchangers with integral shell heads or without test rings, etc.), the integrity of the tube-to-tubesheet joint can be tested either by a tube side hydrostatic test or by a vacuum test of the tube bundle.

- **10.2.3** The purchaser shall specify If tube-to-tubesheet joint integrity shall be tested after final expansion of the tubes by a helium leak test (e.g. per ASME *BPVC, Section V*). See A.6 for additional guidance.

10.2.4 Except for differential-pressure designs, an independent hydrostatic test of the shell side and the tube side shall be performed. The minimum fluid temperature for hydrostatic testing shall be as required by the pressure design code.

10.2.5 The water used for hydrostatic testing shall be potable and the test pressure shall be maintained for at least 1 h.

NOTE This does not preclude the use of anti-corrosion or other additives from being utilized.

10.2.6 The chloride content of the test water shall not exceed 50 mg/kg (50 parts per million by mass). Upon completion of the hydrostatic test, the equipment shall be promptly drained and cleared of residual test fluid.

- **10.2.7** Any additional requirements for equipment drying or preservation shall be specified by the purchaser.

10.2.8 The shell side hydrostatic test shall be conducted with the bonnet or channel cover removed.

10.2.9 Nozzle reinforcement pads and other welded pads provided with vent holes shall be pneumatically tested between 100 kPa (ga) (15 psig) and 170 kPa (ga) (25 psig) after completion of any post weld heat treatment and prior to the final pressure test.

10.2.10 For safety considerations, any supplementary pneumatic test shall be performed at a nominal pressure of 170 kPa (ga) (25 psig).

10.2.11 Flanged joints taken apart after the final pressure test shall be reassembled with new gaskets and pneumatically leak tested between 100 kPa (ga) and 170 kPa (ga) (15 psig and 25 psig). Self-energizing seals can be reused subject to approval of the purchaser.

10.2.12 Paint or other external coatings shall not be applied over welds before the final pressure test, unless otherwise agreed by the purchaser.

10.2.13 Heat exchangers that are to be stacked in service with directly interconnecting nozzles, including any interconnecting pipe spool pieces, shall be hydrostatically tested in the stacked position.

10.2.14 During the hydrotest, the primary membrane stress in any pressure containing component shall not exceed 95 % of the material minimum yield strength.

10.2.15 Hydrostatic pressure testing shall be performed with the same type of gaskets and bolting as those required in service for all body flanges, custom designed flanges and permanently blinded connections.

10.3 Nameplates and Stampings

10.3.1 An austenitic stainless steel nameplate shall be permanently attached to the heat exchanger in such a manner that it is visible after insulation has been installed.

10.3.2 When applicable a warning nameplate shall be provided to identify any restrictions such as differential pressure design and testing, or any limitations on the operation and testing of expansion joints.

10.3.3 The nameplate shall be located on the shell, near the channel end.

10.3.4 The following parts shall be stamped with the manufacturer's serial number:

- a) shell flange;
- b) shell cover flange;
- c) channel or bonnet flange;
- d) channel cover;
- e) stationary tubesheet;
- f) floating tubesheet;
- g) floating-head cover flange;
- h) floating-head backing device;
- i) test ring flange and gland.

11 Preparation for Shipment

11.1 Protection

11.1.1 All liquids used for cleaning or testing shall be drained from heat exchangers before shipment. See A.7.1 for additional information.

11.1.2 Heat exchangers shall be free of foreign matter prior to shipment.

11.1.3 All openings in heat exchangers shall be suitably protected to prevent damage and possible entry of water or other foreign material.

11.1.4 All flange-gasket surfaces shall be coated with an easily removable rust preventative and shall be protected by suitably attached durable covers of such material as wood, plastic, or gasketed steel.

11.1.5 All threaded connections shall be protected by metal plugs or caps of compatible material.

11.1.6 Connections that are beveled for welding shall be suitably covered to protect the bevel from damage.

- **11.1.7** The purchaser shall specify if there are requirements for surface preparation and protection (e.g. painting).

11.1.8 Exposed threads of bolting shall be protected with an easily removable rust preventative to prevent corrosion during testing, shipping, and storage. Tapped holes shall be plugged with grease.

11.1.9 Tie-rods or tie-bars installed on shell expansion joints for protection during shipping shall be painted in a contrasting color and clearly tagged to specify their removal before commissioning.

- **11.1.10** The purchaser shall specify if inert gas (e.g. nitrogen, argon) purge and fill is required.

11.1.11 When an inert gas fill is specified, the design shall comply with the following:

- a) Have the following warning displayed on a tag on all openings:

"DANGER - NON-LIFE SUPPORTING ATMOSPHERE Contents are under <Inert gas> pressure and must be depressurized before opening."

- b) Positive pressure to be indicated by a pressure gage.

- c) Gages to be suitably protected from damage during transportation.

- d) Nozzle shipping covers to be made of metal. For alloy or alloy lined nozzles, the cover to be painted carbon steel or of stainless steel.

- e) Nozzle shipping cover thickness to be the larger of 6 mm (1/4 in.) or thickness required for pressure containment.

- f) Bolting used to be the same materials, diameter, and count as the final service bolting.

- g) The exchanger to be filled to a pressure of 69 kPa (ga) (10 psig).

- h) Detail the method provided to add or remove the inert gas during shipment and storage to maintain required pressure.

11.2 Identification

11.2.1 The item number, shipping mass, and purchaser's order number shall be clearly marked on the heat exchanger. The shipping mass is the mass of the exchanger and any other items attached by the heat exchanger manufacturer for shipment. This does not include the mass of any items added by the shipping vendor.

11.2.2 All boxes, crates, or packages shall be identified with the purchaser's order number and the item number.

11.2.3 The words "DO NOT WELD" shall be stenciled (in at least two places 180° apart) on the parts of equipment that have been postweld heat-treated.

12 Supplemental Requirements

12.1 General

- The purchaser shall specify if the additional requirements in Section 12 shall be applied to one or both sides of the heat exchanger. These supplemental requirements should be considered by the purchaser if the cylinder thickness of a heat exchanger component exceeds 50 mm (2 in.) or if the service is considered critical.

12.2 Design

12.2.1 The attachment of welded nozzles and other connections to components shall have integral reinforcement. The nozzles or other connections shall be attached using full-penetration groove weld with additional fillet or butt welds. They can be set-on, set-in, or integrally reinforced forging type inserts. Set-on type connections shall not be welded to plate that contains laminations or other defects and shall only be used if the component is pipe or forged or if the component plates are ultrasonically examined in the area of attachment. In this case the examination for laminations and other defects shall be carried out for a radial distance of at least twice the thickness of the component.

12.2.2 Tubesheet attachment welds to shell or channel cylinders shall be butt welds. For guidance on the use of integral hubs on tubesheets see A.4.4.

12.3 Examination

12.3.1 All material for formed heads or cylinders exceeding 50 mm (2 in.) in thickness shall be ultrasonically examined. Non-destructive examination procedures and acceptance criteria shall comply with the pressure design code.

12.3.2 All forgings, except standard flanges designed as described in 7.6, shall be ultrasonically examined in accordance with the pressure design code. Non-destructive examination procedures and acceptance criteria shall comply with the pressure design code and shall be agreed by the purchaser.

12.3.3 For ultrasonic examination of welds, plates and forgings, the purchaser shall be supplied with a report providing diagrams of the surfaces scanned and indications obtained, the areas repaired, the nature of defects repaired, and the repair procedures used. The following information shall also be provided:

- a) pulse-echo instrument manufacturer's name and model, and the damping control setting;
- b) search unit manufacturer, model, dimensions, and the substance (such as oil or water) that is used to couple the transducer with the material being inspected;
- c) frequency used and the test angle to the component's surface;
- d) the wedge medium for angle-beam examination.

12.3.4 Magnetic-particle examination shall be performed on all pressure-containing plate edges and openings before welding. Any defects found shall be removed and any necessary repairs performed. Liquid penetrant examination shall be substituted for nonmagnetic materials.

12.3.5 Magnetic-particle examination shall be performed on all pressure-retaining welds and weld repairs, after any postweld heat treatment. If accessible, the back side of the root pass shall be examined after being prepared for final welding. Both sides of accessible completed welds shall be examined. Liquid penetrant examination shall be substituted for nonmagnetic materials.

12.3.6 Magnetic-particle examination shall be performed on all pressure-boundary attachment welds. Liquid penetrant examination shall be substituted for nonmagnetic materials.

12.3.7 After the hydrostatic test, a magnetic-particle examination shall be performed on all external pressure-retaining welds and all internal nozzle welds that are accessible without disassembling the heat exchanger. Liquid penetrant examination shall be substituted for nonmagnetic materials.

12.3.8 For components subject to full radiography that cannot be radiographed shall be examined by a full volumetric examination method. The method and procedures used shall be agreed by the purchaser.

12.3.9 Full radiographic examination shall be performed on all pressure-retaining butt welds.

12.3.10 Ultrasonic examination shall be performed on all pressure-retaining butt welds for components over 50 mm (2 in.) thick, after any postweld heat treatment. Ultrasonic examination shall comply with the pressure design code. The entire volume of deposited weld metal shall be examined from two directions. Before the welds are examined, the adjacent base material shall be examined by means of a longitudinal beam with a 100 % scan for a distance of twice the plate thickness back from the weld. A diagram shall be prepared indicating all areas larger than 12 mm (1/2 in.) in diameter that show a loss of back-reflection of 50 % or more. The acceptance criteria shall be agreed by the purchaser and the vendor.

13 Supplemental Requirements for Services Subject to High Temperature Hydrogen Service (HTHS)

13.1 General

13.1.1 The requirements of this section apply to high temperature hydrogen service. See A.8.1 for additional guidance.

13.1.2 The requirements in previous sections of this document that address hydrogen service shall be applied.

13.2 Design Temperature

13.2.1 The design temperature specified may not be the same as the temperature used for the material selection for high temperature hydrogen service. See A.8.2.

13.2.2 Design temperatures shall not vary across a single heat exchanger shell, or from inlet channel to outlet channel of a single heat exchanger, unless otherwise specified by the purchaser. When variable design temperatures are specified, temperature monitoring points shall be provided at the design temperature breaks. See A.8.2.

13.2.3 For a heat exchanger unit arranged in series, different design temperatures for each shell shall not be used unless otherwise specified by the purchaser. See A.8.2.

13.3 Materials

- **13.3.1** Materials, including pressure-retaining internal components such as tubes and tubesheets, floating-heads, and internal bellows, shall be specified by the purchaser in accordance with API Recommended Practice 941, including any design margins to be applied based on integrity operating window (IOW) limits. See A.8.3 and A.8.4.

13.3.2 When the shell side is in HTHS, the nominal composition of construction of the wetted surfaces shall not vary across a single heat exchanger shell, unless otherwise specified by the purchaser. When the tube side is in HTHS, channels shall be of the same nominal composition unless otherwise specified by the purchaser. When different materials are used, temperature monitoring points shall be provided at the material breaks. See A.8.3.

13.3.3 For heat exchangers arranged in series, nominal material composition shall not vary between the different shells, unless specified by the purchaser.

13.4 Connections for Temperature Indication

13.4.1 For a heat exchanger unit arranged in series, where there are varying design temperatures or metallurgy, flanged connections for temperature indication shall be provided in the interconnecting nozzles or interconnecting piping. These connections shall be provided for the side(s) in HTHS. See A.8.5.

- **13.4.2** The purchaser shall specify if skin (metal) temperature-measuring devices, beyond what is required by 13.2.2, shall be provided. The quantity, location, and type of device shall be agreed with the purchaser.

13.5 Fabrication

13.5.1 Weld details used on components in HTHS, including tubesheet to the cylinder welds and nozzle assembly butt welds, shall be full-penetration, full-fusion butt welds that can be 100 % volumetrically examined for the entire length.

13.5.2 Nozzle connections to cylinders or heads for components in HTHS shall be full-penetration, full-fusion welds.

13.5.3 All components constructed of carbon steel shall be post-weld heat treated after completion of fabrication.

13.5.4 Dissimilar metal welds shall not be allowed in a single heat exchanger shell in HTHS, except as allowed in 13.3.2. When dissimilar materials are used, the welding and PWHT procedure shall be agreed between the vendor and purchaser, taking account of the varying microstructure of the weld.

13.6 Inspection and Testing

13.6.1 Positive material identification of non-carbon steel pressure part components including cladding, overlay, weld consumables, heat transfer tube(s), and completed welds shall be performed. The extent and method shall be agreed between the purchaser and vendor.

13.6.2 All butt welds shall be 100 % volumetrically examined after any required post-weld heat treatment.

13.6.3 Wet fluorescent magnetic particle examination shall be performed on accessible wetted surfaces, for all carbon or low-alloy steel pressure-retaining welds, after any required post-weld heat treatment. When the component is clad, the base material weld shall be wet fluorescent magnetic particle examined prior to restoration of the cladding. When the minimum required preheat temperature is above the limits of wet fluorescent magnetic particle solution, a dry magnetic particle method may be used prior to the restoration of the cladding.

13.6.4 Production hardness testing for all pressure-retaining welds and heat-affected zones exposed to HTHS shall be performed in accordance with 10.1.6.

- **13.6.5** The purchaser shall specify if welds subject to HTHS shall be provided with baseline ultrasonic examination reports. The extent, locations, and type of ultrasonic examination shall be as agreed with the purchaser. See API Recommended Practice 941, for ultrasonic examination inspection methods.

Annex A

(informative)

Recommended Practices

A.1 Introduction

This annex has been prepared to give advice to the designer. The advice is offered for guidance only.

The descriptions and the numbers following are those of sections in the main body of this standard.

A.2 General

A.2.1 Cyclic Design—Guidance to 4.9

A.2.1.1 Pressure design codes will generally require that a fatigue evaluation be performed for any service which has been designated to be cyclic. These codes normally provide requirements related to the magnitude of pressure and thermal fluctuations which need to be considered as part of the evaluation.

NOTE For additional guidance on thermal fatigue see API Recommended Practice 571.

A.2.1.2 For assistance in specifying cyclic conditions, it is suggested that the purchaser follow the guidance of ASME *BPVC, Section VIII, Division 2*, and complete a User Design Specification. Methodologies are also available in other pressure design codes, including EN 13445 (Part 3).

A.2.1.3 Pressure design codes including ASME *BPVC, Section VIII, Division 2* and EN 13445 (Part 3) provide screening methods to determine if a fatigue analysis is required for the given cyclic loading. If required, rules for performing a full fatigue analysis are included. Other methods of screening and evaluation of fatigue analysis may be used based on agreement between the purchaser and the vendor.

A.2.2 Sour or Wet Hydrogen Sulfide Service—Guidance to 4.10 and 7.6.13

A.2.2.1 NACE MR0103 establishes material requirements for resistance to sulfide stress cracking (SSC) in sour petroleum refining and related process environments which contain hydrogen sulfide either as a gas or dissolved in an aqueous (liquid-water) phase, with or without the presence of hydrocarbon. SSC is defined as cracking of a metal under the combined action of tensile stress and corrosion in the presence of water and hydrogen sulfide. Other forms of wet hydrogen sulfide cracking, environmental cracking, and severely corrosive or hydrogen charging conditions that can lead to failures by mechanisms other than SSC are outside the scope of NACE MR0103.

A.2.2.2 Compared to the high pH environments of refinery sour service, oil and gas production sour environments are covered by NACE MR0175 (all parts). This is because many wet sour streams in oil and gas production facilities also contain carbon dioxide and hence exhibit a lower pH. In addition, chloride ion concentrations tend to be significantly lower in refinery sour services than in oil and gas production sour services.

A.2.2.3 NACE MR0175 (all parts) provides requirements and recommendations for the selection and qualification of metallic materials for service in equipment used in oil and gas production and natural gas processing plants in hydrogen sulfide containing environments. Mechanisms of cracking that can be caused by hydrogen sulfide include sulfide stress cracking, stress corrosion cracking (SCC), hydrogen-induced cracking (HIC), step-wise cracking (SWC), stress-oriented hydrogen-induced cracking (SOHIC), soft zone cracking (SZC), and galvanically induced hydrogen stress cracking (GHSC).

A.2.2.4 AMPP SP21534, *Materials and Fabrication Requirements for Carbon Steel Pressure Vessels in Petroleum Refining Wet H₂S Environments*, establishes best-practice guidelines for the selection of materials, welding procedures, non-destructive testing, hardness control, and heat treatment of new carbon-steel pressure vessels and heat exchangers used in petroleum refineries exposed to wet hydrogen sulfide (H₂S). The standard addresses varying severity levels ("low", "moderate", "severe") of wet H₂S service, specifying certification requirements, weld preheat/post-weld heat treatment, welded joint quality criteria, and damage-resistance measures such as resistance

to hydrogen-induced cracking. It targets refiners, fabricators, and engineering/construction contractors seeking to mitigate corrosion and integrity risks in sour environments

A.2.2.5 Factors affecting the susceptibility of metallic materials to cracking in oil and gas production facilities in hydrogen sulfide containing environments include the hydrogen sulfide partial pressure, in-situ pH, the concentration of dissolved chlorides or other halides, the presence of elemental sulfur or other oxidants, temperature, galvanic effects, mechanical stress and the duration of contact with a liquid water phase.

A.2.2.6 There are many process service environments which can expose carbon steel materials and weldments to potentially damaging corrosion cracking mechanisms. For some of these environments (including HIC and SOHIC in hydrogen sulfide and amine services), wet fluorescent magnetic-particle testing (WFMT) was historically the only inspection technique that was considered to be highly effective in detecting crack like flaws that are surface breaking, such flaws are typical to HIC and SOHIC.

A.2.2.7 Advanced ultrasonics such as phased array ultrasonic testing (PAUT) have also been found to be effective for detection of damage considered volumetric including flaws that are laminar or planar oriented, ID or OD connected. UT examination for HIC / SOHIC is normally applied from the OD surface. External surface obstructions can prohibit the use of UT examination of some areas. Examples are: reinforcement pads and saddle supports and areas where laminar oriented flaws such as lamination from the manufacturing of steel and blisters resulting from hydrogen accumulation block the UT sound energy propagation therefore obscuring interrogation of the ID surface. In cases such as the aforementioned WFMT on the internal surface should be considered. In cases where process design optimization limits sizing for internal access or internal access is otherwise restricted alternate examination techniques such as robotics with deployable NDT methods can be considered.

A.2.2.8 Whilst PAUT can be very effective for volumetric inspection, it may not be possible to pick up surface crack initiation as is possible with WFMT. Vice-versa, WFMT detects surface breaking flaws but not volumetric flaws bound between the ID and OD surfaces. Depths of crack like flaws detected with WFMT, can be validated by metal removal and physical measurement of the excavation will not be able to quantify the crack depth. The owner/end user should be consulted as to what techniques are required for post construction inspections. Some users may elect to adhere to exchanger designs which are capable of using both PAUT and WFMT inspection methods to complement each other.

A.2.2.9 In order to be able to apply the WFMT inspection technique (if required), the inside wetted surface of welds must be accessible to the technician after the heat exchanger has been fabricated. When the shell side fluid service is prone to produce cracking in carbon steel, and the purchaser chooses to utilize the WFMT inspection, the following design guidelines can be applied to ensure that the wetted weld surfaces will be accessible for WFMT inspection, both after the initial fabrication and post-weld heat treatment, as well as for future in-service inspections:

- Fixed tubesheet heat exchangers should not be used.
- The minimum shell diameter should be DN 200 (8 NPS).
- For shells with nominal diameters between DN 200 (8 NPS) and DN 600 (24 NPS), the following should apply:
 - rear shell covers should be the removable type;
 - all nozzles should be located at the ends of the shell cylinder;
 - the shell cylinder should be a single length of seamless pipe; rolled plate should only be used if there is sufficient access to the internal surfaces of longitudinal and circumferential welds to allow them to be ground flush.
- Shell side distributor belts should not be used.
- The minimum nozzle size should be DN 50 (2 NPS).
- Nozzles DN 150 (6 NPS) and smaller should be long weld neck type.

A.2.3 Expansion Joints—Guidance to 4.12, 7.8, and 9.6.13

A.2.3.1 Cases other than normal steady state operation may influence the need for, or the design of, an expansion joint in fixed tubesheet or single tube pass floating head exchangers. Such cases could include alternate steady state operation, startup, shutdown, upsets, or maintenance activities.

Consideration should be given to the below questions to determine whether there are conditions which could result in more severe mean metal temperature differentials between the shell and tube than normal operation:

- 1) Can hot fluid be suddenly introduced into a cold exchanger?
- 2) Can cold fluid be suddenly introduced into a hot exchanger?
- 3) Can there be an emergency shutdown, for example due to power or pump failure, so that one fluid is suddenly shut off while the other fluid flows (consider each case)?
- 4) Can a rapid shell side steamout occur?
- 5) Will a tube side steamout, if required, be done without simultaneous shell side steamout?
- 6) Can there be a change in fluid temperature or flow rate due to a system malfunction or change:
 - On hot side, with cold side normal?
 - On cold side with hot side normal?
 - On both sides?
- 7) Is one of the process streams cooling water and it can have a substantially colder inlet in winter?

If the answer to a question above is “yes”, the case details (flow rates, temperatures and fluid properties) should be documented on the expansion joint datasheet in Annex C. Otherwise, operation is assumed to be in accordance with the recommendations of the TEMA Standards, paragraph E3.2, “Operating Procedures”.

A.2.3.2 The following information should be supplied to specify the selection of either a thin-walled or thick-walled expansion joint:

- a) provide all of the conditions that the expansion joint should be evaluated for: operating, start-up, shut-down, upset conditions, loss of flow, steamout, etc.;
- b) design temperatures, pressures and metal temperatures should be supplied for each of these cases.;
- c) if a thin-walled bellows is used, the materials of construction should be specified.

A.2.3.3 Advantages and disadvantages for the use of thin-walled and thick-walled expansion joints are included within Table A.12.

Table A.12 —Comparison Between Thick and Thin-walled Expansion Joints

<u>Thin-walled Expansion Joints</u>	<u>Thick-walled Expansion Joints</u>
<u>Advantages</u>	<u>Advantages</u>
Able to accommodate large differential movements	Robust design with respect to handling
Can incorporate multiple plies with intermediate leak detecting ports to offer early warning of pending pressure containment loss	Metal loss can be monitored and repaired in the same manner as other major pressure boundaries
<u>Disadvantages</u>	<u>Disadvantages</u>
Does not provide for a rigid design	Limited capacity to accommodate large differential movements

Requires an inner liner and an outer cover to protect the bellows	Construction can be labor intensive
Requires expansion stays to limit the travel of the bellows	Means must be provided to mitigate bypass around transverse baffles
Heat treatment can affect material properties (bellows should not be re-heat treated without the approval of the expansion joint supplier).	Iterative design process to determine the available geometry for the flanged only or flanged-and-flued heads.

Note: If there are concerns with the use of an expansion joint or if an expansion joint is not practical, consideration should be given to an alternate type of heat exchanger that is more able to accommodate thermal differential movement between the shell and the tube bundle. It is possible that a "U" tube or floating head exchanger might provide a more economical and/or maintenance friendly design.

A.2.3.4 Limitations of Thick-Walled Expansion Joints

Thick-walled expansion joints provide a more robust design than thin-walled bellows expansion joints; however, this inherent relative rigidity does hinder the thick-walled expansion joint's ability to accommodate the differential movement between the shell and the tubes. The following should be considered:

- a) For a given thickness and material, thick-walled designs are constrained to a maximum diameter that meets the pressure and temperature when fully corroded.
- b) For high pressure designs or for high differential temperatures, a single expansion joint element may not have sufficient flexibility to accommodate the differential movement. In such cases multiple expansion joints may be required. This can become problematic when trying to fit the expansion joint in the shell to avoid fluid bypass around the transverse baffles in the heat exchanger.
- c) The availability of the flexible shell elements is often limited by the capacity of the head manufacturer. Not all head manufacturers can supply the flanged only or flanged-and-flued heads for these thick-walled expansion joints.
- d) The design of thick-walled expansion joints may require an iterative process where the geometry and material requirements for the heads necessary to construct the expansion joint are supplied to a head manufacturer. The head manufacturer will then supply a quote for the geometry that they are able to produce. The manufacturers proposed expansion joint should then be re-evaluated based on this new design.

A.2.3.5 Single tube pass tail-pipe spool designs—Additional Guidance to 4.12 and 7.7.3

A.2.3.5.1 The tail-pipe spool design can incorporate an internally flanged removable bellows if desired, however the risk of leaks from additional internal flanges should be considered.

A.2.3.5.2 Thread relief can be considered for all stud bolt holes in order to minimize galling.

A.2.3.5.3 For designs which may be prone to bundle twist and have potential for bolt hole mis-alignment between the tail-pipe spool flange and shell cover (such as Texas Towers built in the horizontal position and then twist when lifted vertically), the designer should incorporate proven methods to accommodate or prevent any bolt hole misalignment that may occur during installation.

A.2.4 Crossflow Shell Side Design—Guidance to 4.144

For shell side crossflow (TEMA X-shells) exchangers, to ensure distribution of the fluid across the entire length of the tube bundle (with the exception for cases with boiling or condensing of single (pure) component fluids) consider the following:

- a) The number, location, and size of the shell side nozzles should be determined based on effective flow distribution across the tube length.
- b) The entrance and exit window area should be greater than one-half of the respective nozzle flow area.

- c) Perforated plates may be considered to provide fluid distribution over the entire tube bundle length, when approved by the purchaser. Perforated plate holes should be sized such that the rho-V-squared does not exceed TEMA's requirements for shell entrance and exit areas and shall be a minimum of 20 % of the plate area. Impingement rods can be used below the plate where TEMA rho-V-squared criteria cannot be met.
- d) For U-tube bundles a full diameter support should be located beyond the last nozzle.

A.2.5 Flow-Induced and Acoustic Vibration – Guidance to 4.16 and 6.1.2

A.2.5.1 Background

The following guidance is to assist the designer in minimizing the risk of damage resulting from flow-induced vibration in a shell-and-tube heat exchanger.

Tubes and the shell side fluid are susceptible to flow-induced vibrations from the unsteady pressures acting on the surfaces of the tubes.

Tube vibration mechanisms in order of their increasing severity are turbulent buffeting, vortex shedding and fluid-elastic instability. Excessive tube vibrations caused by turbulent buffeting or vortex shedding resonances can result in thinning and eventual failure of tubes at the baffles or leaks at the tube-to-tubesheet joints. Fluid elastic instability tends to have a run-away effect; once cross-flow velocities approach a critical velocity the tubes may vibrate uncontrollably and can fail very rapidly.

The shell side may be subjected to acoustic resonances termed "acoustic vibrations." Acoustic vibrations can vibrate the shell and the exchanger's foundations, but are more often associated with high levels of radiated sound. Where the acoustic frequency approaches the tube natural frequency, tube damage may occur. The prediction and control of the vibration response of a shell-and-tube heat exchanger requires the characterization of the system attributes and the excitation.

A.2.5.2 Characterization of the System Attributes

A.2.5.2.1 Tube Characteristics

The natural frequencies of a tube are functions of:

- the modulus of elasticity of the tube material;
- the moment of inertia of the tubular section;
- the number and length of spans;
- the effective tube mass, which includes the mass of the tube, the tube side fluid and a portion of the shell side fluid.

The lowest natural frequency is termed the "fundamental frequency." In general, higher order natural frequencies are not integer multiples of the fundamental frequency.

In a heat exchanger, vibrational energy dissipation or damping is desirable, but not affected much by conventional design choices. Dampening depends on the fluid density, viscosity, and surface tension, the vapor volume fraction, the tube diameter, number and thickness of tube supports, the tube mass, and the fundamental natural frequency of the tube.

A.2.5.2.2 Shell Side Fluid Volume Characteristics

The natural frequencies of the shell side gas volume are dictated by the speed of sound and the size of the exchanger. The most offending mode is transverse to the longitudinal axes of the tubes and perpendicular to the cross-flow. The wavelength of this mode is twice the shell diameter. As the diameter increases, the resonant frequency of the transverse mode decreases, and opposite sides of the shell approach parallel planes. Both of these

effects increase the likelihood of a standing wave. Higher order diametral modes have natural frequencies that are integer multiples of this fundamental frequency. The geometry of the shell also leads to a large number of higher order cylindrical and longitudinal modes that can be problematic in rare circumstances. The speed of sound in a gas filled exchanger, assuming ideal behavior, is a function of the pressure, the ratio of specific heats, the density, and the geometry of the tube bundle. Sound travels much faster in liquids compared to gases so resonant frequencies are much higher and generally beyond the range of excitation. In two-phase flows, the speed of sound is somewhat ambiguous, but clearly diminished relative to both the all-liquid and all-vapor values; fortunately, the abundance of vapor-liquid interfaces provides acoustic damping.

A.2.5.2.3 Characterization of Excitation

When fluid flows over the tube surfaces, pressures oscillating at the boundary result in unsteady forces that lead to turbulent buffeting. The frequency, content, and intensity of these pressure oscillations is a function of the velocity magnitude and direction and the layout of the tubes. The intensity of the pressure oscillations on the tubes is substantially greater for cross-flow compared to parallel flow.

In cross-flow, the principal excitation frequency is a function of the cross-flow velocity, tube diameter, and the dimensionless Strouhal number. Measurements of Strouhal numbers are available for standard layout angles and pitch ratios. In a nonlinear fluid-structure interaction phenomenon termed “lock-on,” the principal frequency of the unsteady pressures on the tube surface will actually tune itself to the fundamental natural frequency of the tube. This high amplitude, resonant response, is termed “vortex shedding” and is primarily a concern in isolated cylinders or possibly tubes on the bundle perimeter. Potentially, the most destructive tube vibration mechanism and thus the principal design concern is fluid-elastic instability. As the cross-flow velocity is increased, the excitation energy may exceed dissipation so that excessive vibration amplitudes result in tube failures (see A.4.1).

Oscillating pressures in the wakes of the tubes can also excite an acoustic resonance of the shell side gas volume. The lock-on phenomenon is possible if the excitation and resonant frequencies are close in value. In this case, the principal frequency of the oscillating pressures can shift to a resonant acoustic frequency as well.

A.2.5.2.4 Vibration Evaluation

Evaluation of the tube and acoustic vibration potential of an exchanger requires the determination of several important parameters: tube natural frequencies, acoustic natural frequencies, turbulent buffeting frequency, vortex shedding frequency, and damping. Excitation frequencies within a range (typically $\pm 20\%$) of a natural frequency have “lock-on” potential, and may lead to resonance. Additionally, the critical velocity, which is the threshold for the occurrence of fluid-elastic instability, and the highest cross-flow velocities within the bundle are important parameters to understand. Tubes most vulnerable to excessive tube vibration include those at the bundle entrance and exit, tubes in the window adjacent to the baffle cut where tubes are typically supported only by every other baffle, and at the bundle perimeter.

Amplitudes exceeding 10 % of the free distance between tubes is commonly used a threshold at which tube vibration can become problematic.

The physical mechanisms of flow-induced vibration within heat exchangers are actually well understood, but the uncertainties of many of the input flow and structural parameters require allowances in the design.

A.2.5.3 Flow Induced Vibration Mitigation

A.2.5.3.1 General

At the design stage, excessive tube vibrations can be dampened by shortening the span lengths that increase the natural frequencies of the tubes. Intermediate supports in vulnerable areas like the inlet or the U-bend region can be very effective and have minimal thermal hydraulic effect. The selection of the baffle type, e.g. NTIW (no tube in the window) or rod supports, and adequate baffle spacing, can keep tube vibrations within acceptable limits.

For operating units, tube vibration problems can be addressed by stiffening the tubes with stakes, clips, or inserts of various types. Removing vulnerable tubes, reducing shell side flow rate, and replacing the tube bundle are additional mitigation techniques.

Tighter tube-to-baffle hole clearances can potentially minimize the potential for vibration/fretting resulting in tube damage. Tube-to-baffle clearances not exceeding 0.4 mm (1/64 in.) have been commonly used.

A.2.5.3.2 Kettle Reboilers

Several cases of tube vibration/fretting have been observed in kettle reboilers / steam generators. In some cases, there is uncertainty in froth height and fluid velocity in the upper tube rows of kettle reboilers. In addition to provisions listed above, considerations should be given to some or all of the following provisions for kettle reboilers:

- a) Additional tube supports may be considered to reduce the unsupported length in order to increase the tube natural frequency. To minimize the potential for vibration in exchanger types with inherently long unsupported lengths the guidance of TEMA is applied. The distance between the full support baffles should not exceed 80% of the values stated TEMA Table RCB-4.52.
- b) Stiffening the top row region of through the use of intermediate supports, covering at least four tube rows at the top of the bundle, placed at the mid-point between the full support baffles, and trimmed above the top row of tubes so as to not impeded vapor escape (especially in the baffle space adjacent to the tubesheet).
- c) The height of weir plates, when provided, shall be at least 50 mm (2 in) above the top tube row. For fluids with high frothing tendency (e.g. amine), the weir plate should be at least 150 mm (6 in) above the top row of tubes.

A.2.5.4 Acoustic Vibration Mitigation

Detuning baffles can successfully mitigate acoustic vibration problems by detuning the resonant frequencies of the gas volume with respect to the excitation frequencies. The baffles are positioned coplanar with the cross flow and the axes of the tubes. These baffles partition the shell volume into several smaller volumes of higher natural frequencies, provided the baffles have sufficient acoustic impedance (baffle thickness and material density). Since the most common acoustic waves stand across the diameter of the shell, full extension of the detuning baffles wall-to-wall is unnecessary and can lead to flow maldistribution. For higher order cylindrical and longitudinal mode resonances, detuning baffles are also effective because the shell loses its cylindrical symmetry. Refer to TEMA or open literature for information concerning threshold indicators for problematic acoustic vibration. Because resonance at the fundamental frequency of a diametral transverse wave and higher harmonics is possible, the use of multiple asymmetrically spaced detuning baffles are common when the acoustic vibration potential is high. It should be noted that misplaced detuning baffles may tune the system and thus exacerbate acoustic vibration. Proprietary heat exchanger design software and literature sources offer guidance on the design of detuning baffles.

A.2.6 Tube Failure in High Pressure Units—Guidance to 4.17

The effects of potential overpressure on the low pressure side of the heat exchanger caused by tube rupture should be considered.

In certain extreme circumstances, the large volume of vapor that will be released during a tube failure scenario may lead to excessive size of the relief valve or rupture disk. An option in such a case is to consider using smaller diameter tubes (e.g. 16 mm [⁵/₈ in.]).

NOTE For further information, see API 521.

A.3 Outline Drawings and Other Supporting Data

A.3.1 Corrosion Allowance—Guidance to 6.1.1.j) and 7

Corrosion allowance should be in accordance with TEMA, unless otherwise specified by the purchaser. High alloy materials are often specified with, or assumed to have, a zero-corrosion allowance. For these materials, when considering selection of component thickness, adding 0.4 mm (1/64 in.) above the pressure design code minimum required thickness might be beneficial to assist fabrication and/or handling.

A.3.2 Design Calculations—Guidance to 6.1.2

In calculating the effective heat transfer area, the purchaser and vendor should agree as to whether the “U” bend region should be included.

A.4 Design

A.4.1 Metal Temperatures—Guidance to 7.1

Some pressure design codes allow for multiple design temperatures [pressures] at coincident design pressures [temperatures]. The purchaser is responsible for providing any alternate design conditions.

Some pressure design codes (such as ASME VIII, Division I, Nonmandatory Appendix UCS-A, UCS-A-3) may allow for a reduction in metal temperatures below the MDMT without the need for additional impact testing to be considered when these occur only under conditions of reduced stress; e.g. due to operating pressure reduction during auto-refrigeration. When such conditions occur and may be considered in the selection of materials and need for impact testing, the purchaser must provide the coincident reduced pressure and temperature conditions to allow for the manufacturer to determine if an adjusted governing MDMT for the thickness under consideration may allow exemption of impact testing in accordance with the pressure design code. Note that the application of these reductions may prevent the exchanger from immediate re-pressurization without allowing the metal temperatures to warm above the MDMT and should only be applied when allowed by the purchaser.

A.4.2 Cladding Thickness—Guidance to 7.2.3

For the tube side of a tubesheet a 5 mm (3/16 in.) cladding thickness does not allow for cladding thickness underneath the pass partition grooves. When 10 mm (3/8 in.) is specified, 5 mm (3/16 in.) thickness will remain after machining the pass partition grooves and provide adequate corrosion protection of the tubesheet.

A.4.3 Low-fin Tubing—Guidance to 7.5.1.3

A.4.3.1 For integrally-finned tubes made from corrosion-resistant alloys (i.e. alloys in Table 1 other than carbon and low chrome steels), the plain-end thickness should comply with Table 1. The wall thickness under the root of the fins may be reduced, when approved by the purchaser, provided the minimum thickness under the root of the fins meets the pressure design code at the design conditions, including any purchaser-specified corrosion allowance for the tubes.

A.4.3.2 When finned tubes are selected, the influence of shell side corrosion and fouling on heat transfer performance over time should be considered for the metallurgy selection and the fin profile. Crevice corrosion may occur on the shell side of integrally-finned tubes in some services and fin density may need to be adjusted to account for this.

A.4.4 Integral Hubs on Tubesheets—Guidance to 7.5.2.7 and 12.2.2

The use of integral hubs on tubesheets may be considered for conditions including the following.

- a) High pressure designs with cylinder thicknesses greater than or equal to 50 mm (2 in.) as specified in 12.1.
- b) When cyclic service is specified.
- c) Service based requirements
- d) When the radial differential thermal expansion between the tubesheet and integral shell or channel is large.
- e) When full volumetric examination of the tubesheet to cylinder welds are required the hub length should be chosen to accommodate the examination.

A.4.5 Baffles, Support Plates and Impingement Protection

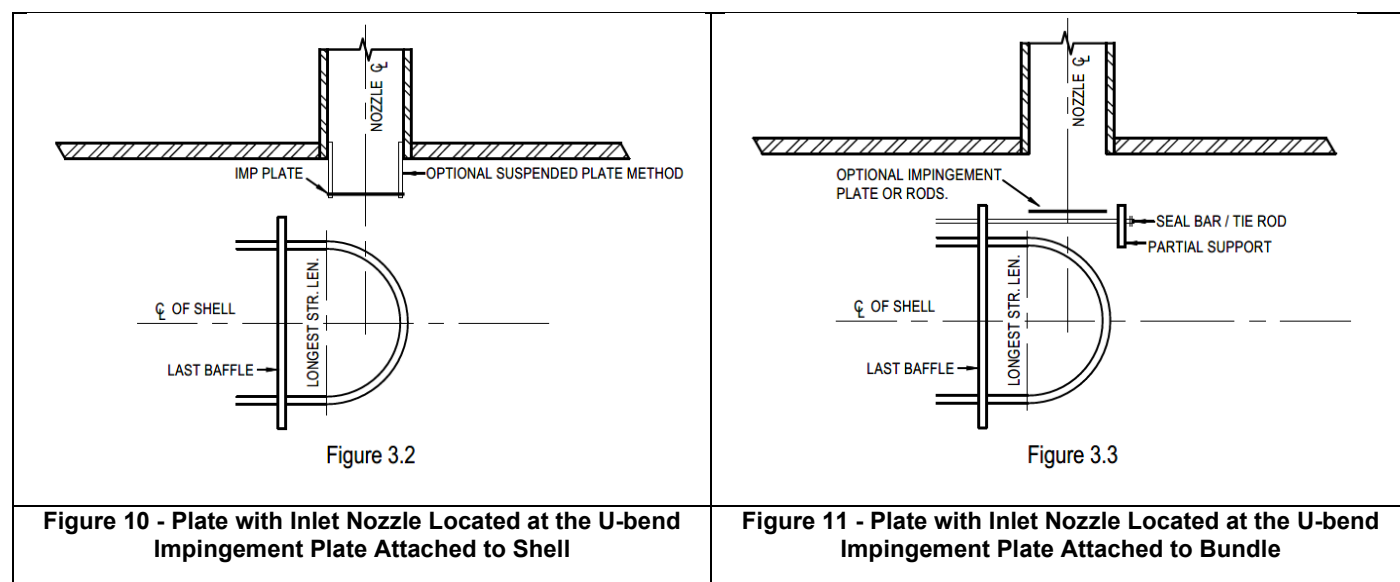
A.4.5.1 Partial Support Plates – Guidance to 7.5.3.3.

When additional support is required at the inlet or outlet baffle spaces, in some cases it may be advantageous for the segmental baffle closest to the nozzle to incorporate “dog-ears” in the window area, or to include window cuts to help direct bypass streams in lieu of or in combination with seal strips.

A.4.5.2 Impingement Protection – Guidance to 7.5.4.7 and 7.5.4.8

When the inlet nozzle is located at the U-bends with insufficient space alternate methods of installing the deflector plate should be agreed between the purchaser and the vendor. Special consideration for impingement protection will be needed. Considerations may include longer tie-rods and/or stiffener bar(s) and, if possible, a partial support plate to support an impingement plate or rods. The u-bend location and the tube layout will dictate the final impingement design. See Figure 10 and Figure 11. Coordination with the end-user may be required.

NOTE Special considerations may include longer tie-rods and/or stiffener bar(s) and if possible, a partial support plate to support an impingement plate or rods. The U-bend location and the tube layout will dictate what is possible for impingement protection. Coordination with the end-user may be required.



A.4.6 Tube Bundle Skid Bars—Guidance to 7.5.6.2.h)

A.4.6.1 For tube bundles with mass and dimensions outside the range of conventional bundle-pulling devices, alternative means of bundle removal should be considered. For example, if the bundle mass exceeds 18,150 kg (40,000 lb), the diameter exceeds 1220 mm (48 in.), or the length exceeds 7.3 m (24 ft), the following options can be considered:

- a) bundle rollers;
- b) skid bars on a rail;
- c) removable shell.

A.4.6.2 Skid bars should not obstruct tube lanes or pass-partition lanes if 45 ° or 90 ° tube layouts are used.

A.4.7 Tube-to-Tubesheet Joints—Guidance to 7.5.7 and 9.10

A.4.7.1 Expanded Tube-to-Tube Joints

A.4.7.1.1 Expanded only tube joints are an interference type joint that relies on the interfacial pressure between the outside diameter of the tube and the diameter of the hole in the tubesheet to provide for sufficient sealing capability.

A.4.7.1.2 Tube holes in the tubesheet are provided with annular ring grooves to improve the joint strength and pull-out load.

A.4.7.1.3 A proper expanded tube joint will result in the tubes being plastically deformed into the tubesheet holes and ring grooves while the tube hole deformation remains elastic, with adequate interfacial pressure after recovery to achieve the desired seal.

A.4.7.1.4 The ability of an expanded only joint to provide a sufficient seal depends upon many factors including the tube dimensions, operating temperatures, metallurgy of the tube and tubesheet including differences in yield stress and thermal expansion coefficients, and the tube layout and ligament.

A.4.7.2 Welded Tube-to-Tubesheet Joints

A.4.7.2.1 When a full strength welded tube-to-tubesheet joint is applied, it is not necessary to perform a tube expansion in addition to the strength weld for joint load strength alone.

A.4.7.2.2 If there is potential for crevice corrosion on the shell side (as is often a concern), in addition to welding the tubes should be expanded to within 3 mm (1/8 in.) from the shell side face of the tubesheet as to seal the gap between the tube and tube hole.

A.4.7.2.3 When conditions exist for which a reliable expanded only joint cannot be achieved, a strength welded joint is required to provide adequate mechanical strength. A strength welded joint will also provide for additional leak tightness compared to an expanded only joint.

A.4.7.2.4 In addition to the conditions under which strength welding is mandatory, the purchaser should consider strength welding for the following conditions:

- a) Where work hardening of the tube material during rolling could result in the potential for environmental cracking, e.g., duplex stainless steel tubes in wet hydrogen sulfide service, a strength-welded tube-to-tubesheet joint should be applied, and a light expansion (3 % wall reduction) should be applied if crevice corrosion from the shell side fluid is a concern.
- b) In services with different tube and tubesheet materials at operating temperatures for which thermal expansion differences can lead to joint relaxation. Examples: where austenitic stainless steel tubes and a ferritic steel tubesheet are used and the metal temperature is above 204 °C (400 °F), or where titanium tubes are used with carbon steel tubesheets.
- c) For fixed tubesheet exchangers in which operating conditions may result in significant tube-to-tubesheet joint loads.
- d) For fixed or floating tubesheet designs with a high differential in mean tube metal temperature between tube passes strength welding should be considered for mechanical end loading unless mockups with pull out tests are performed.
- e) Where titanium tubes are used (provided that the tube wall thickness is adequate for strength welding).
- f) In services where sudden temperature excursions can occur.
- g) In hydrogen service when the design temperature is in excess of 230 °C (450 °F). The joint should be expanded in addition to strength welded.

- h) In heat transfer fluid services with operating temperatures exceeding 204 °C (400 °F).

A.4.7.2.5 The design requirements for an expanded tube joint, including the consideration of the ring grooves, are primarily focused on achieving joint strength. Although a properly expanded joint provides adequate leak tightness for many services and metallurgies, the purchaser should consider the addition of a strength or seal weld when additional leak tightness is required for the following conditions:

- a) Where leakage from the high pressure side to the low pressure side is not acceptable for process reasons (e.g. where contamination of the lower pressure side would be detrimental to the process or pose an unacceptable hazard to utility systems such as heat media systems, steam/condensate, or cooling water).
- b) For exchangers handling steam or boiler feedwater at design conditions exceeding class 150 flange rating.
- c) For f_y ratios as noted in 7.5.7.2.g) that are greater than or equal to 0.6 and less than 1.0.
- d) For design pressures greater than 5200 kPa (ga) (750 psig).
- e) For exchangers in hydrogen and hydrogen charging service.

A.4.7.2.6 If welded tube-to-tubesheet joints are specified for dissimilar tubes and tubesheet material, weld overlay or cladding should be considered on the tubesheet to eliminate bimetallic welds.

A.4.7.2.7 When there is a large temperature difference between the tubesheet faces or dissimilar materials for the tubesheet and tubes, the limits described in Table 3 and in 9.10.1 may not be appropriate.

A.4.8 Bundle Handling Devices—Guidance to 7.5.8

For removable tube bundles with dimensions, mass, or baffle arrangements as those described in 7.5.8, the following options should be considered:

- a) Where allowable by the tube layout, the inclusion of bundle handling devices will provide additional rigidity to the bundle and can potentially allow for multiple support locations for lifting of horizontal bundles without damaging tubes. Bundle handling devices should be secured by welding to the transverse baffles and support plates. The bundle handling devices should be designed not to impede the flow of the inlet and outlet nozzles. The devices should also be designed to facilitate drainage of the shell. Bundle handling devices should not obstruct tube lanes or pass-partition lanes if 45 ° or 90 ° tube layouts are used.
- b) For designs in which it is not practical to incorporate bundle handling devices, the following methods should be considered when handling tube bundles:
 - Use of a specialty shell and tube bundle grapple which span and support multiple baffles without contacting the tubes
 - Use of temporary cylindrical shells which span and support multiple baffles without contacting the tubes
 - Use of temporary blocks or timbers to spread the lifting load to multiple baffles without contacting the tubes

A.4.9 Bolting under Insulation—Guidance to 7.7

Flanges should not be insulated when operating at temperatures which could lead to bolt elongation and potential reduction in gasket sealing stress. Note that the default 0.7 gasket relaxation fraction included in PCC-1 Appendix O may not be valid if bolting at elevated temperatures is covered with insulation.

Additionally, if an insulated flange experiences a small un-noticed leakage it may lead to degradation of the bolting encapsulated by the insulation.

If bolting / flanges operating at relatively high temperatures can be rapidly quenched / cycled due to weather conditions (rainstorms, etc.), it may be prudent to shield the flanges from rain whilst still allowing natural convection by the ambient air. The evaluation of such a condition can be performed on a case-by-case basis.

If personnel protection is a concern, wire mesh around the flanges is recommended over flange insulation for similar concerns mentioned above.

A.4.10 Girth Flange Joint Supplementary Design Requirements—Guidance to 7.7

A.4.10.1 By the very nature of their design, flanged joint assemblies in heat exchangers can result in operating temperature gradients either around the circumference of girth flanges, as a temperature differential between the two flanges or gasket seating surfaces that together make up the joint, or across the tubesheet face(s) or thickness. These temperature differentials influence flange deflection/rotation, tubesheet deflection, radial gasket shear, and/or loss of gasket compression which lead to higher risks of gasket leakage. These influences are not normally considered by conventional pressure vessel code analysis, or by the additional flange analysis that is included in 7.7.

A.4.10.2 Girth flanges may have a higher risk for gasket leakage where process fluid temperature differences exceed 110 °C (200 °F) at any of the following locations:

- Across the tube side face of the stationary tubesheet
- Across the face of a flanged channel cover
- Across the shell side face of the stationary tubesheet
- From the coldest point of the cold side of the stationary tubesheet to the hottest point on the hot side of the stationary tubesheet

A.4.10.3 When large temperature difference exists, the following could be evaluated to reduce the potential for leaks:

- Swapping of the shell inlet/outlet nozzles and re-evaluating the temperature differentials across the tubesheet thickness
- Reducing the number of tube passes and/or shell passes and re-evaluating the temperature differentials across the tubesheet thickness and tubesheet faces
- Eliminating one gasket at the stationary tubesheet by choice of a TEMA type C closure with integral tubesheet hubs
- Use of multiple shells in series and re-evaluating the temperature differentials across the tubesheet thickness and tubesheet faces
- Eliminating both gaskets at the stationary tubesheet by choice of:
 - TEMA type N closures (non-removable bundle) with integral tubesheet hubs, or
 - TEMA type C special closure with integral tubesheet hubs, in which the shell side girth flange can be shifted away from the tubesheet by a minimum distance with a removable bundle configuration, thus eliminating all gaskets at the tubesheet. Due to the weight of the bundle with integral channel and this special shell side flange design, a removable shell in lieu of a removable bundle may be easier to handle when servicing this exchanger style, or
 - Other specialty closure designs
- Incorporating weld-ring type gaskets
- To resolve high temperature differentials across a removable channel cover (if applicable), consider an internally gasketed pass box in the channel such that the removable channel cover operates at a uniform temperature. Care should be taken to minimize thermally induced high stresses areas in the pass box design.
- Revision of process temperatures, and re-evaluation of the temperature differentials.
- Where the high temperature differentials cannot be mitigated by the above methods and/or if the joints are subject to rapid operating temperature changes, additional analysis of the flanged joint may be required in order to assure design adequacy and leak tightness. The method of analysis should be and additional requirements to be applied should be specified or agreed upon with the purchaser.

A.4.11 Considerations and Limitations to Hydraulic Tensioning – Guidance to 7.7.12

A.4.11.1 For additional guidance see ASME PCC-1 Non Mandatory Appendix Q-4.

A.4.11.2 Washers often improve the translation of torque into bolt preload by providing a smooth and low friction bearing surface for the nut. They also act to protect the contact surface of the flange from damage caused by a turning nut, for example where stainless steel flanges and nuts are provided which may otherwise result in galling of the flange and nut after repeated assemblies. There may be cases however, where the use of washers are prohibited by some purchasers.

A.4.12 Gaskets—Guidance to 7.9

A.4.12.1 Spiral-wound Gaskets

Spiral-wound gaskets are not recommended for use in floating-head flanges as the centering rings may result in an increase in the floating-head flange diameter, and the shell diameter for pull-through (TEMA T-type) floating-heads.

A.4.12.2 Jacketed Gaskets

Key reasons why jacketed gaskets can be unreliable in heat exchangers are as follows:

- a) Radial shear susceptibility: Radial thermal expansion and contraction can cause the gasket to become damaged, leading to potential leakage paths.
- b) Poor recovery from compression: Metal jacketed gaskets may have less recover to their original shape after compression compared with other gasket types.
- c) Less tolerance to imperfections on the gasket contact surfaces.

A.5 Materials

A.5.1 Low-Temperature Structural Supports—Guidance to 8.1.9

For structural supports (e.g. saddles, clips, skirts, etc.) attached to exchangers whose shell side MDMT is below -29 °C (-20°F), the following should be considered to ensure adequate toughness:

- a) Materials shall be the same as the pressure parts (i.e. P numbers of ASME Code).
- b) For Stainless Steel equipment any 300 series permanent stainless steel material shall be acceptable as structural elements.
- c) MDMT of the permanent structural supports should be equal to the MDMT of the equipment unless lower temperature is specified by the purchaser
- d) Welding consumables should satisfy MDMT requirements of the equipment.

A.5.2 Gaskets Materials—Guidance to 8.2.5

When using a gasket containing graphite, it should be considered that it is almost the noblest material in the table of galvanic series. Some metals can be sacrificed when combined with graphite in certain environments. If graphite is to be used in contact with the flange in a service where the fluid contains sulfur, chlorides, halogens, or similar impurities, the purchaser should consider alternate filler or facing materials.

A.5.3 Integrally-finned Tubes—Guidance to 8.3.3 and 8.3.4

A.5.3.1 Paragraph UG-8(b) of the ASME B&PV Code, Section VIII, Div. 1 states that the tubes, after finning, shall have a temper or condition that conforms to one of those provided in the governing specifications, or, when specified,

they may be furnished in the “as-fabricated condition” where the finned portions of the tube are in the cold worked temper condition (as-finned) resulting from the finning operation, and the unfinned portions in the temper condition of the tube prior to finning.

A.5.3.2 While manufacturing practices may vary, integrally-finned tubes, and other types of enhanced shape tubes (e.g. twisted tubes), are typically supplied in the as-fabricated (cold-formed) condition, unless heat treatment is specified by the purchaser. Cold-forming results in high residual stresses, and for some alloys, the hardness can exceed the maximum allowable values stated in NACE MR0103 and MR0175 standards. Stress-relieving or solution-annealing type heat treatment may be required, depending on the tube material.

A.5.3.3 Some commonly used heat exchanger tube materials such as austenitic stainless steels, duplex stainless steels, Cu-Ni steels, and high-nickel alloys exhibit work-hardening or strain-hardening behavior during cold-forming of integral fins and other types of enhancements on the tube ID or OD, and as a result, can have an increased risk of environmental cracking as compared to bare tubes, due to an increase in hardness and introduction residual stress where the cold-forming has been applied. Increasing nickel content generally increases the material's work- or strain-hardening susceptibility. For these alloys, solution-annealing, to eliminate residual stresses and reduce material hardness, may be required to comply with NACE MR0103 or MR0175 standards, unless the manufacturer can demonstrate acceptable hardness in the cold-formed condition for integrally-finned tubes and other types of enhanced tube shapes through cross-sectional micro-hardness testing of representative tube samples.

A.5.3.4 Where environmental cracking mechanisms, other than wet H₂S or sour service cracking (sulfide stress cracking, HIC, SOHIC), are applicable (e.g. chloride SCC, carbonate SCC, caustic SCC, amine SCC, ethanol SCC, etc.), factors other than material hardness, such as operating temperature, concentration, and residual stresses influence cracking susceptibility, and therefore integrally-finned tubes and other types of enhanced tube shapes may also require heat treatment to be completed after cold-forming of fins to remove residual stresses. The purchaser shall indicate when heat treatment is required for these tube types considering the service environment and environmental cracking susceptibility of the tube material in the cold-formed condition. Where heat treatment is not practical, the purchaser should consider upgrading the tube material to an alloy that is more resistant to the applicable environmental cracking mechanism(s). API Recommended Practice 571 includes a more complete list of potential environmental cracking mechanisms.

A.6 Leak Testing—Guidance to 10.2.3

Helium leak testing should be applied for tube-to-tubesheet joints in hydrogen service with a design pressure greater than 6900 kPa (g) (1000 psig).

A.7 Preparation for Shipment

A.7.1 Protection—Guidance to 11.1

A.7.1.1 If water residues cannot be tolerated, equipment should be dried by one of the following methods:

- a) Blowing dry air or nitrogen, of relative humidity less than 15 % (usually dehumidified), through the heat exchanger and monitoring the outlet air until the relative humidity falls below 30 %.
- b) Evacuating the heat exchanger with a vacuum pump to an absolute pressure of between 0.4 kPa (0.06 psi) and 0.5 kPa (0.075 psi).

A.7.1.2 After draining and drying, internal surfaces can be protected against corrosion by the addition of a volatile corrosion inhibitor, or by blanketing with an inert gas such as nitrogen. Some jurisdictions treat equipment pressurized above 100 kPa (ga) (14.5 psig) as in service and may result in additional requirements.

A.7.1.3 To ensure integrity of the inert gas blanket during shipping and storage the pressure should be monitored due to changes in ambient conditions.

A.8 Additional Considerations for Specifying Parameters for High Temperature Hydrogen Service

A.8.1 Introduction—Guidance to 13.1

HTHA is a potential material degradation mechanism that can result in surface decarburization, internal decarburization, fissuring, or cracking of carbon and low-alloy steels when subject to high temperature hydrogen service. For more information on HTHA, refer to API Recommended Practice 571.

A.8.2 Setting Maximum Operating or Design Temperature—Guidance to 13.2

A.8.2.1 Material selection temperatures should be made with considerations for normal operating conditions including partial pressure of hydrogen, applied safety margins, alternate operating conditions, and non-uniform fouling.

A.8.2.2 Most exchangers in services that are subject to HTHA are also subject to non-uniform fouling where the highest rate of the fouling occurs at the hot end of the process. In practice, design fouling factors used for initial design do not reflect this phenomena, and fouling is applied as uniform across the exchanger. Operating data, if available, should be evaluated for non-uniform fouling trends and be applied to designs in similar services. Where historical data is not available, the designer should consider the effects of higher rates of hot-end fouling when predicting maximum exposure temperatures.

Similarly, if there are heat exchangers in series in high temperature service, designers sometimes assume operating conditions over the life of the plant based on the maximum expected degree of fouling. However, if the hottest exchanger in the series fouls more than expected and no longer cools the process stream sufficiently, the next exchanger(s) in the series might see higher temperatures than it was designed for. It may then become susceptible to HTHA or other damage mechanisms for which the materials of construction may not be suitable. For additional information on damage mechanisms, see API Recommended Practice 571.

A.8.2.3 Clean exchanger performance should be evaluated in the temperature analysis. When operating in the clean condition, the cold stream would be expected to exit the heat exchanger at temperatures above the expected process operating temperature unless external temperature controls are used.

A.8.2.4 When a hot bypass is required for process conditions (i.e. temperature control around the exchanger), the exchanger on the downstream side of the reintroduction of the bypass of the hot fluid requires detailed analysis to account for the uncooled material flowing through the bypass line. If a flow bypass is not accounted for, the downstream equipment on the hot fluid side may be exposed to unanticipated and undetected material degradation. Analysis of temperatures are required in both clean and fouled operating conditions.

A.8.2.5 When a cold bypass is required for hot stream outlet temperature control, in addition to A.9.2.4, the equipment requires a detailed analysis to account for the additional heating and higher outlet temperature of the cold fluid that passes through the exchanger, prior to remixing with the bypass stream. This can result in the exchanger operating above the material limits since the cooling fluid has been limited. Temperatures should be analyzed in both clean and fouled operating conditions.

A.8.3 Materials Selection for HTHA Services—Guidance to 13.3

A.8.3.1 Materials in contact with the process fluid in HTHS should be selected in accordance with API Recommended Practice 941. All other applicable damage mechanisms should also be reviewed for the final materials selection. For additional information on damage mechanisms, see API Recommended Practice 571.

A.8.3.2 All operating modes should be considered to define the temperature and hydrogen partial pressure to be used in selecting materials per API Recommended Practice 941.

A.8.3.3 A safety margin may be applied when using the API Recommended Practice 941 curves, primarily due to the fact that the curves are based on experience-based data. This margin can, where available, be based on user experience or practices. If a clad or weld overlaid material is used, the base material should be resistant to the hydrogen conditions, with no credit taken for the cladding or overlay.

A.8.3.4 Material breaks are not recommended for exchangers operating in high temperature hydrogen service.

A.8.3.5 For catalytic reforming feed effluent heat exchangers (commonly known as Texas towers) that typically have shell lengths greater than 15 m (50 ft) and a large operating temperature change from inlet to outlet, a material break in the shell is sometimes used. Operators should monitor the shell temperature by use of redundant temperature indicators located on the hot side of the material break. The location, quantity, and type of temperature indicator, e.g. thermowells or skin temperature, should be specified by the purchaser.

A.8.4 Establishing Integrity Operating Window (IOW) for Services Subject to HTHA—Guidance to 13.3.1

A.8.4.1 The establishment, monitoring, and maintaining of IOWs is a vital component of integrity management to control and prevent material degradation and assist in the exchanger inspection planning process, including risk-based inspection. See API Recommended Practice 941 and API Recommended Practice 584.

A.8.4.2 Monitoring of equipment subject to HTHA requires that both chemical and physical operating parameters be examined including operating pressure, hydrogen partial pressure, and temperatures at specific locations of interest based on the exchanger and system design parameters and metallurgy.

A.8.4.3 As HTHA is a time-dependent degradation mechanism, a standard IOW limit is typically applied. If this limit is exceeded over a specified period of time, increased degradation rates or new damage mechanisms could be experienced. Therefore, exceeding standard IOW limits may be acceptable over a specified (limited) period of time. All such cases should be tracked to understand the cumulative effect.

There may be cases where the exchanger mechanical design temperature(s) are higher than the temperatures indicated in the equipment IOWs. Care must be exercised to refer to the IOW temperature limits and not only the mechanical design temperatures when considering operating temperature changes for exchangers in high temperature hydrogen service.

IOW critical limits are also applied; if this limit is exceeded, rapid deterioration could occur. When IOW critical limits are exceeded, an alarm is generated, and immediate action is required to return the process variable(s) to within their IOW.

A.8.4.4 The establishment of IOW parameter limits for new heat exchangers in terms of operating temperature and hydrogen partial pressure for exchangers in HTHS depends upon many factors, including the following:

- a) materials of construction and any heat treatment applied.
- b) previous history of HTHA for the intended service.
- c) safety margins that are applied to the API Recommended Practice 941 guidelines for the selection of materials.
- d) purchaser's philosophy with respect to allowable duration of short-term exceedances of operating parameters used for material selection, e.g. end of run operation or catalyst sulfiding, etc.
- e) the length of time that equipment will be exposed to extreme or off normal operating conditions.
- f) location and accuracy of temperature monitoring devices with respect to critical locations of temperature and/or materials of construction breaks, including influence of hot/cold stream bypasses.
- g) effectiveness and rigor of the heat exchanger/process monitoring program with respect to being able to accurately predict metal temperatures at specific locations where direct temperature indication is not available, including the consideration of non-linear heat exchanger fouling.

For more information on IOWs, refer to API Recommended Practice 584.

A.8.5 Intermediate Temperature Connections—Guidance to 13.4

The installation of temperature recording devices is essential for monitoring temperature trends between shells (or channels) of a heat exchanger unit to ensure operating within the IOW limits.

Annex B (informative)

Shell-and-Tube Heat Exchanger Checklist

The checklist (Table B.1) is used to record the specific requirements the purchaser shall make in response to the sections in this standard alongside which bullets (●) are used to indicate that more information is required or it is necessary to make a decision.

Completion of the checklist is the responsibility of the purchaser.

Table B.1—Checklist for Shell-and-Tube Heat Exchangers

Section	Item	Requirement	
4.1	Specify (or agree) pressure design code.	Complete on datasheet	
4.3	Compliance with applicable local regulations	Complete on datasheet	
4.7	Specify if the shell side fluid characteristics require special considerations? If yes, provide detailed information.	Yes (provide requirements)	No
	Specify if the tube side fluid characteristics require special considerations? If yes, provide detailed information.	Yes (provide requirements)	No
4.8	Is cyclic service design required? If yes provide detailed information.	Yes (provide requirements)	No
4.10	Is the unit subject to sour or wet hydrogen sulfide service on the shell side?	Yes	No
	Is the unit subject to sour or wet hydrogen sulfide service on the tube side?	Yes	No
4.11	Is the shell side in hydrogen service?	Yes	No
	Is the tube side in hydrogen service?	Yes	No
4.12	Is input data required to determine the need for an expansion joint?	Yes (provide details on datasheet)	No
4.15	The design loads and load combinations to be used for the heat exchanger design.	Complete on datasheet	
4.17	Specify if the exchanger has been selected to mitigate a complete tube rupture scenario.	Yes (provide requirements)	No
4.18	Specify if the service is in HTHS. If yes, specify material selection temperatures for each side	Yes (provide requirements)	No
6.1.2	Specify if thermal and hydraulic design calculations including vibration analysis are to be submitted for review?	Yes	No
6.3	Specify the format and quantities for the listed final documents	Provide Requirements	
7.1.1	Specify the maximum design temperature and a minimum design metal temperature (MDMT) for the shell and tube sides.	Complete on datasheet	
7.5.2.7	Specify if integrally-hubbed tubesheets are required.	Yes	No
7.5.7.4	When strength welds are applied, the degree of expansion and use of grooves shall be specified or agreed.	Provide requirements	
7.5.8	Specify if additional methods of support to aid in handling of the bundle.	Yes (provide requirements)	No
7.6.1	Specify the required flange design code	Yes	No
7.6.7	Is insulation required? If yes, specify insulation thickness: Shell.	Yes (provide details on datasheet)	No
	Is insulation required? If yes, specify insulation thickness: Channel.	Yes (provide details on datasheet)	No
7.6.9	Are chemical cleaning connections required?	Yes	No
7.6.11	If nozzle sizes are larger than those listed in Tables 5 and 6. If yes, specify the moments and forces.	Yes (provide details on datasheet)	No
7.6.13	Specify if reinforcing pads shall not be used in sour or wet hydrogen	Yes	No

Section	Item	Requirement	
	sulfide services.		
7.7.19.9	Specify if the joint component approach per ASME PCC-1, App. O shall be applied to the floating head girth flange design.	Yes	No
7.11.2	Specify if any supplemental requirements are required for low chrome steels in high temperature or high pressure hydrogen service. If yes, provide detailed information.	Yes	No
8.1.4	Specify required materials of construction, including bolting and gaskets.	Provide details on datasheet	
8.1.7	Specify if stainless steel grades 321 and 347 shall be stabilized after solution annealing heat treatment.	Yes	No
8.3.4	Specify if heat treatment is required for integrally finned-tubes and other types of enhanced shape tubes	Yes	No
9.6.3	Specify if heat treatment required after bending for U-tubes as it can induce susceptibility to stress corrosion?	Yes	No
9.6.4	Is heat treatment of ferritic and martensitic stainless steels, stabilized austenitic stainless steels, duplex stainless steels, copper, copper nickel, or high nickel alloys (Ni > 30 %) U-tubes required after cold working?	Yes	No
9.6.9	Specify if heat treated portion of U-tubes shall be de-scaled.	Yes	No
9.6.12	Is postweld heat treatment required for process reasons on the shell side?	Yes	No
	Is postweld heat treatment required for process reasons on the tube side?	Yes	No
10.1.14	Specify if carbon steel plate in sour or wet hydrogen sulfide service shall be subjected to an ultrasonic lamination check.	Yes	No
10.1.18	Specify if U-bends shall be metallurgically examined and corrosion tested prior to the start of production bending	Yes	No
10.1.23	Specify if clad and weld overlay construction shall be tested for susceptibility to hydrogen disbonding.	Yes	No
10.1.24	Specify if the material composition and grade for pressure part components including cladding, overlay, weld consumables, heat transfer tube(s), and completed welds shall be verified by positive material identification.	Yes	No
10.2.3	Specify if tube-to-tubesheet joint integrity shall be tested after final expansion of the tubes by a helium leak test.	Yes	No
10.2.7	Are there additional requirements for drying or preservation? If yes, provide detailed information.	Yes	No
11.1.7	Are there requirements for surface preparation and protection? If yes, provide detailed information.	Yes	No
11.1.10	Is inert gas purge and fill required?	Yes	No
12.1	Are the additional requirements specified in Section 12 required for the shell side?	Yes	No
	Are the additional requirements specified in Section 12 required for the tube side?	Yes	No
13.3.1	Specify if the nominal composition of construction of the wetted surfaces of either, or both, the shell side and tube side vary. If yes, provide detailed information.	Either	Both
13.4.2	Specify if skin (metal) temperature measuring devices are required. If yes, detailed information to be agreed.	Yes	No
13.6.5	Specify if welds shall be provided with baseline UT examination reports. If yes, detailed information to be agreed.	Yes	No

Annex C

(informative)

Shell-and-Tube Heat Exchanger Datasheets ⁵

The following datasheets are provided to assist the designer, vendor, and purchaser to specify the data necessary for the design of a shell-and-tube heat exchanger for petroleum, petrochemical and natural gas services.

Completion of the datasheet is a joint responsibility of the purchaser and vendor. The purchaser (owner or contractor) is responsible for the process data, which define the purchaser's explicit requirements.

After the exchanger has been fabricated, the vendor should complete the datasheets to make a permanent record that accurately describes the equipment "as-built."

Additional datasheets may be required to define the shell-and-tube heat exchanger and examples are included in Figure C.1 and Figure C.2.

- Connection schedule (see page 2 of Figure C.1 and Figure C.2), materials of construction (see page 2 of Figure C.1 and Figure C.2): gaskets (see page 2 of Figure C.1 and Figure C.2). These are used if the designer/ user requires such level of detail to define materials for individual components. These sections may not be necessary if page one defines the information sufficiently.
- Additional information, remarks, sketches, etc. (see page 3 of Figure C.1 and Figure C.2). These are used for additional remarks if sufficient space is not available on pages 1 or 2. These are also used for schematic sketches to illustrate required features of the design and tube layouts, if necessary, to provide this information to the manufacturer.
- Additional process information (see page 4 of Figure C.1 and Figure C.2). One or two pages can be necessary if boiling and/or condensing fluids with ranges of physical properties occur.
- Additional process information for thermal expansion design conditions (see page 5 of Figure C.1 and Figure C.2). This is required only for exchangers such as fixed tubesheet or single-pass floating-head type exchangers where thermal expansion must be considered for the mechanical design. This optional page should be included if operating information is supplied to the designer for determination of the metal temperatures necessary for expansion joint reviews.
- Additional cyclic service design data (see page 6 of Figure C.1 and Figure C.2). This is required only for exchangers where the unit is subject to cyclic service.

⁵ Users of this Annex should not rely exclusively on the information contained in this document. Sound business, scientific, engineering, and safety judgment should be used in employing the information contained herein. Where applicable, authorities having jurisdiction should be consulted.

Company		SHELL-AND-TUBE HEAT EXCHANGER				Engineering contractor	
DATA SHEET (SI UNITS)							
P.O. No.:		Doc. No.:				Page 1 of	
1 Client:		Location:					
2 Process unit:		Item No.:					
3 Job No.:		Fabricator:					
4 Service of unit:		No. of units:					
5 Size:		TEMA Type:		Connected in:		Parallel Series	
6 Effective surface per unit (plain/finned):		m		Shells/unit:		Effective surface per shell: m ²	
7 PERFORMANCE OF ONE UNIT				SHELL SIDE		TUBE SIDE	
				Inlet		Outlet	
9 Fluid name:							
10 Fluid quantity, total:				kg/h			
11 Vapor (relative molecular mass):				kg/h			
12 Liquid:				kg/h			
13 Steam:				kg/h			
14 Water:				kg/h			
15 Non-condensable (relative molecular mass):				kg/h		/	
16 Temperature:				°C			
17 Density (vapor/liquid):				kg/m ³			
18 Viscosity (vapor/liquid):				mPa·s			
19 Specific heat (vapor/liquid):				kJ/(kg·K)			
20 Thermal conductivity (vapor/liquid):				W/(m·K)			
21 Specific latent heat:				kJ/kg @ °C		@	
22 Inlet pressure:				kPa (ga)			
23 Velocity:				m/s			
24 Pressure drop (allowable/calculated):				kPa		/	
25 Fouling resistance:				m ² ·K/W			
26 Average film coefficient:				W/(m ² ·K)			
27 Heat exchanged:				kW		Mean temperature difference (MTD): °C corrected/weighted	
28 Heat transfer rate:				W/(m ² ·K)		Required: Fouled: Clean:	
29 ρV ² :				kg/(m ² ·s ²)		Inlet nozzle: Bundle entrance: Bundle exit:	
30 Hydrogen Service:				Tube Side (Y/N) Shell Side (Y/N) Sour Service:		Tube Side (Y/N) Shell Side (Y/N) Cyclic Service: (Y/N)	
31 CONSTRUCTION PER SHELL							
32 Tube Number:		OD: mm		NOZZLES – Number, Size and Rating			
33 Tube Wall Thickness:		mm minimum wall/average wall		SHELL SIDE		TUBE SIDE	
34 Tube Pitch:		mm Tube layout:		Inlet:			
35 Tube Length:		m Tube type: plain/finned/other		Outlet:			
36 Tube to tubesheet joint:				Intermediate:			
37 Shell diameter:				mm (ID/OD) /		Vent:	
38 Cross-baffle type:				Drain:			
39 Baffle Spacing: c/c				mm No. of cross passes:		Pressure relief:	
40 Baffle Cut:				% vertical/horizontal		Design pressure: kPa (ga)	
41 Tube support type:				Vacuum: kPa (abs)			
42 Longitudinal baffle seal type:				Design temp. (Max/MDMT): °C		/ /	
43 Bypass seal type:				Number of passes per shell:			
44 Impingement protection: (Y/N) Type:				Corrosion allowance: mm			
45 MATERIALS OF CONSTRUCTION							
46 Shell:		Tubes:		GASKETS			
47 Shell cover:				Shell side:			
48 Channel or bonnet:				Tube side:			
49 Channel cover:				Floating head:			
50 Floating head cover/bolts:				Spare Gaskets required:			
51 Tubesheet: Stationary:		Floating:		Test ring required: (Y/N)			
52 Baffles: Cross:		Longitudinal:		Insulation - shell: mm			
53 Tube support material:				Insulation - channel: mm		inlet/outlet	
54 Expansion joint type:				Expansion joint material:			
55 Pressure design code:		Stamp:		Calculate MAWP: (Y/N)		TEMA Class:	
56 REMARKS:							
57							
58							
Rev. No.		Description		Date		Prepared by Reviewed by	

Company		SHELL-AND-TUBE HEAT EXCHANGER CONNECTIONS AND MATERIALS DATA SHEET (SI UNITS)			Engineering contractor	
P.O. No.:		Doc. No.:			Page 2 of	
1 Client:		Location:				
2 Process unit:		Item No.:				
3 Job No.:		Fabricator:				
4 CONNECTION SCHEDULE (Optional)						
5	Mark	Number required	Size	Rating	Facing	Description
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						Chemical Cleaning
17 MATERIALS OF CONSTRUCTION (Optional)						Corrosion allowance
18	Shell:					mm
19	Head:					mm
20	Pipe/stub ends:					mm
21	Nozzle necks:					mm
22	Nozzle flanges:					mm
23	Body flanges:					mm
24	Expansion joint:					mm
25	Support:					mm
26	Bolting (internal):					mm
27	Bolting (external):					mm
28	Nozzle reinforcement:					mm
29	Tubes:					
30	Tubesheets:					mm
31	Bonnet/channel:					mm
32	Bonnet head(s):					mm
33	Channel cover(s):					mm
34	Body flanges:					mm
35	Pipe/stub ends:					mm
36	Bolting (internal):					mm
37	Bolting (external):					mm
38	Nozzle reinforcement:					mm
39	Nozzle necks:					mm
40	Nozzle flanges:					mm
41	Baffles, spacers, tie rods:					mm
42						
43						
44						
45						
46						
47						
48						
49 GASKETS (Optional)			MECHANICAL DATA (Optional)			
50	Shell side:	Thickness:	mm	MAWP (hot and corroded): kPa (ga)		
51	Y =	Pa	m =	MAP (new and cold): kPa (ga)		
52	Tube side:	Thickness:	mm	Hydrotest pressure:		
53	Y =	Pa	m =	Field:	kPa (ga)	Shop: kPa (ga)
54	Floating head:	Thickness:	mm	Mass:	Empty:	kg
55	Y =	Pa	m =	Full of water:	kg	
Rev. No.		Revision		Date		Prepared by
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Company		SHELL-AND-TUBE HEAT EXCHANGER ADDITIONAL INFORMATION DATA SHEET (SI UNITS)		Engineering contractor	
P.O. No.:		Doc. No.:		Page 3 of	
1	Client:		Location:		
2	Process unit:		Item No.:		
3	Job No.:		Fabricator:		
4	ADDITIONAL REMARKS, SKETCHES, ETC. (Optional)				
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					Reviewed by

Company	SHELL-AND-TUBE HEAT EXCHANGER ADDITIONAL PROCESS INFORMATION DATA SHEET (SI UNITS)				Engineering contractor								
P.O. No.:	Doc. No.:				Page 5 of								
1	Client:			Location:									
2	Process unit:			Item No.:									
3	Job No.:			Fabricator:									
4													
5													
6													
7	DESIGN CONDITIONS FOR EXPANSION JOINT (Optional)												
8	Case^a	SHELL SIDE				TUBE SIDE							
9		Flow condition^b	Fluid temperature		Pressure^d	Mean metal temp.^e	Number of cycles^f	Flow condition^b	Fluid temperature		Pressure^d	Mean metal temp.^e	Number of cycles^f
10			Inlet	Outlet^c						Inlet			
11		°C	°C	kPa (ga)	°C			°C	°C	kPa (ga)	°C		
12													
13													
14													
15													
16													
17													
18													
19													
20													
21													
22	Determine the mean shell and tube metal temperatures at the listed cases. Evaluate the need for an expansion joint based on the metal temperatures at these conditions with either or both sides clean or with specified fouling.												
23	Unless otherwise stated, operation in accordance with the recommendations of the TEMA Standards, paragraph E3.2, "Operating Procedures", is assumed.												
24	^a A case which may affect design (e.g. steam-out, upset, etc.).												
25	^b F = flowing (specify flow rate), S= stagnant, E = empty.												
26	^c Outlet temperature if known, thermal designer determines for other conditions.												
27	^d Specify design pressure for operating conditions. Use maximum actual pressure at other conditions.												
28	^e Mean metal temperature to be specified by thermal designer.												
29	^f Number of cycles that will occur during life of the exchanger for each case.												
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Rev. No.		Revision		Date		Prepared by		Reviewed by					

Company		SHELL-AND-TUBE HEAT EXCHANGER CYCLIC SERVICE INFORMATION DATA SHEET (SI UNITS)		Engineering Contractor	
P.O. No.:		Doc. No.:		Page 6 of	
1	Client:		Location:		
2	Process unit:		Item No.:		
3	Job No.:		Fabricator:		
4					
5					
6	Description of Cyclic Service Operation				
7					
8					
9					
10	Condition	Time	Duration	Composition	Flow Rate
11		(h/m)	(h/m)		(kg/h)
12	Initial	0.0			
13					
14					
15					
16					
17					
18					
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21					
22					
23					
24					
25					
26					
27					
28					
29					
30					
31					
32	Condition	Time	Duration	Temperature	Pressure
33		(h/m)	(h/m)	(°C)	kPa (ga)
34	Initial	0.0			
35					
36					
37					
38					
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52	NOTES:				
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Rev. No.		Revision		Date	Prepared by
					Reviewed by

Company		SHELL-AND-TUBE HEAT EXCHANGER				Engineering contractor	
DATA SHEET (US CUSTOMARY UNITS)							
P.O. No.:		Doc. No.:				Page 1 of	
1 Client:		Location:					
2 Process unit:		Item No.:					
3 Job No.:		Fabricator:					
4 Service of unit:		No. of units:					
5 Size:		TEMA Type:		Connected in:		Parallel Series	
6 Effective surface per unit (plain/finned):		ft ²		Shells/unit:		Effective surface per shell: ft ²	
7 PERFORMANCE OF ONE UNIT		SHELL SIDE		TUBE SIDE			
		Inlet		Outlet		Inlet	
9 Fluid name:		lb/h					
10 Fluid quantity, total:		lb/h					
11 Vapor (relative molecular mass):		lb/h					
12 Liquid:		lb/h					
13 Steam:		lb/h					
14 Water:		lb/h					
15 Non-condensable / relative molecular mass:		lb/h		/		/	
16 Temperature:		°F					
17 Density (vapor/liquid):		lb/ft ³					
18 Viscosity (vapor/liquid):		cP					
19 Specific heat (vapor/liquid):		Btu/(lb·°F)					
20 Thermal conductivity (vapor/liquid):		Btu/(h·ft·°F)					
21 Specific latent heat:		Btu/lb @ °F		@		@	
22 Inlet pressure:		psig					
23 Velocity:		ft/s					
24 Pressure drop (allowable/calculated):		psi		/		/	
25 Fouling resistance:		h·ft ² ·°F/Btu					
26 Average film coefficient:		Btu/(h·ft ² ·°F)					
27 Heat exchanged:		Btu/h		Mean temperature difference (MTD):		°F corrected/weighted	
28 Heat transfer rate:		Btu/(h·ft ² ·°F)		Required:		Fouled: Clean:	
29 ΔV ² :		lb/(ft·s ²)		Inlet nozzle:		Bundle entrance: Bundle exit:	
30 Hydrogen Service:		Tube Side (Y/N) Shell Side (Y/N)		Sour Service:		Tube Side (Y/N) Shell Side (Y/N) Cyclic Service: (Y/N)	
31 CONSTRUCTION PER SHELL							
32 Tube Number:		OD: in		NOZZLES – Number, Size and Rating			
33 Tube Wall Thickness:		in minimum wall/average wall				SHELL SIDE TUBE SIDE	
34 Tube Pitch:		in Tube layout:		Inlet:			
35 Tube Length:		ft Tube type: plain/finned/other		Outlet:			
36 Tube to tubesheet joint:				Intermediate:			
37 Shell diameter:		in (ID/OD) /		Vent:			
38 Cross-baffle type:				Drain:			
39 Baffle Spacing: c/c		in No. of cross passes:		Pressure relief:			
40 Baffle Cut:		% vertical/horizontal		Design pressure:		psig	
41 Tube support type:				Vacuum:		psia	
42 Longitudinal baffle seal type:				Design temp. (Max/MDMT):		°F / /	
43 Bypass seal type:				Number of passes per shell:			
44 Impingement protection: (Y/N) Type:				Corrosion allowance:		in	
45 MATERIALS OF CONSTRUCTION							
46 Shell:		Tubes:		GASKETS			
47 Shell cover:				Shell side:			
48 Channel or bonnet:				Tube side:			
49 Channel cover:				Floating head:			
50 Floating head cover/bolts:				Spare Gaskets required:			
51 Tubesheet: Stationary:		Floating:		Test ring required:		(Y/N)	
52 Baffles: Cross:		Longitudinal:		Insulation - shell:		in	
53 Tube support material:				Insulation - channel:		in inlet/outlet	
54 Expansion joint type:				Expansion joint material:			
55 Pressure design code:		Stamp:		Calculate MAWP: (Y/N)		TEMA Class:	
56 REMARKS:							
57							
58							
Rev. No.		Revision		Date		Prepared by Reviewed by	

Company		SHELL-AND-TUBE HEAT EXCHANGER CONNECTIONS AND MATERIALS DATA SHEET (US CUSTOMARY UNITS)			Engineering contractor	
P.O. No.:		Doc. No.:			Page 2 of	
1	Client:			Location:		
2	Process unit:			Item No.:		
3	Job No.:			Fabricator:		
4	CONNECTION SCHEDULE (Optional)					
	Mark	Number required	Size	Rating	Facing	Description
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						Chemical cleaning
16	MATERIALS OF CONSTRUCTION (Optional)					Corrosion allowance
17	Density (vapor/liquid):					in
18	Viscosity (va Head):					in
19	Specific heat Pipe/stub ends:					in
20	Thermal con: Nozzle necks:					in
21	Nozzle flanges:					in
22	Body flanges:					in
23	Expansion joint:					in
24	Support:					in
25	Bolting (internal):					in
26	Bolting (external):					in
27	Nozzle reinforcement:					in
28	Tubes:					
29	Tubesheets:					in
30	Bonnet/channel:					in
31	Bonnet head(s):					in
32	Channel cover(s):					in
33	Body flanges:					in
34	Pipe/stub ends:					in
35	Bolting (internal):					in
36	Bolting (external):					in
37	Nozzle reinforcement:					in
38	Nozzle necks:					in
39	Nozzle flanges:					in
40	Baffles, spacers, tie rods:					in
41						
42						
43						
44						
45						
46						
47						
48	GASKETS (Optional)			MECHANICAL DATA (Optional)		
49	Shell side:	Thickness:	in	MAWP (hot and corroded):	psig	
50	Y =	psi	m =	MAP (new and cold):	psig	
51	Tube side:	Thickness:	in	Hydrotest pressure:		
52	Y =	psi	m =	Field:	psig	Shop: psig
53	Floating head:	Thickness:	in	Mass: Empty:	lb	Bundle: lb
54	Y =	psi	m =	Full of water:	lb	
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Company		SHELL-AND-TUBE HEAT EXCHANGER ADDITIONAL INFORMATION DATA SHEET (US CUSTOMARY UNITS)		Engineering contractor					
P.O. No.:		Doc. No.:		Page 3 of					
1	Client:		Location:						
2	Process unit:		Item No.:						
3	Job No.:		Fabricator:						
4	ADDITIONAL REMARKS, SKETCHES, ETC. (Optional)								
5									
6									
7									
8									
9									
10									
11						Vapor (relative molecular mass):			
12									
13									
14									
15									
16									
17						Density (vapor/liquid):			
18						Viscosity (vapor/liquid):			
19						Specific heat (vapor/liquid):			
20						Thermal conductivity (vapor/liquid):			
21									
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Company	SHELL-AND-TUBE HEAT EXCHANGER ADDITIONAL PROCESS INFORMATION DATA SHEET (US CUSTOMARY UNITS)	Engineering Contractor
P.O. No.:	Doc. No.:	Page 4 of

1 Client:	Location:
2 Process unit:	Item No.:
3 Job No.:	Fabricator:
4	
5	
6	
7	
8	
9 Fluid Name: _____	Reference Pressure 1: _____ psia
10 Pressure _____	Temp. _____
11 Enthalpy _____	Vapor Mass Fraction _____
12	
13	
14	
15	
16	
17	
18	
19	
20	
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24	
25	
26 Density _____	Density _____
27 Vapor _____	Liquid _____
28 lb/ft ³ _____	lb/ft ³ _____
29 Viscosity _____	Viscosity _____
30 Vapor _____	Liquid _____
31 cP _____	cP _____
32 Thermal _____	Thermal _____
33 Cond. Vap. _____	Cond. Liq. _____
34 Btu/h-ft-°F _____	Btu/h-ft-°F _____
35 Sp. Heat _____	Sp. Heat _____
36 Vapor _____	Liquid _____
37 Btu/lb-°F _____	Btu/lb-°F _____
38 Surface _____	Liquid _____
39 Tension _____	Crit. Press. _____
40 dyne/cm _____	Crit. Press. _____
41 psia _____	°F _____
42	
43	
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58 Fluid Name: _____	Reference Pressure 1: _____ psia
59 Pressure _____	Temp. _____
60 Enthalpy _____	Vapor Mass Fraction _____
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11 Density _____	Density _____
12 Vapor _____	Liquid _____
13 lb/ft ³ _____	lb/ft ³ _____
14 Viscosity _____	Viscosity _____
15 Vapor _____	Liquid _____
16 cP _____	cP _____
17 Thermal _____	Thermal _____
18 Cond. Vap. _____	Cond. Liq. _____
19 Btu/h-ft-°F _____	Btu/h-ft-°F _____
20 Sp. Heat _____	Sp. Heat _____
21 Vapor _____	Liquid _____
22 Btu/lb-°F _____	Btu/lb-°F _____
23 Surface _____	Liquid _____
24 Tension _____	Crit. Press. _____
25 dyne/cm _____	Crit. Press. _____
26 psia _____	°F _____

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Date	Prepared by
Reviewed by	

Company		SHELL-AND-TUBE HEAT EXCHANGER EXPANSION JOINT INFORMATION DATA SHEET (US CUSTOMARY UNITS)					Engineering contractor						
P.O. No.:		Doc. No.:					Page 5 of						
1	Client:					Location:							
2	Process unit:					Item No.:							
3	Job No.:					Fabricator:							
4													
5													
6													
7	DESIGN CONDITIONS FOR EXPANSION JOINT (Optional)												
8	Case ^a	SHELL SIDE					TUBE SIDE						
9		Flow condition ^b	Fluid temperature		Pressure ^d	Mean metal temp. ^e	Number of cycles ^f	Flow condition ^b	Fluid temperature		Pressure ^d	Mean metal temp. ^e	Number of cycles ^f
10		Vapor (relative molecular mass).	Inlet	Outlet ^c	psig	°F			Inlet	Outlet ^c	psig	°F	
11			°F	°F					°F	°F			
12													
13													
14													
15													
16													
17	Density (vapor/liq uid):												
18	Viscosity (vapor/liq uid):												
19	Specific heat (vapor/liq uid):												
20	Thermal conductiv ity (vapor/liq uid):												
21													
22	Determine the mean shell and tube metal temperatures at the listed cases. Evaluate the need for an expansion joint based on the metal temperatures at these conditions with either or both sides clean or with specified fouling.												
23	Unless otherwise stated, operation in accordance with the recommendations of the TEMA Standards, paragraph E3.2, "Operating Procedures", is assumed.												
24	^a A case which may affect design (e.g. steam-out, upset, etc.). ^b F = flowing (specify flow rate), S = stagnant, E = empty. ^c Outlet temperature if known, thermal designer determines for other conditions. ^d Specify design pressure for operating conditions. Use maximum actual pressure at other conditions. ^e Mean metal temperature to be specified by thermal designer. ^f Number of cycles that will occur during life of the exchanger for each case.												
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Company		SHELL-AND-TUBE HEAT EXCHANGER CYCLIC SERVICE INFORMATION DATA SHEET (US CUSTOMARY UNITS)		Engineering Contractor	
P.O. No.:		Doc. No.:		Page 6 of	
1	Client:		Location:		
2	Process unit:		Item No.:		
3	Job No.:		Fabricator:		
4					
5					
6	Description of Cyclic Service Operation				
7					
8					
9					
10	Condition	Time	Duration	Composition	Flow Rate
11	Vapor (relative rr	(h/m)	(h/m)		(lb/h)
12	Initial	0.0			
13					
14					
15					
16					
17					
18					
19					
20					
21					
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23					
24					
25					
26					
27					
28					
29					
30					
31					
32	Condition	Time	Duration	Temperature	Pressure
33		(h/m)	(h/m)	(°F)	(psig)
34	Initial	0.0			
35					
36					
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51	NOTES:				
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Rev. No.		Revision		Date	Prepared by
					Reviewed by

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- [3] EN 13445 (all parts) ⁶, *Unfired pressure vessels*
- [4] ASME *Boiler and Pressure Vessel Code (BPVC), Section II, Materials*
- [5] WRC BUL 537 ⁷, *Precision Equations and Enhanced Diagrams for Local Stresses in Spherical and Cylindrical Shells Due to External Loadings for Implementation of WRC Bulletin 107*
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⁷ The Welding Research Council, P.O. Box 201547, Shaker Heights, Ohio 44122, www.forengineers.org.

⁸ ASTM International, 100 Barr Harbor Drive, West Conshohocken, Pennsylvania 19428, www.astm.org.

[18] API Recommended Practice 581, *Risk Based Inspection Technology*

[19] API Recommended Practice 584, *Integrity Operating Windows*